QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

corda @ 80033580ecfd · R3 — public repository (passive review)

Organization	QSI
Report date	2026-11-06 15:38 UTC
Report version	1
Audit run	ab2a7138-099f-4102-a6ca-533220ad3fb9 · Document audit · 2026-11-06 15:38 UTC
Analyzer engine	qsi-engine 1.3.0
Risk profile	Crypto exchange, custodian or wallet
Content SHA-256	`09dd82bf776b5adb6af7367e5cccfe5b259eb10fde085b2eef6d9bb38f43d5c6`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

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This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

10	0	3	6	1
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	0
Known vulnerable component	0
Confirmed configuration error	3
Potential cryptographic risk	0
Long-term post-quantum risk	7
Improvement recommendation	0
Insufficient data	0

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-19547	High	Certificate has expired (Corda Node Root CA)	61.4
QSI-2026-19549	High	Possible hard-coded secret in core/src/main/kotlin/net/corda/core/CordaOID.kt preliminary	47.1
QSI-2026-19556	High	Possible hard-coded secret in node/src/main/kotlin/net/corda/node/utilities/registration/HTTPNetworkRegistrationService.kt preliminary	47.1
QSI-2026-19553	Medium	Quantum-vulnerable ECDSA: transaction and message signing (1 file(s)) preliminary	35.6
QSI-2026-19550	Medium	Quantum-vulnerable RSA: role not determined (2 file(s)) preliminary	34.2

Mosca's inequality (x + y > z): x = 10, y = 5, z = 8 → migration would finish too late under this scenario. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography. Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Scope and boundaries

Mode: Document audit

Repository: https://github.com/corda/corda @ 80033580ecfd66a22ab5e9cf5a1319d1111ade92 — evidence links open the lines at this commit.

Supplied inputs

1117 files, 5785712 bytes in total, 3 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.3.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Crypto exchange, custodian or wallet" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
source_code	checked	1116	build.gradle: 776 lines scanned with 16 rule patterns and 267 registry patterns; buildCacheSettings.gradle: 16 lines scanned with 16 rule patterns and 267 registry patterns; buildSrc/build.gradle: 56 lines scanned with 16 rule patterns and 267 registry patterns
certificates_and_keys	checked	1	config/dev/corda_dev_ca.cer: certificates/keys parsed
tls_configuration	checked	1	node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/SSLHelper.kt: apache TLS directives parsed
secrets	checked	1	node/src/main/kotlin/net/corda/node/utilities/registration/NetworkRegistrationHelper.kt: 1 redacted value(s) not reported (comments, test modules, fixture declarations, placeholders, values the source annotates as published, or hex values not bound to a secret name)
entropy	checked	1	random-number generation: 1064 code files read with 2 macro definitions from the tree; 0 ineffective guard(s), 0 weak random-bytes provider(s), 0 weak key-generation site(s)

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 1118 (input files: C)

Architecture overview

Static review of the source tree corda at 80033580ecfd66a22ab5e9cf5a1319d1111ade92. 1117 files analysed; of 3034 files, 1293 are test, fuzzing, benchmark or example code and were left out, 624 are not read by any analyzer, 0 exceed the per-file limit and 0 did not fit the total limit. Files no analyzer reads include: .ci/Dockerfile, .ci/JenkinsfileStaticAnalysis, .ci/README.md, .ci/api-current.txt, .ci/check-api-changes.sh. 33 files in build-output or dependency-cache directories (.git, .hg, .venv, __pycache__, build, dist, node_modules, target, vendor, venv) were not read. No requests were sent to the owner's systems.

Overall security posture

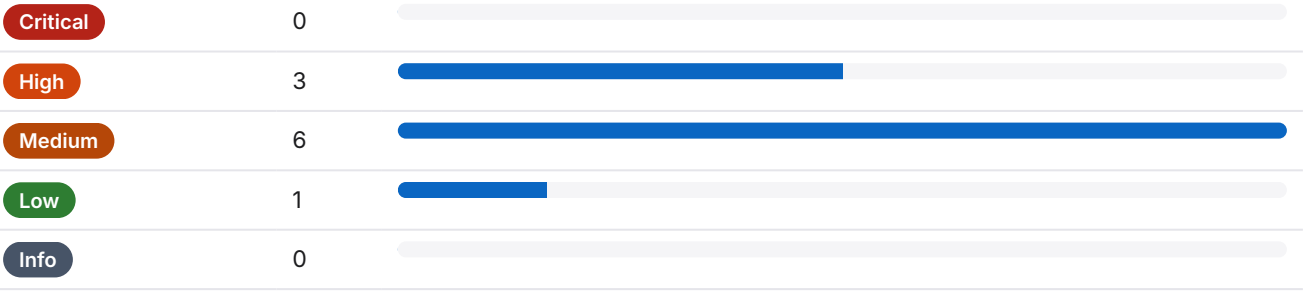
74
Indicative posture index (0–100)

68.0%
Quantum-vulnerable share of inventory

25
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-19547	Certificate has expired (Corda Node Root CA)	Confirmed configuration error	High	High
QSI-2026-19549	Possible hard-coded secret in core/src/main/kotlin/net/corda/core/CordaOID.kt	Confirmed configuration error	High	Medium, preliminary
QSI-2026-19556	Possible hard-coded secret in node/src/main/kotlin/net/corda/node/utilities/registration/HTTPNetworkRegistrationService.kt	Confirmed configuration error	High	Medium, preliminary
QSI-2026-19553	Quantum-vulnerable ECDSA: transaction and message signing (1 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19550	Quantum-vulnerable RSA: role not determined (2 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19551	Quantum-vulnerable ECDSA: role not determined (4 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19552	Quantum-vulnerable Ed25519: role not determined (1 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19554	Quantum-vulnerable ECDSA: authentication and identity (certificates, tokens, device attestation) (1 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19555	Quantum-vulnerable Ed25519: authentication and identity (certificates, tokens, device attestation) (1 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-19548	Quantum-vulnerable ECDSA key in certificate (Corda Node Root CA)	Long-term post-quantum risk	Low	High

Detailed findings

QSI-2026-19547 · CERT-EXPIRED

High

Certificate has expired (Corda Node Root CA)

Confirmed configuration error | Confidence: High

The X.509 certificate with subject Corda Node Root CA expired on 2026-07-17 (112 days ago).

Affected asset	certificate:Corda Node Root CA
Data source	config/dev/corda_dev_ca.cer
Detection method	X.509 parsing (pyca/cryptography), comparison of notAfter with the analysis time (UTC).
Analyzer	certificates 1.3.0 · 2026-11-06 15:38 UTC
Priority score	61.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Clients that validate certificates will refuse connections; users may be trained to ignore warnings.

REPRODUCTION STEPS

```
openssl x509 -in config/dev/corda_dev_ca.cer -noout -enddate
```

EVIDENCE

certificate · config/dev/corda_dev_ca.cer · line 1 · SHA-256 fcb0f5a41fc213fa… · File 5862e13fe4779283…

```
Subject: C=UK,L=London,OU=corda,O=R3,CN=Corda Node Root CA
Issuer: C=UK,L=London,OU=corda,O=R3,CN=Corda Node Root CA
Serial: 368d507782c9d515
Not before: 2016-07-19T00:00:00+00:00
Not after: 2026-07-17T00:00:00+00:00
Public key: ECDSA secp256k1
Signature algorithm: 1.2.840.10045.4.3.2 (ECDSA)
Signature hash: sha256
CA: True
SAN: -
SHA-256 fingerprint: 46266a82eebee45e1497a0a53ea80b4d6d382c505bd9ada0c134340f88ef7a72
```

LIMITATIONS OF THE CHECK

Document-based analysis: only the supplied file was examined. Deployed configuration may differ; the result reflects the file content at the recorded SHA-256.

RECOMMENDATION

Renew the certificate and automate renewal (e.g. ACME) with expiry monitoring.

RETEST PROCEDURE

Re-parse the deployed certificate and confirm notAfter is in the future.

NIST SP 800-57 Part 1 Rev. 5 — Recommendation for Key Management

QSI-2026-19549 · CODE-HARDCODED-SECRET

High

Possible hard-coded secret in core/src/main/kotlin/net/corda/core/CordaOID.kt

Confirmed configuration error | Confidence: Medium | preliminary — requires further verification

1 secret-like value(s) of type generic_assignment (first at line 21). Values were redacted by QSI.

Affected asset	file:core/src/main/kotlin/net/corda/core/CordaOID.kt
Data source	core/src/main/kotlin/net/corda/core/CordaOID.kt
Detection method	Regular-expression matching of language-specific API patterns.
Analyzer	source_code 1.3.0 · 2026-11-06 15:38 UTC
Priority score	47.1 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Credentials in source code leak through repositories, builds and logs.

REPRODUCTION STEPS

```
Open line 21 in the supplied file core/src/main/kotlin/net/corda/core/CordaOID.kt and inspect the quoted lines (see evidence).
```

EVIDENCE

file_excerpt · core/src/main/kotlin/net/corda/core/CordaOID.kt · line 21 · SHA-256 15e2a40ba613452d... · File 68cef7fe4d8dd644...

```
21 |     const val ALIAS_PRIVATE_KEY = "[REDACTED]"
```

LIMITATIONS OF THE CHECK

Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Move secrets to a secrets manager/environment; rotate the exposed values.

RETEST PROCEDURE

Re-scan the source and review the call sites listed in the evidence.

OWASP Application Security Verification Standard

QSI-2026-19556 · CODE-HARDCODED-SECRET

High

Possible hard-coded secret in node/src/main/kotlin/net/corda/node/utilities/registration/HTTPNetworkRegistrationService.kt

Confirmed configuration error | Confidence: Medium | preliminary — requires further verification

1 secret-like value(s) of type generic_assignment (first at line 31). Values were redacted by QSI.

Affected asset	file:node/src/main/kotlin/net/corda/node/utilities/registration/HTTPNetworkRegistrationService.kt
Data source	node/src/main/kotlin/net/corda/node/utilities/registration/HTTPNetworkRegistrationService.kt
Detection method	Regular-expression matching of language-specific API patterns.
Analyzer	source_code 1.3.0 · 2026-11-06 15:38 UTC

Priority score	47.1 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Credentials in source code leak through repositories, builds and logs.

REPRODUCTION STEPS

```
Open line 31 in the supplied file node/src/main/kotlin/net/corda/node/utilities/registration/
HTTPNetworkRegistrationService.kt and inspect the quoted lines (see evidence).
```

EVIDENCE

file_excerpt · node/src/main/kotlin/net/corda/node/utilities/registration/HTTPNetworkRegistrationService.kt · line 31 · SHA-256 cd5f56090a0380ac… · File c3cab1b88698dc3f…

```
   31 |         private const val CENM_SUBMISSION_TOKEN = "[REDACTED]"
```

LIMITATIONS OF THE CHECK

Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Move secrets to a secrets manager/environment; rotate the exposed values.

RETEST PROCEDURE

Re-scan the source and review the call sites listed in the evidence.

OWASP Application Security Verification Standard

QSI-2026-19553 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: transaction and message signing (1 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

6 reference(s) to ECDSA across 1 file(s), e.g. core/src/main/kotlin/net/corda/core/crypto/internal/Secp256k1SupportProvider.kt:32. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:ECDSA:transaction_signing
Data source	core/src/main/kotlin/net/corda/core/crypto/internal/Secp256k1SupportProvider.kt
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 15:38 UTC
Priority score	35.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open core/src/main/kotlin/net/corda/core/crypto/internal/Secp256k1SupportProvider.kt:32 and inspect
the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · core/src/main/kotlin/net/corda/core/crypto/internal/Secp256k1SupportProvider.kt · line 32 · SHA-256 1cc21aed7f8b5ffe… · File 0e66c3cbf611ceb7…

```
   31 |  */
   32 | class Secp256k1SupportProvider : Provider("Secp256k1Support", "1.0", "Augmenting SunEC with
support for the secp256k1 curve via BC") {
   33 |     init {
```

file_excerpt · core/src/main/kotlin/net/corda/core/crypto/internal/Secp256k1SupportProvider.kt · line 66 · SHA-256 7ba0ac1a0c89777d… · File 0e66c3cbf611ceb7…

```
   65 |                 if (!::bc.isInitialized) {
   66 |                     bc = Signature.getInstance("SHA256withECDSA", cordaBouncyCastleProvider)
   67 |                 }
```

file_excerpt · core/src/main/kotlin/net/corda/core/crypto/internal/Secp256k1SupportProvider.kt · line 76 · SHA-256 b78449aeb3bcaf9e… · File 0e66c3cbf611ceb7…

```
   75 |             if (!::sunEc.isInitialized) {
   76 |                 sunEc = Signature.getInstance("SHA256withECDSA", sunEcProvider)
   77 |             }
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19550 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable RSA: role not determined (2 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

3 reference(s) to RSA across 2 file(s), e.g. core/src/main/kotlin/net/corda/core/crypto/Crypto.kt:92. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:RSA:general
Data source	core/src/main/kotlin/net/corda/core/crypto/Crypto.kt
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 15:38 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open core/src/main/kotlin/net/corda/core/crypto/Crypto.kt:92 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · core/src/main/kotlin/net/corda/core/crypto/Crypto.kt · line 92 · SHA-256 fedad570e631922d… · File a0bc267096ca6cb7…

```
   91 |             "RSA",
   92 |             "SHA256withRSA",
   93 |             null,
```

file_excerpt · node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt · line 224 · SHA-256 d2836749b7a6179e… · File 85a9aa6e746282b3…

```
  223 |             Crypto.ECDSA_SECP256K1_SHA256 ->
keyPairGenerator.initialize(ECGenParameterSpec("secp256k1"))
  224 |             Crypto.RSA_SHA256 -> keyPairGenerator.initialize(scheme.keySize!!)
  225 |             else -> throw IllegalArgumentException("No mapping for scheme ID $
{scheme.schemeNumberID}")
```

file_excerpt · node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt · line 232 · SHA-256 9c9d73cbab4b1b99… · File 85a9aa6e746282b3…

```
  231 |         Crypto.ECDSA_SECP256R1_SHA256, Crypto.ECDSA_SECP256K1_SHA256 -> "EC"
  232 |         Crypto.RSA_SHA256 -> "RSA"
  233 |         else -> throw IllegalArgumentException("No algorithm for scheme ID $
{scheme.schemeNumberID}")
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19551 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: role not determined (4 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

15 reference(s) to ECDSA across 4 file(s), e.g. core/src/main/kotlin/net/corda/core/crypto/Crypto.kt:103. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:ECDSA:general
Data source	core/src/main/kotlin/net/corda/core/crypto/Crypto.kt
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 15:38 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open core/src/main/kotlin/net/corda/core/crypto/Crypto.kt:103 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · core/src/main/kotlin/net/corda/core/crypto/Crypto.kt · line 103 · SHA-256 ddc1e3c37f91880d… · File a0bc267096ca6cb7…

```
  102 |             "ECDSA_SECP256K1_SHA256",
  103 |             AlgorithmIdentifier(X9ObjectIdentifiers.ecdsa_with_SHA256,
SECObjectIdentifiers.secp256k1),
  104 |             listOf(AlgorithmIdentifier(X9ObjectIdentifiers.id_ecPublicKey,
SECObjectIdentifiers.secp256k1)),
```

file_excerpt · core/src/main/kotlin/net/corda/core/crypto/Crypto.kt · line 104 · SHA-256 0cbd5b8d15e381e9… · File a0bc267096ca6cb7…

```
  103 |             AlgorithmIdentifier(X9ObjectIdentifiers.ecdsa_with_SHA256,
SECObjectIdentifiers.secp256k1),
```

```
  104 |             listOf(AlgorithmIdentifier(X9ObjectIdentifiers.id_ecPublicKey,
SECObjectIdentifiers.secp256k1)),
  105 |             cordaBouncyCastleProvider.name,
```

file_excerpt · core/src/main/kotlin/net/corda/core/crypto/Crypto.kt · line 107 · SHA-256 b74a0bc63bf81f60… · File a0bc267096ca6cb7…

```
  106 |             "EC",
  107 |             "SHA256withECDSA",
  108 |             ECNamedCurveTable.getParameterSpec("secp256k1"),
```

file_excerpt · node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/NOT_YET_REGISTERED_MARKER_KEYS_AND_CERTS.kt · line 12 · SHA-256 bdbdc27ab1cd2296… · File d0241aa19e69653a…

```
   11 | object NOT_YET_REGISTERED_MARKER_KEYS_AND_CERTS {
   12 |     val ECDSAR1_KEYPAIR by lazy
{ Crypto.deriveKeyPairFromEntropy(Crypto.ECDSA_SECP256R1_SHA256, BigInteger.valueOf(0)) }
   13 |     val ECDSAR1_CERT by lazy { createSelfSignedCACertificate(X500Principal("CN=DUMMY"),
ECDSAR1_KEYPAIR) }
```

file_excerpt · node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/CryptoService.kt · line 127 · SHA-256 54855cad7202e394… · File 4ea4fcfa338bf704…

```
  126 |      */
  127 |     fun defaultWrappingSignatureScheme(): SignatureScheme = Crypto.ECDSA_SECP256R1_SHA256
  128 |
```

file_excerpt · node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt · line 222 · SHA-256 2369170574ed87d8… · File 85a9aa6e746282b3…

```
  221 |         when (scheme) {
  222 |             Crypto.ECDSA_SECP256R1_SHA256 ->
keyPairGenerator.initialize(ECGenParameterSpec("secp256r1"))
  223 |             Crypto.ECDSA_SECP256K1_SHA256 ->
keyPairGenerator.initialize(ECGenParameterSpec("secp256k1"))
```

file_excerpt · node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt · line 223 · SHA-256 b2a0b1eb7158aa9f… · File 85a9aa6e746282b3…

```
  222 |             Crypto.ECDSA_SECP256R1_SHA256 ->
keyPairGenerator.initialize(ECGenParameterSpec("secp256r1"))
  223 |             Crypto.ECDSA_SECP256K1_SHA256 ->
keyPairGenerator.initialize(ECGenParameterSpec("secp256k1"))
  224 |             Crypto.RSA_SHA256 -> keyPairGenerator.initialize(scheme.keySize!!)
```

file_excerpt · node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt · line 231 · SHA-256 4d4226acca743a76… · File 85a9aa6e746282b3…

```
  230 |     private fun keyAlgorithmFromScheme(scheme: SignatureScheme): String = when (scheme) {
  231 |         Crypto.ECDSA_SECP256R1_SHA256, Crypto.ECDSA_SECP256K1_SHA256 -> "EC"
  232 |         Crypto.RSA_SHA256 -> "RSA"
```

LIMITATIONS OF THE CHECK

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RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19552 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: role not determined (1 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

11 reference(s) to Ed25519 across 1 file(s), e.g. core/src/main/kotlin/net/corda/core/crypto/Crypto.kt:4. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:Ed25519:general
Data source	core/src/main/kotlin/net/corda/core/crypto/Crypto.kt
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 15:38 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open core/src/main/kotlin/net/corda/core/crypto/Crypto.kt:4 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · core/src/main/kotlin/net/corda/core/crypto/Crypto.kt · line 4 · SHA-256 099c04d861bc4fde… · File a0bc267096ca6cb7…

```
3 | import net.corda.core.CordaOID
4 | import net.corda.core.crypto.Crypto.EDDSA_ED25519_SHA512
5 | import net.corda.core.crypto.internal.AliasPrivateKey
```

file_excerpt · core/src/main/kotlin/net/corda/core/crypto/Crypto.kt · line 134 · SHA-256 84bf586c243b1831… · File a0bc267096ca6cb7…

```
133 |     @JvmField
134 |     val EDDSA_ED25519_SHA512: SignatureScheme = SignatureScheme(
135 |             4,
```

file_excerpt · core/src/main/kotlin/net/corda/core/crypto/Crypto.kt · line 136 · SHA-256 4c3400ed0a6b6564… · File a0bc267096ca6cb7…

```
135 |             4,
136 |             "EDDSA_ED25519_SHA512",
137 |             AlgorithmIdentifier(EdECObjectIdentifiers.id_Ed25519, null),
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19554 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: authentication and identity (certificates, tokens, device attestation) (1 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

1 reference(s) to ECDSA across 1 file(s), e.g. node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/X509Utilities.kt:75. Role, judged from paths, function names and nearby code: authentication and identity (certificates, tokens, device attestation). Short-lived authentication stays safe until a CRQC exists; long-lived identities (device attestation, CA and SSH keys) need a plan earlier.

Affected asset	source-code:ECDSA:authentication
Data source	node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/X509Utilities.kt
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 15:38 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/X509Utilities.kt:75 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/X509Utilities.kt · line 75 · SHA-256 796a593d0cda664e… · File 48f8f548369b64b5…

```
74 |     val DEFAULT_IDENTITY_SIGNATURE_SCHEME = Crypto.EDDSA_ED25519_SHA512
75 |     val DEFAULT_TLS_SIGNATURE_SCHEME = Crypto.ECDSA_SECP256R1_SHA256
76 |
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19555 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: authentication and identity (certificates, tokens, device attestation) (1 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

1 reference(s) to Ed25519 across 1 file(s), e.g. node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/X509Utilities.kt:74. Role, judged from paths, function names and nearby code: authentication and identity (certificates, tokens, device attestation). Short-lived authentication stays safe until a CRQC exists; long-lived identities (device attestation, CA and SSH keys) need a plan earlier.

Affected asset	source-code:Ed25519:authentication
Data source	node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/X509Utilities.kt
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 15:38 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/X509Utilities.kt:74 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/X509Utilities.kt · line 74 · SHA-256 450c55652268000b… · File 48f8f548369b64b5…

```
73 |     // schemes (for instance `UtimacoCryptoService.DEFAULT_IDENTITY_SIGNATURE_SCHEME`).
74 |     val DEFAULT_IDENTITY_SIGNATURE_SCHEME = Crypto.EDDSA_ED25519_SHA512
75 |     val DEFAULT_TLS_SIGNATURE_SCHEME = Crypto.ECDSA_SECP256R1_SHA256
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-19548 · CERT-QV-KEY

Low

Quantum-vulnerable ECDSA key in certificate (Corda Node Root CA)

Long-term post-quantum risk | Confidence: High

The certificate Corda Node Root CA uses a ECDSA secp256k1 public key, valid until 2026-07-17. ECDSA is broken by Shor's algorithm on a CRQC.

Affected asset	certificate:Corda Node Root CA
Data source	config/dev/corda_dev_ca.cer
Detection method	X.509 public key algorithm identification.
Analyzer	certificates 1.3.0 · 2026-11-06 15:38 UTC
Priority score	36.5 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=0.0, y=5.0, z=8.0 → time available under this scenario

POTENTIAL IMPACT

For authentication, risk materializes only once a CRQC exists while the key/certificate is still trusted. Certificates are short-lived, so the main concern is the PKI and long-lived roots/ intermediates.

REPRODUCTION STEPS

```
openssl x509 -in config/dev/corda_dev_ca.cer -noout -text | grep -A2 'Public Key Algorithm'
```

EVIDENCE

certificate · config/dev/corda_dev_ca.cer · line 1 · SHA-256 fcb0f5a41fc213fa… · File 5862e13fe4779283…

```
Subject: C=UK,L=London,OU=corda,O=R3,CN=Corda Node Root CA
Issuer: C=UK,L=London,OU=corda,O=R3,CN=Corda Node Root CA
Serial: 368d507782c9d515
Not before: 2016-07-19T00:00:00+00:00
Not after: 2026-07-17T00:00:00+00:00
Public key: ECDSA secp256k1
Signature algorithm: 1.2.840.10045.4.3.2 (ECDSA)
Signature hash: sha256
CA: True
SAN: -
SHA-256 fingerprint: 46266a82eebee45e1497a0a53ea80b4d6d382c505bd9ada0c134340f88ef7a72
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Track in the PQC inventory; prefer short certificate lifetimes; follow CA/Browser Forum and NIST PQC transition guidance for ML-DSA certificates.

RETEST PROCEDURE

Re-audit once PQC or hybrid certificates are deployed.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

Evidence

ID	Type	Data source	SHA-256
QSI-2026-19547	certificate	config/dev/corda_dev_ca.cer (line 1)	fcb0f5a41fc213fa274755226a8330f59b
QSI-2026-19549	file_excerpt	core/src/main/kotlin/net/corda/core/CordaOID.kt (line 21)	15e2a40ba613452d9b5f52b33a8a6c42e0
QSI-2026-19556	file_excerpt	node/src/main/kotlin/net/corda/node/utilities/registration/HTTPNetworkRegistrationService.kt (line 31)	cd5f56090a0380acb8b9d189aa07886bb6
QSI-2026-19553	file_excerpt	core/src/main/kotlin/net/corda/core/crypto/internal/Secp256k1SupportProvider.kt (line 32)	1cc21aed7f8b5ffe0ffeedc36b2344d08e
QSI-2026-19553	file_excerpt	core/src/main/kotlin/net/corda/core/crypto/internal/Secp256k1SupportProvider.kt (line 66)	7ba0ac1a0c89777dc58c9b13d171c61b03
QSI-2026-19553	file_excerpt	core/src/main/kotlin/net/corda/core/crypto/internal/Secp256k1SupportProvider.kt (line 76)	b78449aeb3bcaf9e2aec6fef912446b666
QSI-2026-19550	file_excerpt	core/src/main/kotlin/net/corda/core/crypto/Crypto.kt (line 92)	fedad570e631922d0d8aeddf511a64b7db
QSI-2026-19550	file_excerpt	node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt (line 224)	d2836749b7a6179e0326f75d1d60241374
QSI-2026-19550	file_excerpt	node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt (line 232)	9c9d73cbab4b1b994cdec67bb3c1464d39
QSI-2026-19551	file_excerpt	core/src/main/kotlin/net/corda/core/crypto/Crypto.kt (line 103)	ddc1e3c37f91880d7b9ee05f3f92042177
QSI-2026-19551	file_excerpt	core/src/main/kotlin/net/corda/core/crypto/Crypto.kt (line 104)	0cbd5b8d15e381e9030d404c63c1b8c0fc
QSI-2026-19551	file_excerpt	core/src/main/kotlin/net/corda/core/crypto/Crypto.kt (line 107)	b74a0bc63bf81f600cf4dfe9064d3b056d
QSI-2026-19551	file_excerpt	node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/NOT_YET_REGISTERED_MARKER_KEYS_AND_CERTS.kt (line 12)	bdbdc27ab1cd2296d6837cc2810b785b81
QSI-2026-19551	file_excerpt	node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/CryptoService.kt (line 127)	54855cad7202e394b18d1beb70ed5aae46
QSI-2026-19551	file_excerpt	node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt (line 222)	2369170574ed87d8d55ee2b51e8ae9357f
QSI-2026-19551	file_excerpt	node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt (line 223)	b2a0b1eb7158aa9f9c8bf2b0050285e85d
QSI-2026-19551	file_excerpt	node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt (line 231)	4d4226acca743a76d511d9ef4d8bd5110e
QSI-2026-19552	file_excerpt	core/src/main/kotlin/net/corda/core/crypto/Crypto.kt (line 4)	099c04d861bc4fde1ab02770b565be9441

QSI-2026-19552	file_excerpt	core/src/main/kotlin/net/corda/core/crypto/Crypto.kt (line 134)	84bf586c243b183196895373ad877db448
QSI-2026-19552	file_excerpt	core/src/main/kotlin/net/corda/core/crypto/Crypto.kt (line 136)	4c3400ed0a6b6564fcc0a8dd1e44ad2a57
QSI-2026-19554	file_excerpt	node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/X509Utilities.kt (line 75)	796a593d0cda664eeeb5394fecebc03132
QSI-2026-19555	file_excerpt	node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/X509Utilities.kt (line 74)	450c55652268000b11d9f2c010decf2eee
QSI-2026-19548	certificate	config/dev/corda_dev_ca.cer (line 1)	fcb0f5a41fc213fa274755226a8330f59b

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	10	10
Migration time y	5	5
Assumed years to CRQC z	8 (early)	8
x + y − z	7 → migration would finish too late under this scenario	7 → migration would finish too late under this scenario

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Crypto exchange, custodian or wallet”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
transaction and message signing	ECDSA (1)	QSI-2026-19553	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them
authentication and identity (certificates, tokens, device attestation)	ECDSA (1), Ed25519 (1)	QSI-2026-19554, QSI-2026-19555	Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them
role not determined	ECDSA (4), RSA (2), Ed25519 (1)	QSI-2026-19550, QSI-2026-19551, QSI-2026-19552	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count
reduced margin (Grover), approved	1
Grover: acceptable to NIST, below ECCG/ANSSI guidance	6
n/a	1
vulnerable (Shor)	17

Detailed findings: QSI-2026-19553, QSI-2026-19550, QSI-2026-19551, QSI-2026-19552, QSI-2026-19554, QSI-2026-19555, QSI-2026-19548

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit
Shor's algorithm — factoring Q1: quantum computation offline, classical access to the victim	breaks	RSA (2)	Breaks RSA at every key length: a longer modulus only raises the cost polynomially. Plan migration to ML-KEM (or a hybrid) for key establishment and ML-DSA / SLH-DSA for signatures; start with keys protecting data that must stay confidential for years (harvest now, decrypt later) and with long-lived signatures.
Shor's algorithm — elliptic-curve discrete logarithms Q1: quantum computation offline, classical access to the victim	breaks	ECDSA (12), Ed25519 (2), X25519 (1)	Elliptic-curve keys are expected to fall before RSA keys of comparable classical strength. On blockchains, coins whose public key is already on chain (reused addresses, pay-to-public-key and Taproot outputs) are exposed first; plan the move to post-quantum signatures and avoid key reuse in the meantime.
Grover's algorithm — key search and preimages Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	AES-128 (2), HMAC (1), SHA-2 (4)	Not a break. 256-bit keys (AES-256, ChaCha20) keep a comfortable margin; AES-128 meets NIST category 1, while some authorities ask for 256-bit keys in quantum-resistant contexts (see the AES-128 entry). Prefer 256-bit keys for long-lived data; 256-bit hashes need no change for preimage resistance.
Brassard–Høyer–Tapp — collision search Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	SHA-2 (4)	256-bit hashes (SHA-256, SHA3-256) remain adequate for collision resistance; 160-bit hashes are already weak classically (SHA-1 collisions are practical). Use 384/512-bit hashes where NIST categories 4–5 are required.

Simon's algorithm — superposition attacks on symmetric constructions Q2: superposition queries to the victim's keyed primitive	only with superposition queries	GCM mode (1)	No action for ordinary deployments of AES-GCM or CBC-MAC — the attack model does not apply. Record it as a model assumption; see the offline variant for Even–Mansour-type designs.

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Remediation plan

QSI-2026-19547 · P1 · 30 days root-cause fix

Renew the certificate for Corda Node Root CA and enable automated renewal with alerting ≥ 30 days before expiry.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	None expected.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	The deployed certificate is valid and renewal is automated.

QSI-2026-19549 · P1 · 30 days root-cause fix

Rotate and remove secrets found in core/src/main/kotlin/net/corda/core/CordaOID.kt.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible compatibility impact on old clients.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Secret scanner reports no secrets; old values revoked.

QSI-2026-19556 · P1 · 30 days root-cause fix

Rotate and remove secrets found in node/src/main/kotlin/net/corda/node/utilities/registration/HTTPNetworkRegistrationService.kt.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible compatibility impact on old clients.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Secret scanner reports no secrets; old values revoked.

QSI-2026-19553 · P3 · 90 days short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19550 · P3 · 90 days short-term risk reduction

Add RSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19551 · P3 · 90 days short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19552 · P3 · 90 days short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19554 · P3 · 90 days short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19555 · P3 · 90 days short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19548 · P3 · 90 days short-term risk reduction

Add ECDSA usages at certificate:Corda Node Root CA to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-19553 · P4 · 90 days root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-19550 · P4 · 90 days root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-19551 · P4 · 90 days root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-19552 · P4 · 90 days root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-19554 · P4 · 90 days root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	

	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-19555 · P4 · 90 days

root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-19548 · P4 · 90 days

root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
P1	QSI-2026-19547	Renew the certificate for Corda Node Root CA and enable automated renewal with alerting ≥ 30 days before expiry.	root-cause fix
P1	QSI-2026-19549	Rotate and remove secrets found in core/src/main/kotlin/net/corda/core/CordaOID.kt.	root-cause fix
P1	QSI-2026-19556	Rotate and remove secrets found in node/src/main/kotlin/net/corda/node/utilities/registration/HTTPNetworkRegistrationService.kt.	root-cause fix

60 days

Priority	ID	Action	Type
None			

90 days

Priority	ID	Action	Type
P3	QSI-2026-19553	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19550	Add RSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19551	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19552	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19554	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19555	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-19548	Add ECDSA usages at certificate:Corda Node Root CA to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P4	QSI-2026-19553	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

P4	QSI-2026-19550	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19551	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19552	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19554	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19555	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-19548	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-19547	Re-parse the deployed certificate and confirm notAfter is in the future.
QSI-2026-19549	Re-scan the source and review the call sites listed in the evidence.
QSI-2026-19556	Re-scan the source and review the call sites listed in the evidence.
QSI-2026-19553	Re-run the inventory after refactoring.
QSI-2026-19550	Re-run the inventory after refactoring.
QSI-2026-19551	Re-run the inventory after refactoring.
QSI-2026-19552	Re-run the inventory after refactoring.
QSI-2026-19554	Re-run the inventory after refactoring.
QSI-2026-19555	Re-run the inventory after refactoring.
QSI-2026-19548	Re-audit once PQC or hybrid certificates are deployed.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.2.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.3.0
inventory_document	1.1.0
solidity	1.3.0
source_code	1.4.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.3.0
OSV.dev API	v1

Sources and normative documents

Reviewed: 2026-10-13

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11. FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA) — NIST. https://csrc.nist.gov/pubs/fips/204/final (final (August 2024))
12. FIPS 205 — Stateless Hash-Based Digital Signature Standard (SLH-DSA) — NIST. https://csrc.nist.gov/pubs/fips/205/final (final (August 2024))
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28. S. Jaques, M. Naehrig, M. Roetteler, F. Virdia — Implementing Grover oracles for quantum key search on AES and LowMC (EUROCRYPT 2020; arXiv:1910.01700) — Springer / arXiv. https://arxiv.org/abs/1910.01700 (peer-reviewed article: lower cost estimates for the AES key searches that define the NIST categories)

Appendices

A. Cryptographic inventory

Algorithm	Parameter	Location	Quantum status	Classical status
HMAC	HMAC-SHA-512	core/src/main/kotlin/net/corda/core/crypto/Crypto.kt:910	reduced margin (Grover), approved	acceptable
AES-128	AES-128	node-api/src/main/kotlin/net/corda/nodeapi/internal/ArtemisTcpTransport.kt:22	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
AES-128	AES-128	tools/loadtest/src/main/kotlin/net/corda/loadtest/Disruption.kt:71	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
SHA-2	SHA-256	config/dev/corda_dev_ca.cer (line 1)	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
SHA-2	SHA-256	core/src/main/kotlin/net/corda/core/internal/InternalUtils.kt:192	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
SHA-2	SHA-256	node/src/main/kotlin/net/corda/node/services/messaging/InterceptingActiveMQJAASSecurityManager.kt:186	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
SHA-2	SHA-256	node/src/main/kotlin/net/corda/node/services/persistence/AesDbEncryptionService.kt:129	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
GCM mode		node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/AesEncryption.kt:32	n/a	acceptable
ECDSA	secp256k1	config/dev/corda_dev_ca.cer (line 1) — Corda Node Root CA	vulnerable (Shor)	acceptable
ECDSA	secp256k1	core/src/main/kotlin/net/corda/core/crypto/Crypto.kt:103	vulnerable (Shor)	acceptable

ECDSA		core/src/main/kotlin/net/corda/core/crypto/Crypto.kt:107	vulnerable (Shor)	acceptable
ECDSA	P-256	core/src/main/kotlin/net/corda/core/crypto/Crypto.kt:118	vulnerable (Shor)	acceptable
ECDSA	secp256k1	core/src/main/kotlin/net/corda/core/crypto/internal/Secp256k1SupportProvider.kt:32	vulnerable (Shor)	acceptable
ECDSA		core/src/main/kotlin/net/corda/core/crypto/internal/Secp256k1SupportProvider.kt:66	vulnerable (Shor)	acceptable
ECDSA	P-256	node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/NOT_YET_REGISTERED_MARKER_KEYS_AND_CERTS.kt:12	vulnerable (Shor)	acceptable
ECDSA	P-256	node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/X509Utilities.kt:75	vulnerable (Shor)	acceptable
ECDSA	P-256	node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/CryptoService.kt:127	vulnerable (Shor)	acceptable
ECDSA	P-256	node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt:222	vulnerable (Shor)	acceptable
ECDSA		node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt:223	vulnerable (Shor)	acceptable
ECDSA	secp256k1	node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt:223	vulnerable (Shor)	acceptable
Ed25519	Ed25519	core/src/main/kotlin/net/corda/core/crypto/Crypto.kt:4	vulnerable (Shor)	acceptable
Ed25519	Ed25519	node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/X509Utilities.kt:74	vulnerable (Shor)	acceptable
RSA		core/src/main/kotlin/net/corda/core/crypto/Crypto.kt:92	vulnerable (Shor)	acceptable
RSA		node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt:224	vulnerable (Shor)	acceptable
X25519		core/src/main/kotlin/net/corda/core/crypto/internal/Curve25519.kt	vulnerable (Shor)	acceptable

B. JSON

qsi.report/1 · SHA-256 09dd82bf776b5adb6af7367e5cccfe5b259eb10fde085b2eef6d9bb38f43d5c6

C. Input files

File	Size	SHA-256
build.gradle	37471	`0b5fb42156188be73f1bfc00eb5e7b300b095d3dbe02a93a9d6a5a`
buildCacheSettings.gradle	525	`c28dca0923ff6b910d706e494d96fa30fa9d032ab220407a89930f`
buildSrc/build.gradle	2142	`c4f6a554de07ae19161691a83273a94d72d983e552c29692497158`

buildSrc/src/main/groovy/corda.common-publishing.gradle	9127	90327b14defff8d894057125084a2d201ad9d686cf6b57d57fae3e
buildSrc/src/main/groovy/corda.kotlin-1.2.gradle	4440	89f902c0caeeeaa716874c1dc54e59db5162ec7240ca19025aed78
buildSrc/src/main/groovy/corda.root-publish.gradle	294	196d9f7543be51e192b09ba482e56bba66e4d97ed7c2b76c1a0def
buildSrc/src/main/groovy/net.corda.build.docker/ImageBuilder.groovy	9681	e932bc2359c6b6d874414091ed583af84dd6c62f57afffcd3efd2f
client/jackson/build.gradle	2391	6a125a6904a6c888a4747b0b7148413bd74a8fb15bc5920b2263fe
client/jackson/src/main/kotlin/net/corda/client/jackson/JacksonSupport.kt	26525	56b181af8cc60f94959588ccac114f4e41b5e5c1ebaeee61091a25
client/jackson/src/main/kotlin/net/corda/client/jackson/StringToMethodCallParser.kt	13077	0a228141d344f2fa1c619eb4af47d8fa9cd03368107a3e801cc90e
client/jackson/src/main/kotlin/net/corda/client/jackson/internal/CordaModule.kt	31095	7af57bb1d906c12a6097e42c0ee5e3e1e609412cb8882a5d29bc85
client/jackson/src/main/kotlin/net/corda/client/jackson/internal/JacksonUtils.kt	1273	1f4caf9566aab949119761e13c98ef5f518be14d5bdeb010135db9
client/jfx/build.gradle	3373	b8e09c89443588d8bc54244daa529d4557db3bddf812284de2dab7
client/jfx/src/main/java/net/corda/client/jfx/utils/AggregatedList.java	842	41d2a84b4caa3f7ea6a694df2f579dfdf98cf6851c6b724db72d74
client/jfx/src/main/java/net/corda/client/jfx/utils/ConcatenatedList.java	645	6bb9ebc83720720af28cb17f9e84b6f0a78ac61d273b063c5341ff
client/jfx/src/main/java/net/corda/client/jfx/utils/FlattenedList.java	646	ce8d0d89281c3e81fc59a53deb52f2b10fcebda48346382e66cc0d
client/jfx/src/main/java/net/corda/client/jfx/utils/MappedList.java	686	ceee888a1eca49019ce0865e3c17f49de82f9ff15f9e07e1bcc07b
client/jfx/src/main/java/net/corda/client/jfx/utils/ReplayedList.java	617	5e4957f52abe4b3b400808a5cce9e075800681685c62e0dd623a34
client/jfx/src/main/kotlin/net/corda/client/jfx/model/ContractStateModel.kt	1998	be37ef327a62a3708021cea50ea712bb9b878eceac006729c977a1
client/jfx/src/main/kotlin/net/corda/client/jfx/model/ExchangeRateModel.kt	1876	8c135cfe3aa801dff16327b2380bcc47b97a08cbbae9c9753d820c
client/jfx/src/main/kotlin/net/corda/client/jfx/model/Models.kt	3004	ad1f852a3e9c0cca67cbe70e9bc1b18b331d702918e3b294c136e7
client/jfx/src/main/kotlin/net/corda/client/jfx/model/ModelsUtils.kt	2122	c2df8792b9fdb243de7cdfbef5bce72292f030128ccb42daff5c86
client/jfx/src/main/kotlin/net/corda/client/jfx/model/NetworkIdentityModel.kt	2027	7b28e62aa5dbad9889673ac6eb0284f8cf2ced03aeb951f20cfd56
client/jfx/src/main/kotlin/net/corda/client/jfx/model/NodeMonitorModel.kt	6492	45277d9e8d04448cb30f2c4e03365f10f5ce84d56fac696032b502
client/jfx/src/main/kotlin/net/corda/client/jfx/model/TrackedDelegate.kt	3821	d071a7a846d16500f5a0d88f72950349fa89d7629ddb310dcc87bd

client/jfx/src/main/kotlin/net/corda/client/jfx/model/TransactionDataModel.kt	5333	c524f9cf8e710c1301d15d5d18b655f98dc17150b4a8213c8213fb
client/jfx/src/main/kotlin/net/corda/client/jfx/utils/AbstractAggregatedList.kt	6469	c197df0b8fe313614c1717559e40d8f5048299f822f0c9f956ba01
client/jfx/src/main/kotlin/net/corda/client/jfx/utils/AbstractConcatenatedList.kt	14387	dfcc82353a517bb79fb735662c7099c796f04d66f378a97b52ee6a
client/jfx/src/main/kotlin/net/corda/client/jfx/utils/AbstractFlattenedList.kt	5394	aa4d298e3774be8fd018db6904f5c5074dc2feaa50ab0577ed5479
client/jfx/src/main/kotlin/net/corda/client/jfx/utils/AbstractMappedList.kt	2842	980c6a462e9e0674c4161a470b27a20205a868fffd176ff0fc8ef8
client/jfx/src/main/kotlin/net/corda/client/jfx/utils/AbstractReplayedList.kt	2322	2c1ce340055059218405972cba7af6d3b602e4d9aefe3aaf5b3564
client/jfx/src/main/kotlin/net/corda/client/jfx/utils/AmountBindings.kt	2131	f449a477f1501552d4e72c2ca0ae6c60874919006c711f94715c0f
client/jfx/src/main/kotlin/net/corda/client/jfx/utils/AssociatedList.kt	2690	7d60cdbe9ac700b3dbd532d986ac02424e1ef379fd04ce5090e605
client/jfx/src/main/kotlin/net/corda/client/jfx/utils/ChosenList.kt	2023	f33211f84e7202374d6348bf6a5b8e28e23c00a6d65644e37c7dbf
client/jfx/src/main/kotlin/net/corda/client/jfx/utils/LeftOuterJoinedMap.kt	1982	1674138ca61bd964b5e742e9be65704e542d8346d610bdc21d8a3d
client/jfx/src/main/kotlin/net/corda/client/jfx/utils/MapValuesList.kt	2603	7c30acca0cfe9b60273e1fc2cfa1093afedd3458b2a638dc193507
client/jfx/src/main/kotlin/net/corda/client/jfx/utils/ObservableFold.kt	3025	f713e5b67770923a477454d25eb7b3df5327da63248971994d6ae4
client/jfx/src/main/kotlin/net/corda/client/jfx/utils/ObservableUtilities.kt	14607	4627d257cc4ccc3b0be0bc4fc54f5a8b82621540b415b8bf40a0cb
client/jfx/src/main/kotlin/net/corda/client/jfx/utils/ReadOnlyBackedObservableMapBase.kt	3748	9fe8147ee52e3148f0a964eb04dec20298d7a54917ce0cd3c678b2
client/rpc/build.gradle	5433	dec0bdf717a853ab99f594a25b04c5e44eb99a1964ce72a9f5b3f3
client/rpc/src/main/kotlin/net/corda/client/rpc/CordaRPCClient.kt	35650	0024cc4331abe2370925116931c11a27c6c7741123640bf7eec60f
client/rpc/src/main/kotlin/net/corda/client/rpc/PermissionException.kt	465	13115985bb7e98780325e85aa9e9d35ecb55454f3b0f6a18a2bbab
client/rpc/src/main/kotlin/net/corda/client/rpc/RPCConnection.kt	1978	e9496c9999ee139ce3c481bde9f760aedb7cf2e0479f6673550055
client/rpc/src/main/kotlin/net/corda/client/rpc/RPCException.kt	1264	bcb185c6d640753d2da252ef53444f3808ef55778902c1bcf3eca0
client/rpc/src/main/kotlin/net/corda/client/rpc/RPCSinceVersion.kt	199	3d8c36c90e36ba295c54517e196d9d1d3e3538c395c27038a55436
client/rpc/src/main/kotlin/net/corda/client/rpc/Utils.kt	702	4d943eedbf51ab631f96dfb6077b58cb1d136e9b8597d17df7bbd0

client/rpc/src/main/kotlin/net/corda/client/rpc/ext/MultiRPCClient.kt	18172	6451fa5fbbc9fb2d59f10ee57d515646098cbe46bfba3876c64a9f
client/rpc/src/main/kotlin/net/corda/client/rpc/ext/RPCConnectionListener.kt	1882	2ee53aac0ada7a55d5b5e70cbfee4656399da52442eba62cb4e83b
client/rpc/src/main/kotlin/net/corda/client/rpc/internal/ClientCacheFactory.kt	736	bb41c09993b42d0c8d74ac149071241ff803c8b81a4e1cf25654b1
client/rpc/src/main/kotlin/net/corda/client/rpc/internal/DistributionMux.kt	2116	1b9cb0751e11edba3c96ca69e2f2f449b2706021be6c1a06795ab3
client/rpc/src/main/kotlin/net/corda/client/rpc/internal/RPCClient.kt	11302	11e9d6cda37358444ab5674bd576f78a1fd6f5ec97107e05768e9c
client/rpc/src/main/kotlin/net/corda/client/rpc/internal/RPCClientProxyHandler.kt	35282	cf7f45a0f891c3a4fd91db35797e0bc23ff9f8f1e901fb1cb84d2a
client/rpc/src/main/kotlin/net/corda/client/rpc/internal/RPCClientTelemetry.kt	1755	6c9474abcc51f32b03b172fd2da1295f577c5690b3d1e16b7e6c54
client/rpc/src/main/kotlin/net/corda/client/rpc/internal/RPCUtils.kt	715	9a4f45ae669d363abcd2c62a1679604e268262c40faf383004d973
client/rpc/src/main/kotlin/net/corda/client/rpc/internal/ReconnectingCordaRPCOps.kt	21954	4b01753ddd35262bcf6bcbf2d5001128a24b9595069187aa7a6b89
client/rpc/src/main/kotlin/net/corda/client/rpc/internal/ReconnectingObservable.kt	3655	67ac9c8603dbcb155a90ba5bc5829aef93f00981adf7d0cec5b0d7
client/rpc/src/main/kotlin/net/corda/client/rpc/internal/SerializationEnvironmentHelper.kt	2381	244bdde714083bc23e75c98c7326b4657dcc18b71be79570978dc3
client/rpc/src/main/kotlin/net/corda/client/rpc/reconnect/CouldNotStartFlowException.kt	465	8d5aca4473203de6224d47a2ff5c637f02f461257f8a8636509faa
common/configuration-parsing/build.gradle	1209	99a706c784e902f80df33c1e18365a8aec64f0b3f3517b45e59a30
common/configuration-parsing/src/main/kotlin/net/corda/common/configuration/parsing/internal/Configuration.kt	30596	d259bffa77bb17b3fd0aa39d4ee1999067e238c409334ec4ab3a8a
common/configuration-parsing/src/main/kotlin/net/corda/common/configuration/parsing/internal/Properties.kt	16752	6e8d9df0929078abe772528f07fc8006e7799b33c9e32c2382509f
common/configuration-parsing/src/main/kotlin/net/corda/common/configuration/parsing/internal/Schema.kt	3061	52d1aac2231f244f09e8eb040836b3016486391494c6d4b8021124
common/configuration-parsing/src/main/kotlin/net/corda/common/configuration/parsing/internal/Specification.kt	10873	2fc6a67dcaec7d83af1fbca98e0ab65ea1e704ef057c6d55874ee7
common/configuration-parsing/src/main/kotlin/net/corda/common/configuration/parsing/internal/Utils.kt	4632	8ae994378ec777e655349ca695f2bf76b4e6f83bc6231f92670ca0
common/configuration-parsing/src/main/kotlin/net/corda/common/configuration/parsing/internal/versioned/VersionExtractor.kt	1182	2d81dee258e9aa6d235ea70f05ad89c5f6c047ab73d372cfde50b0

common/logging/build.gradle	1515	842173f33e5102df46c2736ae165d9a9674b7bc1a6826d853d5926
common/logging/src/main/kotlin/net/corda/common/logging/Constants.kt	397	907d15d2e53480ec6b4b4af52cd6bcb50cd652fe0b688edc01b63d
common/logging/src/main/kotlin/net/corda/common/logging/CordaVersion.kt	1286	d52e1209c710e5171dc9cccf84c929ee96610e13e7436bd54e36b9
common/logging/src/main/kotlin/net/corda/common/logging/ExceptionsErrorCodeFunctions.kt	2760	ae5e78bea6ec8fb0219d1b77416f71c6041fc677a5438115f688de
common/logging/src/main/kotlin/net/corda/common/logging/errorReporting/CordaErrorContextProvider.kt	571	76658903bb4d956822d11a492d7f514ab84d1d764f1f897d694fb3
common/logging/src/main/kotlin/net/corda/common/logging/errorReporting/ErrorCode.kt	1438	52cb7aad3b3f4a79b9eb386309cddeec7c80c5a7de36e8ae031f83
common/logging/src/main/kotlin/net/corda/common/logging/errorReporting/ErrorCodes.kt	470	e70f189981a80c34c278aae9dbf6ca9106218e2b9c7d87d7f16735
common/logging/src/main/kotlin/net/corda/common/logging/errorReporting/ErrorContextProvider.kt	788	daf6e6f0cdc3a6e6d372d0bae6a7025238d15e1bb9e1c1cc6b9c18
common/logging/src/main/kotlin/net/corda/common/logging/errorReporting/ErrorReporter.kt	414	d58780c28188f19a0a0338a00e396854aa6aa824c7846e54f1966e
common/logging/src/main/kotlin/net/corda/common/logging/errorReporting/ErrorReporterImpl.kt	1903	7f31b343423722f7f392777ad5c97b04eae24f1d835f7a0f562dc6
common/logging/src/main/kotlin/net/corda/common/logging/errorReporting/ErrorReporting.kt	2531	80e6fe9746d04a53e578b1b665ffc06a2bac03605fdb4cd9cb1a87
common/logging/src/main/kotlin/net/corda/common/logging/errorReporting/ErrorReportingExceptions.kt	1168	f0f271d5dfc40e8cc6c82488a86b42cfc4a44a1061ad14465a88da
common/logging/src/main/kotlin/net/corda/common/logging/errorReporting/ErrorReportingUtils.kt	696	815773c9bba3a9a0bf4799fabc6c4e8cbd8de214e6957a9df5ff7c
common/logging/src/main/kotlin/net/corda/common/logging/errorReporting/ErrorResource.kt	2584	d8a792bf61c1cb6305a5eea64229986a19f9e55a3b2dce2b05697c
common/logging/src/main/kotlin/net/corda/common/logging/errorReporting/Errors.kt	696	828c5d2a6b0c7c3d7fe727d9ff1263385508d8d2184fee3f5c8663
common/logging/src/main/kotlin/net/corda/common/logging/errorReporting/ResourceBundleProperties.kt	351	a638b3b3126788b3251d70fad12268c6e71d6d5746a2074a6c42e9
common/logging/src/main/template/kotlin/net/corda/common/logging/Constants.kt	407	2308d2ec16ffd75919ad16831d2a512247aa2f916f12592134cd93
common/validation/build.gradle	420	da4ca20aadb657efe263f26a0f72b4f4164b636e40c703574197c3
common/validation/src/main/kotlin/net/corda/common/validation/internal/Validated.kt	5801	7741b21d42bf530348af96c43ba0a26d2482333da048822cfde389
common/validation/src/main/kotlin/net/corda/common/validation/internal/Validator.kt	453	e1cbeca4db58a8b1930eeafc0ba186c6ca77bac881680ea2642564
confidential-identities/build.gradle	2214	86451360c76b0708e18f3b01b7080ec9cce8d58fc8db06f2fb7717
confidential-identities/src/main/kotlin/net/corda/confidential/IdentitySyncFlow.kt	6194	6732426603577e6dbdce86393560c7d55fc63d55bca863c759c580

confidential-identities/src/main/kotlin/net/corda/confidential/SwapIdentitiesFlow.kt	7506	2338e6ccd8415c4dda0e68efa38d4a6b66f2d819f7d25cee6d1917
config/dev/corda_dev_ca.cer	769	5862e13fe47792831ef3776b46d6f66bbe1b8b8224eec2fa9cf121
core-1.2/build.gradle	927	830c780df4692e2a7ffa7c3a7b78ac53bf9addf0be7e641bacf905
core-1.2/src/main/kotlin/kotlin/CharCode1.2.kt	164	c5d26b3d133edb831539b3432ca882c2f8b36a9f214e6f90d51d94
core-1.2/src/main/kotlin/kotlin/collections/KotlinCollections1.2.kt	1187	126cdcaf17dc94330c018fffeb77f0ce88b5f868a87170324fa776
core-1.2/src/main/kotlin/kotlin/io/path/KotlinIoPath1.2.kt	1465	e6a1ed3275e771026a039a9c30895939e634c47cd5866ea1feece8
core-1.2/src/main/kotlin/kotlin/text/CharJVM1.2.kt	984	4367c18af66e99058f86cfb12adc16e1bf7e12a75355a213fbb15f
core-1.2/src/main/kotlin/kotlin/text/StringBuilder1.2.kt	452	0efe4be4e0d0ac3f64e8957bb077ace795b0ed8e1a538fbbfb3f72
core-1.2/src/main/kotlin/kotlin/text/Strings1.2.kt	289	16ddaab0dc29b75779270e4574a3ca617136dbe720a62db2b5aa6c
core-1.2/src/main/kotlin/kotlin/text/StringsJVM1.2.kt	611	c2d07d67764d2d52605eaa14bc33c7fda2cc02b0dad3ebcd344473
core/build.gradle	6365	d8130b27da9fc1a7073c3e1543f23af31a454d2368ad8644886a50
core/src/main/java/net/corda/core/crypto/AddressFormatException.java	249	22ecdd94b6bb7f0f6aad61755b5bb73800bce513cc8903549f33bc
core/src/main/java/net/corda/core/crypto/Base58.java	7736	7c28c85e63c83670a9a9ddb9c5dcc49be506447db11d0b468a48ed
core/src/main/java/net/corda/core/flows/IdentifiableException.java	485	b5b0a2746ec919076d359371b8fc0f2403deed5369a5d3140e4f70
core/src/main/java/net/corda/core/internal/package-info.java	166	8abc50bca1e9549cb2b26d57e2b365a0e04bedccc688b99dcd05d8
core/src/main/kotlin/net/corda/core/ClientRelevantError.kt	301	dd6f1dc00786d1fd9e2ab3ee290a769b8978fdea34aef6cebfdfd2
core/src/main/kotlin/net/corda/core/CordaException.kt	3750	823b7c3073d049275030da51af38a37f4b2d12548d346a05844ebf
core/src/main/kotlin/net/corda/core/CordaInternal.kt	516	1cc2bc6def0504e108fab2b57947ca2b198d18c1a5172de0ef4869
core/src/main/kotlin/net/corda/core/CordaOID.kt	925	68cef7fe4d8dd644d9bd7441b298212c3ab00a99ce552d7ab78f72
core/src/main/kotlin/net/corda/core/DoNotImplement.kt	686	63acdf05a63bc3336ca98c7fcf5337a7153f0750283c56513f3bdc
core/src/main/kotlin/net/corda/core/Utils.kt	2208	cb5c4c3558474a21e101d1440713e226e42495aa6879efb6c475b1
core/src/main/kotlin/net/corda/core/concurrent/ConcurrencyUtils.kt	1772	05a530a20f7efdcb9e2bcaea1a6419300f6c4b2d950bd3f916097c
core/src/main/kotlin/net/corda/core/concurrent/CordaFuture.kt	958	c2f6be6eee36b40affba7d683483726c413354a14d8bd61f4da81d
core/src/main/kotlin/net/corda/core/context/InvocationContext.kt	9085	3b3065c37183d399e226bf5c353706a731fd2af9f15e1e98e8655f
core/src/main/kotlin/net/corda/core/context/Trace.kt	1965	69760652ff92ac9b34acf985d4fa0db8ed8f04f0144d7e955c0751
core/src/main/kotlin/net/corda/core/contracts/Amount.kt	24845	79a2903b3c383eb6219d7fbb8fd2f03e916595e3b2f832c3d72f5d

core/src/main/kotlin/net/corda/core/contracts/Attachment.kt	3040	029db995f9a15c3a6f1681f90952be0aca4df1f41dae096cca584b
core/src/main/kotlin/net/corda/core/contracts/AttachmentConstraint.kt	6717	04b726dfb89885262634b6e392abab0704588df270e691b91af14d
core/src/main/kotlin/net/corda/core/contracts/BelongsToContract.kt	1285	5eca9edf113b02c83e5ed75d687bc4a66c79f23d5d3554037df6bc
core/src/main/kotlin/net/corda/core/contracts/ComponentGroupEnum.kt	876	90b7218b5f1cbdb1336f9a713602644e02260deb503f907c3fea10
core/src/main/kotlin/net/corda/core/contracts/ContractAttachment.kt	2060	48687a32841de17064e4d1cb510b24bb3316af90dd7568c8f5bdbc
core/src/main/kotlin/net/corda/core/contracts/ContractState.kt	1375	80e300e51a621b850769eac20ab0d41110ad1484fd3698d198c845
core/src/main/kotlin/net/corda/core/contracts/ContractsDSL.kt	5508	398d98823ef59a63e353e4f00b35cd82a5051454dfe0d4c25b7c5a
core/src/main/kotlin/net/corda/core/contracts/FungibleAsset.kt	2423	4ac443c2e174e128afafc25ddb2202993b3c17ee93a7daa598f833
core/src/main/kotlin/net/corda/core/contracts/FungibleState.kt	1844	1ab9f063982b612c2525035f7d17bf3a297d709f2c5cc4f41db485
core/src/main/kotlin/net/corda/core/contracts/NotaryInstruction.kt	904	ec9c561e43ade961a03157d8e40beac9a0c314ee8025bb651c4a55
core/src/main/kotlin/net/corda/core/contracts/RotatedKeys.kt	5708	f4166ef42c01e5725d8a9929790d5c475adacc41bd97e5ac78c840
core/src/main/kotlin/net/corda/core/contracts/StatePointer.kt	9622	d96c0ee2556050f37f9d0079b17c4acacce24ab33771ea414dedbe
core/src/main/kotlin/net/corda/core/contracts/Structures.kt	18253	a5380eb9109fd2697df7d172824a08ad8f2c3a133b1fbfa6ec9c1a
core/src/main/kotlin/net/corda/core/contracts/TimeWindow.kt	4743	eba59138c8ee147c893d5a5a958c7395152c2bdcf1d229b58e56fe
core/src/main/kotlin/net/corda/core/contracts/TransactionState.kt	4624	604a2bbc6c1fd735dd68e3f25a6feb653c8775a98b960b63fa254d
core/src/main/kotlin/net/corda/core/contracts/TransactionVerificationException.kt	20310	87f2d2d5c71099945f428d3b1c2c57e9094f632909b0ba1e38e030
core/src/main/kotlin/net/corda/core/contracts/UniqueIdentifier.kt	1870	1cff08eb91439cb219377edcf447021c0eb781548654164e889f39
core/src/main/kotlin/net/corda/core/cordapp/ConfigException.kt	195	543b442df7ea66b0878731cdb0df3ebf9ecb6546128e6b56d6a57b
core/src/main/kotlin/net/corda/core/cordapp/Cordapp.kt	7086	4cfdbc75ed2f266856bd48dcb9b01a47f7b80fff7291daf61cef88
core/src/main/kotlin/net/corda/core/cordapp/CordappConfig.kt	2047	5d9f343e6e7bdad9445a0f4d3589a80def342184fd56682522f6d3

core/src/main/kotlin/net/corda/core/cordapp/CordappContext.kt	2981	beb6496da13ed8f35907d4575fc3c0f78197702f9c96f27b1886f9
core/src/main/kotlin/net/corda/core/cordapp/CordappInfo.kt	1312	2d8b3ee7403ed0438b7112ed7f538158a33adb4273626899314296
core/src/main/kotlin/net/corda/core/cordapp/CordappProvider.kt	1072	6fa292bc18afacf941ea0055c211732def68451dfe57ea8fcdd454
core/src/main/kotlin/net/corda/core/crypto/CompositeKey.kt	14610	c0915ff0cb476c68774c4be080cf5822988508fe453b7d9393420d
core/src/main/kotlin/net/corda/core/crypto/CompositeKeyFactory.kt	1386	21f7f9b89755669c83820742da56a99f0c15ee4fe6327312f440bf
core/src/main/kotlin/net/corda/core/crypto/CompositeSignature.kt	3686	e67a5f9fe0d51d054d0a925fb7246953b8417e826b10013b8263b5
core/src/main/kotlin/net/corda/core/crypto/CompositeSignaturesWithKeys.kt	431	afbb31fb97e79cb3de2f6c95c52d12adb510eac23ab87fcc520c04
core/src/main/kotlin/net/corda/core/crypto/CordaSecurityProvider.kt	1954	e62758c2c1e62996c2c15158926aeec47c0ae2e3f3ad06c742e0d1
core/src/main/kotlin/net/corda/core/crypto/Crypto.kt	59754	a0bc267096ca6cb7725bea9df50565226dc0f2c2bebe7b816873c9
core/src/main/kotlin/net/corda/core/crypto/CryptoUtils.kt	15974	2d67521306ac43f2269d42243b7a2e8588f09bf755cb8fc35e1f63
core/src/main/kotlin/net/corda/core/crypto/DigestAlgorithm.kt	1308	4ea49124c6769f0f8074830ac1a70ccce1a0790e0057f96e378775
core/src/main/kotlin/net/corda/core/crypto/DigestService.kt	5654	6fee1bbc2b26bfa13697bf8d5c0ee629ddc0f745463ab5038849fb
core/src/main/kotlin/net/corda/core/crypto/DigitalSignature.kt	2434	187ff92a80494d5cd206e5330aa9e859c2d01311a0e48bc00d4c15
core/src/main/kotlin/net/corda/core/crypto/MerkleTree.kt	3567	f72d81f883f63167779777c3649aa2a4568a87321a6a6ed7a02810
core/src/main/kotlin/net/corda/core/crypto/NullKeys.kt	755	a25a058510dc0d16ad3b0d921b5f3ed4f0d13f4929d07586d5f83f
core/src/main/kotlin/net/corda/core/crypto/PartialMerkleTree.kt	10481	1457527e910bb0686eaec611f131eb2bf621d0463ab9d88e4c90a1
core/src/main/kotlin/net/corda/core/crypto/SecureHash.kt	15828	865a4387463ba8d1e5c765de149d8d32438f30d742b5d7004636f1
core/src/main/kotlin/net/corda/core/crypto/SignableData.kt	730	fc238338bde66706d0c59c19dc07d208b0a7976471e7450e446dac
core/src/main/kotlin/net/corda/core/crypto/SignatureMetadata.kt	1169	00420b5906b5c386a93636bde61b31ce1d8f1108752523f8583918
core/src/main/kotlin/net/corda/core/crypto/SignatureScheme.kt	2023	9d6b6095e8dbf073c06916733c99494accd4ea64a3dfd530a5f6bc
core/src/main/kotlin/net/corda/core/crypto/SignedData.kt	1460	a33f2194358dcf081c08962351374c465404b6be0f6df18307beff

core/src/main/kotlin/net/corda/core/crypto/TransactionSignature.kt	3536	282cc8067ec538d0ca9cbaa66bd26c93e5de606b66c28db3ba4db9
core/src/main/kotlin/net/corda/core/crypto/internal/AliasPrivateKey.kt	1531	84db5515c07c3d7256270be406b493852c60c532da1fb1ff72c768
core/src/main/kotlin/net/corda/core/crypto/internal/Curve25519.kt	2031	7e037bd928013c8304d9456f97bd2510b5265f796bc658f7cf8168
core/src/main/kotlin/net/corda/core/crypto/internal/DigestAlgorithmFactory.kt	2872	bc22d8f13f279cae0cfdfe3ba2f168e69e450e7a1e96885c211c19
core/src/main/kotlin/net/corda/core/crypto/internal/Instances.kt	2417	72be4a45b1a3853a721c8a8df468dc841efaa3d49747c571d948d0
core/src/main/kotlin/net/corda/core/crypto/internal/PlatformSecureRandom.kt	2805	e4ac676a5275ea92dbd30ad378782aa32c945990483ec6d80bedea
core/src/main/kotlin/net/corda/core/crypto/internal/ProviderMap.kt	1832	c38a4de5f6a7855efca172ae61594358fd9a277d778cedc557a302
core/src/main/kotlin/net/corda/core/crypto/internal/PublicKeyCache.kt	2159	4e07692d56d4bdab93a25727043667e8ac7bb06b72a077f1cba31f
core/src/main/kotlin/net/corda/core/crypto/internal/Secp256k1SupportProvider.kt	6670	0e66c3cbf611ceb74e699ac67bed3a2517a89c3a42f9d7798afd49
core/src/main/kotlin/net/corda/core/crypto/internal/keyrotation/crossprovider/ProofProvider.kt	728	7bd59da1283761c20e7b25e403121e2d030cbd30577293d7a8bd72
core/src/main/kotlin/net/corda/core/crypto/keyrotation/crossprovider/KeyRotationProof.kt	1204	7011b28b8fe37526c7534bad1a60d97203b69e07e006c5a25e49fd
core/src/main/kotlin/net/corda/core/crypto/keyrotation/crossprovider/KeyRotationProofChain.kt	6657	41e38cff08b44788c7ce2b8fed3d5658adf811bf04e5b16e978d5b
core/src/main/kotlin/net/corda/core/crypto/keyrotation/crossprovider/KeyRotationUtils.kt	527	5e87cd41c2136990ef66fb301f08ac6fd55fa07cdbbc9287bbf818
core/src/main/kotlin/net/corda/core/crypto/keyrotation/crossprovider/PartyIdentityResolved.kt	1876	cf24dfeffc2dd154a56aa38161444f5249aaeb6f4fdf4327da12c6
core/src/main/kotlin/net/corda/core/crypto/keyrotation/crossprovider/PartyIdentityResolver.kt	4814	3cff2add564dbeb12e86a752913d013daae8f9344e825b8b1e65c0
core/src/main/kotlin/net/corda/core/flows/AbstractStateReplacementFlow.kt	9249	fc6a7c214833075603e4488d6560f431763fa71ce0b39f7bb96868
core/src/main/kotlin/net/corda/core/flows/CollectSignaturesFlow.kt	19364	661f7200dbe97157e938a803c0191ed6fd63a7312ab52531c4b5da
core/src/main/kotlin/net/corda/core/flows/ContractUpgradeFlow.kt	3880	bc43a5bff5898c5880c2362093abaa174af4050942c60f557944d7
core/src/main/kotlin/net/corda/core/flows/FinalityFlow.kt	28208	c1be14e80f9e007e7338080bd429f6eb59701777dea125980d1c00
core/src/main/kotlin/net/corda/core/flows/FinalityRecoveryFlow.kt	4581	9180e9aa6cb9d63048e0edb192b5085882a10713cb581e6f205ded

core/src/main/kotlin/net/corda/core/flows/FlowException.kt	2791	7ffc7244334e343b2a72801de6c452abb132e1e405dcfa310aeddf
core/src/main/kotlin/net/corda/core/flows/FlowExternalOperations.kt	5834	571d5edd0eb4c64ab0dfaf02a71e88fc749886756beeb726f9e06e
core/src/main/kotlin/net/corda/core/flows/FlowInitiator.kt	3490	0a80ed8aaf57b652c351d1b7c469454746c0eae9662a582494853e
core/src/main/kotlin/net/corda/core/flows/FlowLogic.kt	35285	7b30721cc603f90dea61176e74121e05cee29c96ce0e9afda9907f
core/src/main/kotlin/net/corda/core/flows/FlowLogicRef.kt	2598	e9c59ab1454b2ccd140c9c87e78574474c939d9163c68307fdce52
core/src/main/kotlin/net/corda/core/flows/FlowSession.kt	10650	774b63e93204acb365f2406c1085c81cca1eebc0784fbf6adf6866
core/src/main/kotlin/net/corda/core/flows/FlowStackSnapshot.kt	830	9ade388d058bd47c2cb07c4e6a6ca1ebaa7c93a8b267ae1441bcec
core/src/main/kotlin/net/corda/core/flows/HospitalizeFlowException.kt	760	bd851b77cb66926b9eb4f7c9fda309579452f43d9604287738d807
core/src/main/kotlin/net/corda/core/flows/InitiatedBy.kt	793	bb3b20e360300db958a0ac20dceec42a58204a36a3e3e6bbcb9167
core/src/main/kotlin/net/corda/core/flows/InitiatingFlow.kt	1658	e910f380f5e05c340a2f3f539f00319449bd9a5dbc3f8b3386d1cb
core/src/main/kotlin/net/corda/core/flows/KilledFlowException.kt	569	9cfad36adfe5a14668a0f843580b38d4b56ca9a4b379ae87bc33d7
core/src/main/kotlin/net/corda/core/flows/LedgerRecoverFlow.kt	5630	0c3442b11b19c4a69b756165d70723b5b1fd607a73028f1401ed21
core/src/main/kotlin/net/corda/core/flows/MoveNotaryFlow.kt	10288	d7dade9fabdf2368a7c402b875d871b726ce3fee07748aeca35028
core/src/main/kotlin/net/corda/core/flows/NotaryChangeFlow.kt	3050	fbed06af5ef1e278d3d8e796cd8870d67592ae66dd2a3ded48b60e
core/src/main/kotlin/net/corda/core/flows/NotaryError.kt	4454	a6ad43a5e7a10503ec5616e1167426a3e2fb251af08b7461b4c8f9
core/src/main/kotlin/net/corda/core/flows/NotaryFlow.kt	10220	f0a20f3d0502565d7d176a9cc1780cc2b84cbdb05a7c30ca3b0189
core/src/main/kotlin/net/corda/core/flows/NotaryWireFormat.kt	5309	788298068fe203034d698c67e6ebbb3f67a8e4b82423236cf6cf0f
core/src/main/kotlin/net/corda/core/flows/ReceiveTransactionFlow.kt	10457	f364acb2a2e718fa8543e04729c8444e34bd797bef75a2938a54a4
core/src/main/kotlin/net/corda/core/flows/RecoveryTypes.kt	4654	a80d7018147b8a22bd7649f5d5860111d9f7ba4eb2e3bdba9bc8ef
core/src/main/kotlin/net/corda/core/flows/ResultSerializationException.kt	425	8802925c168e5b1ee3ba1c3987b3988b10c046e58b4d3443ba970e
core/src/main/kotlin/net/corda/core/flows/SchedulableFlow.kt	541	ca3657893eb0503d5d9869a6bffcd33562c6924f039a12cd6a2fd4

core/src/main/kotlin/net/corda/core/flows/SendTransactionFlow.kt	19139	c12279119912eea08d392360d200ec0dc0c7297ba05c2e73667a68
core/src/main/kotlin/net/corda/core/flows/StartableByRPC.kt	539	bc93e12bae06f2b26f4242bd83974f393881e5373151864867343d
core/src/main/kotlin/net/corda/core/flows/StartableByService.kt	379	0e9b099b38a4188f2c95e4164bcd3030b8d196ea88f7c008051230
core/src/main/kotlin/net/corda/core/flows/StateMachineRunId.kt	554	237f0283c790f59c6419a31a04b8611b6e2bf0e39bda69506f44b5
core/src/main/kotlin/net/corda/core/flows/SystemFlow.kt	337	39e5e0390e8e9863bbbd7a368860b8d5c394d3f023c1788894a685
core/src/main/kotlin/net/corda/core/flows/WithReferencedStatesFlow.kt	5463	2c4ea8537acf1873b0fd9ab934ae23a96ee5f5798949b0beab5c8d
core/src/main/kotlin/net/corda/core/flows/scheduler/mapper/FlowSchedulerMapper.kt	582	9672412a41116b76c0ad43da0c4bb29f41b909487e93c2193d8c5e
core/src/main/kotlin/net/corda/core/flows/scheduler/mapper/FlowSchedulerMapperImpl.kt	2116	10e4425a6f6e147651b52df4fec0125a39d3c9e36b97505beb4c85
core/src/main/kotlin/net/corda/core/flows/scheduler/mapper/FlowThreadPool.kt	478	5cfa1b09d1753b40360f7c5c1feeaebaecfc7af2ba8e549e792b9b
core/src/main/kotlin/net/corda/core/identity/AbstractParty.kt	2127	1483f8a4bf9a4a2f94d4a1bcc7ee8be30ee298ca1cee55e849ff43
core/src/main/kotlin/net/corda/core/identity/AnonymousParty.kt	1333	c397c86fc60a80e3ee4a1dfd15b3be57951fe6b93b1d8fdda7ec4d
core/src/main/kotlin/net/corda/core/identity/CordaX500Name.kt	5602	a8a468db5c0d76341ae32fc8a5043b3cecab1f99a1c63aa7c5ab00
core/src/main/kotlin/net/corda/core/identity/IdentityUtils.kt	4356	e08cf04ed30c62ab0632b233d3e783812aaebf406828e8818e71ba
core/src/main/kotlin/net/corda/core/identity/Party.kt	2959	537e1a200eba23d4987ed45d9ce0d96256015337d2fdb84b6b3f96
core/src/main/kotlin/net/corda/core/identity/PartyAndCertificate.kt	3386	4bc6e6130f5a50b4d51e0e4661970afa2bc1571c09088cf0e7de5f
core/src/main/kotlin/net/corda/core/internal/AbstractAttachment.kt	4214	1a178423466639f5fcabf0295e532e89131b14dcc1eb67789a86a5
core/src/main/kotlin/net/corda/core/internal/AttachmentTrustCalculator.kt	1549	e4d71ab23ddbe69292a9fff79841a226c5d09109a80988df6778ad
core/src/main/kotlin/net/corda/core/internal/AttachmentWithContext.kt	893	07aa7f12e3b9b5092efdb2f1880b1ca647a6a27345330c3a296afc
core/src/main/kotlin/net/corda/core/internal/BackpressureAwareTimedFlow.kt	2268	0928a6b906f7ed26cc6e5706e45fed92200a0aedeb95520811f2b5
core/src/main/kotlin/net/corda/core/internal/CertRole.kt	4631	f46d731acaa856fdbf63b7781d7e2891124616b36ced919c294242
core/src/main/kotlin/net/corda/core/internal/ClassGraphUtils.kt	554	ac60103f93c8f225e7ad9a96387d0652b200ea576daaba7910f766

core/src/main/kotlin/net/corda/core/internal/ClassLoadingUtils.kt	3955	9ff4686df72523b89cad189a39b3e7769cce769cbcfa638cf71d87
core/src/main/kotlin/net/corda/core/internal/ConstraintsUtils.kt	6822	22dd76b3669bb9da1210c375e0bcb7469f493f07a5340f9a3ab7bb
core/src/main/kotlin/net/corda/core/internal/ContractStateClassCache.kt	589	63d5dc355042108ede54922c1a249eb2501ba6c2aedfcea631b9be
core/src/main/kotlin/net/corda/core/internal/ContractUpgradeUtils.kt	1897	971536b8f74765bc308b47b38e7df59af9626ce07b5696f5b7de40
core/src/main/kotlin/net/corda/core/internal/CordaUtils.kt	4358	1538318d2c2a349ccaee6a3bf4fd19b096434cfebbbc7fba4fd1f0
core/src/main/kotlin/net/corda/core/internal/DigitalSignatureWithCert.kt	2505	acec2a3aef9bc9301a2a1c988804e1c62db1bbcd42cdda56fb8399
core/src/main/kotlin/net/corda/core/internal/Emoji.kt	4915	cab9501072c208c9cc4084d3309ff3b4feceaf59d6988b04c6594a
core/src/main/kotlin/net/corda/core/internal/FetchDataFlow.kt	17137	c18eed8a1b9b37cf598ae4bc6fcb29da34227b3d30069f0819da3a
core/src/main/kotlin/net/corda/core/internal/FlowAsyncOperation.kt	2135	4d6ffc4ef77f1c6855d0edb08ff43658b3b0e6497e1dcd76f3836d
core/src/main/kotlin/net/corda/core/internal/FlowIORequest.kt	3957	84439c8afaa8e1db3cd7869417075360dde353e308023d5ea06322
core/src/main/kotlin/net/corda/core/internal/FlowStateMachine.kt	2003	607cb7336013215972b428d11a7bd80c47cb6d7496abf091e13ada
core/src/main/kotlin/net/corda/core/internal/IdempotentFlow.kt	1590	47b51d78cbce05d44edcf3ef949b9fb6311300b63fe4a900a971b6
core/src/main/kotlin/net/corda/core/internal/InternalUtils.kt	25833	4a4b73a6292acaff5dfad753fd8f7ff3252366cd98a083883eadb4
core/src/main/kotlin/net/corda/core/internal/JarSignatureCollector.kt	4385	34facbb61bb80214235976e2a205bc88857cacccc9acb0d414a15e
core/src/main/kotlin/net/corda/core/internal/LazyPool.kt	2956	3f98ad6c3fa6161322634a67afe147957fe4c111f6336c80fec1e2
core/src/main/kotlin/net/corda/core/internal/LazyStickyPool.kt	2322	ba7b80fd64e34929392d43ea1452ff537e20beeaa443a14750eab8
core/src/main/kotlin/net/corda/core/internal/LegalNameValidator.kt	7969	496c41c2f54a7214a03480c5c4eaf38a5345a09c44a25c98123630
core/src/main/kotlin/net/corda/core/internal/LifeCycle.kt	2321	317faa41cba32b25fa729ffb158e9e6a94fd7d3193df77b4c3e27a
core/src/main/kotlin/net/corda/core/internal/NamedCache.kt	1326	ddb200bcad2a25ebb8d4cb5615f80023f6afa17cc5e4a1f41db790
core/src/main/kotlin/net/corda/core/internal/NetworkParametersStorage.kt	1847	1d8154271b6f0849d1700d8ad438aa5d46f784b422ef679b7ba3d9
core/src/main/kotlin/net/corda/core/internal/NodeInfoConstants.kt	144	251550a03bfb69d79011e6057e5d3492e4f9b94c283677959ea9d1

core/src/main/kotlin/net/corda/core/internal/PathUtils.kt	3558	1f569f617411d18eb5639961955f466e4467469b1c8a1913fd065b
core/src/main/kotlin/net/corda/core/internal/PlatformVersionSwitches.kt	1135	2171bcee20910e6cfd239e9fa6656ef370d3b026e5f070262b6ad0
core/src/main/kotlin/net/corda/core/internal/ProgressTracker.kt	105	10fef032bb8ac92e092c223a0b198c67a5368dc97575cf77bba702
core/src/main/kotlin/net/corda/core/internal/ResolveTransactionsFlow.kt	6738	d65aa6a6270e155e75dc1421fd8a03f38d166295c1852e1a12155d
core/src/main/kotlin/net/corda/core/internal/ServiceHubCoreInternal.kt	3484	53ec3d3d31960ad7318e0813cb814d1f70199a0a52759ad4169fbb
core/src/main/kotlin/net/corda/core/internal/StatePointerSearch.kt	2818	092cb0dc13fb7d811933411e244860d8acac882079f144454b5421
core/src/main/kotlin/net/corda/core/internal/ThreadBox.kt	1403	fc00e654d42749f4e233148d7e67bf95a4a8b15b96039ce42f8681
core/src/main/kotlin/net/corda/core/internal/ToggleField.kt	3541	30ec668a04e9ce9738439c82c396568bbaf6a49876e0e658800892
core/src/main/kotlin/net/corda/core/internal/TransactionUtils.kt	20542	ce56370e74d48e1cee1434f83ea5048edcf9d9afd3edd87bdfa399
core/src/main/kotlin/net/corda/core/internal/WaitForStateConsumption.kt	1828	c7036d5a7e098c46829b074655fffd0b409f31d117b1f668bffdc7
core/src/main/kotlin/net/corda/core/internal/X500Utils.kt	2735	81342a7f15eee190d968d9195af44804f0db59276c8ee9ca05636b
core/src/main/kotlin/net/corda/core/internal/concurrent/CordaFutureImpl.kt	7839	baf800c1e4f886e0198bbf7636a881307f9d0d2e4628b8e5d86887
core/src/main/kotlin/net/corda/core/internal/cordapp/CordappImpl.kt	5130	fe347e0683c93d96aa85290480d1aeacf3794b0613efa93b002ce5
core/src/main/kotlin/net/corda/core/internal/cordapp/CordappProviderInternal.kt	1118	208deb80681353a23d44eec49990af1d06cfa427bcf7cd13b9122b
core/src/main/kotlin/net/corda/core/internal/cordapp/KotlinMetadataVersion.kt	1268	e8b5609b5f81b3c80fbfefe07acf377c0cf6525995a278333b0fe0
core/src/main/kotlin/net/corda/core/internal/cordapp/LanguageVersion.kt	2293	71bedebe49686167f99c546a0d51562af3ee864dd61038a5fddfe9
core/src/main/kotlin/net/corda/core/internal/cordapp/ManifestUtils.kt	776	ac621755c73eec3a7523668c64db92a43aa99c2fbffd43b300996c
core/src/main/kotlin/net/corda/core/internal/errors/AddressBindingException.kt	765	1853929edf8b5725f557ece1a7e932f714dccb04c8a0efabf23dee
core/src/main/kotlin/net/corda/core/internal/messaging/AttachmentTrustInfoRPCOps.kt	302	732db3635ded5a5a0aa4038eba635d4a7f2e8e54b110a7f98bc41a
core/src/main/kotlin/net/corda/core/internal/messaging/FlowManagerRPCOps.kt	805	f5ba216e66f1f875265bfe80cf5ec36c933aeef2996b89304ca6c2

core/src/main/kotlin/net/corda/core/internal/notary/NotaryService.kt	2259	2636652727a0cd5621c76a2a6a1f99cb9c4d3036f49ea0338c2163
core/src/main/kotlin/net/corda/core/internal/notary/NotaryServiceFlow.kt	9397	53e107ce9f08cb9f624c4a95c24cde2787a013f32927842feadff6
core/src/main/kotlin/net/corda/core/internal/notary/NotaryUtils.kt	3388	3ed781827ec9e6ed32b8e78749b4dd40e805a72fcf08810636f0cc
core/src/main/kotlin/net/corda/core/internal/notary/SinglePartyNotaryService.kt	4549	d793e668b61c6bd160c7b018106140201291c85a3beab1973c3b34
core/src/main/kotlin/net/corda/core/internal/notary/UniquenessProvider.kt	2543	c0eb460a600ef2bee720416954581728a6031deb1189880fb14c8e
core/src/main/kotlin/net/corda/core/internal/rules/TargetVersionDependentRules.kt	1870	7355c7c9af516899d013f0849f7d60f8250297f7295f1c86662386
core/src/main/kotlin/net/corda/core/internal/telemetry/SimpleLogTelemetryComponent.kt	5620	c23ec3ced28fe10abf9f4521fc547ee8b407b43e2feefbd8a670f3
core/src/main/kotlin/net/corda/core/internal/telemetry/TelemetryServiceImpl.kt	9227	58e10dcd2860f66cdda8d4dc5c47ad32ba3212cad86ee5787c9458
core/src/main/kotlin/net/corda/core/internal/utilities/Internable.kt	565	990539c7a129b6206cd4a0836e0d3ea1b0d8bc8df269b6985ab996
core/src/main/kotlin/net/corda/core/internal/utilities/InvocationHandlerTemplate.kt	606	40217e147392e50c98d73018c45667a1f3c7f7f0a4dc7ebf71899b
core/src/main/kotlin/net/corda/core/internal/utilities/PrivateInterner.kt	3556	65085784a4ea9bfda859632efb78aa86151007cf115269b2110d61
core/src/main/kotlin/net/corda/core/internal/utilities/TimingUtils.kt	267	d6c098e76a534b7d70c665e055e7b19a6b11f28203cae27af5c486
core/src/main/kotlin/net/corda/core/internal/utilities/ZipBombDetector.kt	2327	6871e84d1e0861edd0a68f46f60b677902a077b78ca3860faead44
core/src/main/kotlin/net/corda/core/internal/verification/AttachmentFixups.kt	3438	26fa59d2329b8a1e2b4376c25f3d0e4cb4615d2d939a4b77e495ea
core/src/main/kotlin/net/corda/core/internal/verification/ExternalVerifierHandle.kt	199	745cc43357ba2540ba9333a97164442664ecc55702c9d1a6335211
core/src/main/kotlin/net/corda/core/internal/verification/NodeVerificationSupport.kt	15168	17d4f2994d728466136fe7f4fd0f6babfde077df1d268d9499bf45
core/src/main/kotlin/net/corda/core/internal/verification/VerificationResult.kt	2638	fc2319e0ed2664b3225cc7592a771b1f607ba2b206691258594b49
core/src/main/kotlin/net/corda/core/internal/verification/VerificationSupport.kt	2031	5ea1c5f0cb9963c20294496439a8cde7570b0d21567d76f34229fb
core/src/main/kotlin/net/corda/core/internal/verification/Verifier.kt	27157	c782c45d0acd3541fe1adc65a21449de7484f1fe25df30f39b1387
core/src/main/kotlin/net/corda/core/internal/verification/VerifyingServiceHub.kt	4118	1351bc2ec1f6c274778e85eae13ead7fdf7da50eb06b088888bc57

core/src/main/kotlin/net/corda/core/messaging/ClientRpcSslOptions.kt	321	142f1b3e39f0c7365c3bbc0f2c8ab3fdf7305800ec35085540cd4a
core/src/main/kotlin/net/corda/core/messaging/CordaRPCOps.kt	34663	b47b2c36e74e7fbb13f0972f4fdd78d479d2744b2f28bd60b1423f
core/src/main/kotlin/net/corda/core/messaging/FlowHandle.kt	3569	0d147fc2bba40d1e004f5134fb88f4acee0a6ff379c04b9e233bd6
core/src/main/kotlin/net/corda/core/messaging/Messaging.kt	646	63f672070467a8f949f94cceba92028f0867aa68202bfb6178ac2a
core/src/main/kotlin/net/corda/core/messaging/RPCOps.kt	686	970de7d83dec497e0f2c17557afa2117d4b96b265b7cd002fb5c8f
core/src/main/kotlin/net/corda/core/messaging/RPCReturnsObservables.kt	431	cc7892f79225d73f223e58c172b4730629ba9e6d628a793cc633d6
core/src/main/kotlin/net/corda/core/messaging/flows/FlowManagerRPCOps.kt	604	f15d0c1c4b8dba51918175d6bdd562f35ed687dc1a5eea15e28baf
core/src/main/kotlin/net/corda/core/node/AppServiceHub.kt	3005	9e6271c3186ef05dc5d8a7f2a226775add1a0e809a7ca5442c78ce
core/src/main/kotlin/net/corda/core/node/AutoAcceptable.kt	500	e1df2bc0711f2bc5682c4068585ba4b6b31b964305463194e3d2a7
core/src/main/kotlin/net/corda/core/node/NetworkParameters.kt	14377	2415faf5e8fc778f481acd193012ea5759420f94709409c6bfd20f
core/src/main/kotlin/net/corda/core/node/NodeDiagnosticInfo.kt	1263	3f122f6ecad646fa56b89ed113888b9e3b57cc68be277b1af8c68a
core/src/main/kotlin/net/corda/core/node/NodeInfo.kt	3650	b4949e4542c8abfd114e2f15e165a5e168e810d878cb987dc0f31d
core/src/main/kotlin/net/corda/core/node/ServiceHub.kt	29113	ffc27597a509898f58059f25446347c3f3f0421a7a1c75c370ae6c
core/src/main/kotlin/net/corda/core/node/services/AttachmentStorage.kt	4430	d6a2aba4370d98fe8ff825ac7edad87db1309eb621752492e6ec4d
core/src/main/kotlin/net/corda/core/node/services/ContractUpgradeService.kt	994	f5f9a12acc2b3b78c3b2636768420cd5ddd639bf09c438c5863e02
core/src/main/kotlin/net/corda/core/node/services/CordaService.kt	1323	5aae32d1eb64dce30847dba60aa6dcd4ec486e3f415f3d8547f20c
core/src/main/kotlin/net/corda/core/node/services/IdentityService.kt	10042	8780b51970700f40f52428b06416ab43899d4bf548e6c0cd1818c8
core/src/main/kotlin/net/corda/core/node/services/KeyManagementService.kt	4992	11010e505ebfcd922ab1f7c99fd87d34d1dc31cd902293a2f9f92a
core/src/main/kotlin/net/corda/core/node/services/NetworkMapCache.kt	6089	7f50b1294b24e82cc121141b6a76667d5e1cce316b39813b432f78
core/src/main/kotlin/net/corda/core/node/services/NetworkParametersService.kt	807	5c740b4f4bcee55481c93291615f1743697b4f3908b5723d77f021
core/src/main/kotlin/net/corda/core/node/services/PartyInfo.kt	467	48061b4cb3802aea4777887c36d712c0857f55105d7630322c3f4c

core/src/main/kotlin/net/corda/core/node/services/ServiceLifecycleObserver.kt	2163	e34fffb23021a5f2f4645e6f123b1806180146686a89633690a7f2
core/src/main/kotlin/net/corda/core/node/services/TelemetryService.kt	184	095f8f5fea1ca67c49f1ada12bcb09cc8494a70eb34e734a0e6076
core/src/main/kotlin/net/corda/core/node/services/TimeWindowChecker.kt	404	8d19d2318c59419d759e5bd44cc1d8a3c5c0fb04721c76c0b5b8a9
core/src/main/kotlin/net/corda/core/node/services/TransactionStorage.kt	1715	b5ca5622cb1ffd6f77b596641ed8284c6ab2ba61eb48c9e49862f4
core/src/main/kotlin/net/corda/core/node/services/VaultService.kt	31731	dde536d4843539d2f536128a140950fe33646b22c3377fc28eb5fb
core/src/main/kotlin/net/corda/core/node/services/diagnostics/DiagnosticsService.kt	503	587012b50a616fd7914f007770a97c00ff5cc93c2ae18c08693243
core/src/main/kotlin/net/corda/core/node/services/diagnostics/NodeVersionInfo.kt	715	108a3d7aef1793d6f4a65ee255e2e6d65f4b2dad413b3973b578ee
core/src/main/kotlin/net/corda/core/node/services/vault/CordaTransactionSupport.kt	386	24876696c571c3ab58cb5b34a9e673608dc75049695c6aaffb2622
core/src/main/kotlin/net/corda/core/node/services/vault/QueryCriteria.kt	39045	0e11863caf4523726de41716100cae1ef3a711e6b5e029c93aba41
core/src/main/kotlin/net/corda/core/node/services/vault/QueryCriteriaUtils.kt	24616	da3f20643386c1a6c7d4f2ac92d328fa16ab2ba3560e9cb1394e9c
core/src/main/kotlin/net/corda/core/node/services/vault/SessionScope.kt	283	3e95038ecb60d583b6552e7cadbe7f76e2fc2894fbdfe88253322d
core/src/main/kotlin/net/corda/core/observable/Observables.kt	637	efd72249edcf3062d6cc9dc9c64f4c51b57e7ce9a156bb94267f8b
core/src/main/kotlin/net/corda/core/observable/internal/ResilientSubscriber.kt	5332	10a075c9c7d6d88250c824922c9b6adb4bca4f8661de36ed1a34f2
core/src/main/kotlin/net/corda/core/schemas/CommonSchema.kt	2548	ba95c254392a267d06bb46b4d4037c02aeb04d2e0dde5df55e2d15
core/src/main/kotlin/net/corda/core/schemas/PersistentTypes.kt	7012	6865759ab861bfea4b55315f7835b651030d1e7f523920ad67336a
core/src/main/kotlin/net/corda/core/serialization/ConstructorForDeserialization.kt	280	bf9a563cd92d35f0babe4b725ba2e8210340b153d06591c3abed81
core/src/main/kotlin/net/corda/core/serialization/CordaSerializationTransformEnumDefault.kt	3384	3b5660f1a535769c8cc50ac554a5f85f515d40406f8f9c39a2c7f7
core/src/main/kotlin/net/corda/core/serialization/CordaSerializationTransformRename.kt	1549	139f0f1899ae51722a35c2640f2562e17a1fa3df82216013618e00
core/src/main/kotlin/net/corda/core/serialization/CustomSerializationScheme.kt	1448	ebd996505d82d53547da469995caba6af6b77f44c718ae1348b62f
core/src/main/kotlin/net/corda/core/serialization/DeprecatedConstructorForDeserialization.kt	526	8f52114465063f038a5105ac404e0d527731e5a350fbbed6407ecc

core/src/main/kotlin/net/corda/core/serialization/MissingAttachmentsException.kt	425	c37ca9fb1540bb16ffc73a809e09d5e0a20ae5ecf786fdedb0a962
core/src/main/kotlin/net/corda/core/serialization/MissingAttachmentsRuntimeException.kt	507	c1f35bf8792a732b6cf0f646d8af1521e0752bfe90577b564e0809
core/src/main/kotlin/net/corda/core/serialization/SerializationAPI.kt	15527	d47020d10de26b38786016e92c06f13ea676e2156a1bd10f940a3d
core/src/main/kotlin/net/corda/core/serialization/SerializationAnnotations.kt	1363	34cb0ef68df03ad770ec08763340b3ade44723f230fec385d079b2
core/src/main/kotlin/net/corda/core/serialization/SerializationCustomSerializer.kt	1813	0516f0fc8d762c72b7fbf64a129c39037662b188ee91863dc8b43d
core/src/main/kotlin/net/corda/core/serialization/SerializationSchemeContext.kt	1562	ed57653946cff5600667ff7471fb5b6a355e872cb614149ca8867d
core/src/main/kotlin/net/corda/core/serialization/SerializationToken.kt	2628	9e5bb583b24e994c7c5f5ca1e93c2fe3376fcdb4a448d166000351
core/src/main/kotlin/net/corda/core/serialization/SerializationWhitelist.kt	802	59f91de8b5df7a4d951b919354ea27c6ff725749f5f35782ea2b0c
core/src/main/kotlin/net/corda/core/serialization/internal/AttachmentsClassLoader.kt	30328	0da53afd302e6de25bf09c55bcbad2ac857a98c2fff63eefd80f9e
core/src/main/kotlin/net/corda/core/serialization/internal/CheckpointSerializationAPI.kt	5218	3256d24aceeb4edbfb23403b5499f9a9e519a52368a626522083e9
core/src/main/kotlin/net/corda/core/serialization/internal/CustomSerializationSchemeUtils.kt	870	f59ef936288cb97bf45b3d03d59a5c8f524d5a55913a855f09aea9
core/src/main/kotlin/net/corda/core/serialization/internal/MissingSerializerException.kt	605	192e4d92154eb6cac7a89ec8c695f1c3601b2c3c5b185877fb1d6b
core/src/main/kotlin/net/corda/core/serialization/internal/SerializationEnvironment.kt	5174	5bece15cbb3a7e3285eae4b78c3ecb23efc757d617d41904c4976b
core/src/main/kotlin/net/corda/core/solana/AccountMeta.kt	1253	bd43fc2e7c57317236dfe850e846aa6c26b2f1ab8d888edb5d992e
core/src/main/kotlin/net/corda/core/solana/Pubkey.kt	893	b7ef000f21d5e186a2ab3ec90184062320e19439ac3b8f7820b16f
core/src/main/kotlin/net/corda/core/solana/SolanaInstruction.kt	1183	52217c4c7a7c5207e3ff9272e74c50530da2326894e11fcf154281
core/src/main/kotlin/net/corda/core/transactions/BaseTransaction.kt	8548	5ded6c9439d29e19bc938cb18ddd6d66c58b57f153a3fe5436ffeb
core/src/main/kotlin/net/corda/core/transactions/BaseTransactions.kt	2860	11aba5003725067db2d0966a61f87663eeef9a671372289d2b33dc
core/src/main/kotlin/net/corda/core/transactions/ContractUpgradeTransactions.kt	23149	a7353563296123b4517469825dc7ae0274ece23675f0c5deb81f45
core/src/main/kotlin/net/corda/core/transactions/LedgerTransaction.kt	48255	5f10be6a49824c7237bb2dfafb2c0e62afd32da3c5455b853504bc
core/src/main/kotlin/net/corda/core/transactions/MissingContractAttachments.kt	1043	23bd83f753578a47d57e286ba894a41f01ba2efc5aca9b6ad50b73

core/src/main/kotlin/net/corda/core/transactions/NotaryChangeTransactions.kt	12699	dfd98937e2af9747fcee93dbfb7780230c7ace0272b183f698e3ed
core/src/main/kotlin/net/corda/core/transactions/SignedTransaction.kt	18720	5e0d988ff758dabbb4ea46450c9d67c40d36c4eac7dd23034728b7
core/src/main/kotlin/net/corda/core/transactions/TransactionBuilder.kt	54678	a58291c7fbefbf7c57e6b562ea38eaa30bfbef0b130cd37a982a9a
core/src/main/kotlin/net/corda/core/transactions/TransactionWithSignatures.kt	5266	174580257de41cad9c6a32bbbde5f54d01b3733c6a344c2efc6256
core/src/main/kotlin/net/corda/core/transactions/TraversableTransaction.kt	28048	58d194e5924756e5d5d4a9b036af20e7fb19a6bd02282cbf344f19
core/src/main/kotlin/net/corda/core/transactions/WireTransaction.kt	34036	90dc7764bd631df2dd04f152250a5ce3019bdd1008bc5fe1183a4e
core/src/main/kotlin/net/corda/core/utilities/ByteArrays.kt	9398	d9737cfd12137945d95f1499822f0d409ff68fbe93900257700c40
core/src/main/kotlin/net/corda/core/utilities/EncodingUtils.kt	4069	902d11b6a8292aa8c3df64331ea0fd2875e42d7057e21398c33037
core/src/main/kotlin/net/corda/core/utilities/Id.kt	1260	a6d8bc4b43d031d4e4604eb17fc47ec87aa70fc3bd03344f2a6f62
core/src/main/kotlin/net/corda/core/utilities/KotlinUtils.kt	4922	f1a16c539ad1dbfbea53802931330d84c56973d9fa48d18c84907d
core/src/main/kotlin/net/corda/core/utilities/NetworkHostAndPort.kt	2066	b40c8551ff2e2948c04791126dec649bbc7afc175c00afbf11d9f9
core/src/main/kotlin/net/corda/core/utilities/NonEmptySet.kt	2624	436ea9b720a2576842d815b18b136d11ef72cb99f8744811b8e0f4
core/src/main/kotlin/net/corda/core/utilities/ProgressTracker.kt	14191	8ef3274c2c422d867d63b0fb99e690878a93d72b46683a240f2bbd
core/src/main/kotlin/net/corda/core/utilities/ThreadDumpUtils.kt	2817	b42e90670522a040a6cab832eb2bf7ec82a0694c944770847676f8
core/src/main/kotlin/net/corda/core/utilities/Try.kt	3704	38eb70cc5d3536c2cd94d418334f6a1dcf2670c0f5397a292769f0
core/src/main/kotlin/net/corda/core/utilities/UntrustworthyData.kt	1330	a0a2bea2b741a1ef018bc4926ef81867a208aabaef07273f2ea8d9
core/src/main/kotlin/net/corda/core/utilities/UuidGenerator.kt	152	73910fdf539fef6954ad8cb27357c4b9f34677424457a5df6967e8
core/src/obfuscator/kotlin/net/corda/core/internal/utilities/TestResourceWriter.kt	2338	fdd00a1853402e354710639fcb6aa16b5e97dc9ddd789c15d8472b
core/src/obfuscator/kotlin/net/corda/core/obfuscator/XorInputStream.kt	763	563a7da6ec8069903024a1e07013ce8ef9d9c8ccc99d2025e0bfca
core/src/obfuscator/kotlin/net/corda/core/obfuscator/XorOutputStream.kt	756	a5dee49ae4deabfd52edda4d1d4c2a146e2d2483ed2185ef2d9846
detekt-plugins/build.gradle	380	da6ca99768ad3ba61e3f8ab7683b80fe36d8eba85fb5e3671a4100
detekt-plugins/src/main/kotlin/net/corda/detekt/plugins/CordaDetektProvider.kt	593	3c20ed70b27b23f5a17cc3e8ac8ef2e6c4c181839973c9194c6905

detekt-plugins/src/main/kotlin/net/corda/detekt/plugins/rules/TestWithMissingTimeout.kt	1898	b7081d434d79cc65a8b78ca0bc10553691c4baeb02b5697fa78c5f
docker/build.gradle	8953	3f95bf68482ea45f5e11fa896cba23421d8280c7d13d2e5c3e6224
docker/src/main/kotlin/net.corda.core/ConfigExporter.kt	4156	41e029b62b2553342568c28e0ac0691ac28e804b240fedd38d006c
docs/build.gradle	4638	76939e1941a76463b9bdf5c7c3242bf9ae2b40bd3a50ae77a33a5a
experimental/avalanche/build.gradle	815	a959df656779b877d882a4a64ec8f1aaf72dbdff95948bc466b10e
experimental/avalanche/src/main/kotlin/net/corda/avalanche/Main.kt	7942	0a8026383002386015a1b9c2de5f47c15e27220e616c2d66caf941
experimental/avalanche/src/main/kotlin/net/corda/avalanche/Parameters.kt	1546	80506afce8a93d93b6d97c87bba6f7046aad02c1c8a316ede59bfa
experimental/blobwriter/build.gradle	890	94459b0b3375c7b9f9b27797f2ca8991317ee26adf7c54ef9bed7b
experimental/blobwriter/src/main/kotlin/net/corda/blobwriter/BlobWriter.kt	5158	6e4788639043d3d26ab3b1c8d7c01f722c520464eae00750eab959
experimental/build.gradle	1263	161cec38a080787c2be5789397aafc8a3cb34af8cc9e678ff7fc83
experimental/corda-utils/build.gradle	1086	bfda8d1b897bfad618b97a2704a4fe328835d44b01bede8bf53891
experimental/corda-utils/src/main/kotlin/io/cryptoblk/core/StatusTransitions.kt	4956	66026ff77fc858403a9ff4655fe53e3d1170f3341ca8e27716bde6
experimental/corda-utils/src/main/kotlin/io/cryptoblk/core/Utils.kt	2781	4b1f1bb278a0c74fae2d517357cec4a41ad073ac501d7669394a75
experimental/cpp-serializer/bin/blob-inspector/BlobInspector.h	471	13b577526de5943ad5cb337c1437dfb3684201af029671bed562e3
experimental/cpp-serializer/bin/blob-inspector/CordaBytes.h	723	2485f7b4e0427849ac01ea0511cabe8b492f778344f250fd99e862
experimental/cpp-serializer/include/amqp/AMQPDescribed.h	291	a002839929da397b1fb6f3e4db6eec2e3a476aba93c8c138844285
experimental/cpp-serializer/include/amqp/AMQPHeader.h	482	a14189768bff2dfee83b8baf8b1cde30a0155fb4c913622bbafe7e
experimental/cpp-serializer/include/amqp/AMQPSectionId.h	400	60aa70c6a24296c69baf1da897d1c8eaf7cdd7e3cfeb72cf383fae
experimental/cpp-serializer/include/amqp/ICompositeFactory.h	927	4b8c0b11735895df71304008c311cb3cb8d2d26457d83487b8967d
experimental/cpp-serializer/include/amqp/reader/IReader.h	2060	b83b81c3e4357dff48d4b5c146d1160f9c19fcc4afb6e20b16b0ae
experimental/cpp-serializer/include/amqp/schema/Descriptors.h	799	5faca7e38c1bea4cb795d618d3c6b44039f4291d4c7b90699b82da
experimental/cpp-serializer/include/amqp/schema/ISchema.h	317	2a5294b203b72f9af271bab73038e5914702c01baf440cb2248fcb
experimental/cpp-serializer/include/amqp/serializable/ISerializable.h	148	7f39ad8461f2ae25a1753c76bd9de0819e4d2fb9e08d84b8d75084

experimental/cpp-serializer/include/colours.h	454	7ca00d927a98c431bb79811cccc00e43d95950476bbc82c85c618b
experimental/cpp-serializer/include/debug.h	430	7c078c37064434b143b320dfc8a852802749fcf2ccf2646df0653c
experimental/cpp-serializer/include/serialiser/Serialiser.h	238	88103a8e7f3f920f547e6a324aecef739eaa9465a2792995d05184
experimental/cpp-serializer/include/types.h	685	083f99a9355b2af24603c1d80cc1d75eb97c2bc1c1cd2cd2e9f3af
experimental/cpp-serializer/src/amqp/CompositeFactory.h	2362	1c6ff4b9b2902611d6da185b8e133c59abe465ccbd5c5f794a2301
experimental/cpp-serializer/src/amqp/reader/CompositeReader.h	1522	546388766f5d7e9c2115092a3d89aedb30f5d06fe6ef1caf499876
experimental/cpp-serializer/src/amqp/reader/PropertyReader.h	1554	ead07478db52bbdea702b81ec1f7e6c49fd80abc521e540f23fb8d
experimental/cpp-serializer/src/amqp/reader/Reader.h	7281	7b3428f4ee9ae51644556d5b4aac3f80ef3b8ff7ec2b8e00461836
experimental/cpp-serializer/src/amqp/reader/RestrictedReader.h	1201	2495983239e69e2295e911bf848b90443f2153c1fe959d0da7c743
experimental/cpp-serializer/src/amqp/reader/property-readers/BoolPropertyReader.h	1066	67e567348e8f1492c04dd5dfcce5052e005dc801d24d5260ca0045
experimental/cpp-serializer/src/amqp/reader/property-readers/DoublePropertyReader.h	1069	effbbac0c194522683f7481a61fa47ef744994488cc60105854481
experimental/cpp-serializer/src/amqp/reader/property-readers/IntPropertyReader.h	1065	2457c6f1e4e76f15a03d11bd9740089adbb7f2db727e7f70213b66
experimental/cpp-serializer/src/amqp/reader/property-readers/LongPropertyReader.h	1066	2f03d080bf7e41583e159cdcf7530be052f940b6199ae555674718
experimental/cpp-serializer/src/amqp/reader/property-readers/StringPropertyReader.h	1069	687a79cace1a00a518dfdba676e289ee70ca5bf558343170771c02
experimental/cpp-serializer/src/amqp/reader/restricted-readers/ArrayReader.h	1275	9c3883d94761248aa5f3830622be9e3bf33b287dae6208bc9b855c
experimental/cpp-serializer/src/amqp/reader/restricted-readers/EnumReader.h	764	b18b4da7307077d98e39346fcad43f7ffeb635feaf856d7c0bc837
experimental/cpp-serializer/src/amqp/reader/restricted-readers/ListReader.h	1302	e5e42653d1d4e442815c0ff5e1290e18746f1a6f9bb54bce51a3e9
experimental/cpp-serializer/src/amqp/reader/restricted-readers/MapReader.h	1472	4edb4aec65ca425a0635d75c6f95b82968479d47b75145c7853be9
experimental/cpp-serializer/src/amqp/schema/AMQPTypeNotation.h	1723	051e285a0908a3d154b53964e7fdca632b9320dc546d69f29d6531
experimental/cpp-serializer/src/amqp/schema/OrderedTypeNotations.h	5903	e62dccbe40265dbf4a725c09931b17ea83f21b335485af92285b72
experimental/cpp-serializer/src/amqp/schema/described-types/Choice.h	616	88b0c4d22798aba4bdb016d7afb24830e62acf65906f687d5b872f
experimental/cpp-serializer/src/amqp/schema/described-types/Composite.h	2632	120ee2140dddbed7c0f761ad39fef3e801b30224ecc899de8127ed

experimental/cpp-serializer/src/amqp/schema/described-types/Descriptor.h	686	a2c536f864d05b275e671a9e5ca3455db99b82a72b8757488b7e11
experimental/cpp-serializer/src/amqp/schema/described-types/Envelope.h	924	e29ff98938596b5fc288afef93b675ed2ab481c8ae79fe566646d3
experimental/cpp-serializer/src/amqp/schema/described-types/Schema.h	1751	2aa444eab520abd4b446307193ce7827804b70812f3e236ea12936
experimental/cpp-serializer/src/amqp/schema/descriptors/AMQPDescriptor.h	2253	eb4d91a79586811b84ac42d9334ef169980482d10db917d23a84e6
experimental/cpp-serializer/src/amqp/schema/descriptors/AMQPDescriptorRegistory.h	1222	c9edafe8147bd58a4dbf61e8f03d1257cd4f36205557747eb68e02
experimental/cpp-serializer/src/amqp/schema/descriptors/AMQPDescriptors.h	3622	ccf78c3d9f0f199cad51aad16539d20cf8eb620b07bca639e3116a
experimental/cpp-serializer/src/amqp/schema/descriptors/corda-descriptors/ChoiceDescriptor.h	696	fea9430081f400e10681bace4ac283a841b0a7da4d5fda5f7fb687
experimental/cpp-serializer/src/amqp/schema/descriptors/corda-descriptors/CompositeDescriptor.h	735	4e0e6cd3daa517be3643c75ab5756d0d2f7371e145cf2dfe876ede
experimental/cpp-serializer/src/amqp/schema/descriptors/corda-descriptors/EnvelopeDescriptor.h	1087	81cbd87f7458d0fa4feded0639862360a5320e5d0478beb989163e
experimental/cpp-serializer/src/amqp/schema/descriptors/corda-descriptors/FieldDescriptor.h	859	79973512d4547a4d0fec2565bb33bc1e155b790885fc9ff43c9540
experimental/cpp-serializer/src/amqp/schema/descriptors/corda-descriptors/ObjectDescriptor.h	883	0b7e57e0c121b26d0a81d2fe328e1acf8b9a1724a869a963739d61
experimental/cpp-serializer/src/amqp/schema/descriptors/corda-descriptors/RestrictedDescriptor.h	1125	6b019d00173f53513a98f88cca16a1160d8698884e760b4ba734ab
experimental/cpp-serializer/src/amqp/schema/descriptors/corda-descriptors/SchemaDescriptor.h	799	f84ab5ba47be7d25dd77bf8b04359c1d5941289938b3aca407145e
experimental/cpp-serializer/src/amqp/schema/field-types/ArrayField.h	658	0414d12d0cf5dbae469fa2898a1503c5b5ac909a7ab5f649afd751
experimental/cpp-serializer/src/amqp/schema/field-types/CompositeField.h	691	1aced4bee775f9fc7b14666bf9c616f6fe73c20654dd16d698a75b
experimental/cpp-serializer/src/amqp/schema/field-types/Field.h	1966	1f28464e98ee673f92ad24c1c8b79055f229b418d0347317630f6e
experimental/cpp-serializer/src/amqp/schema/field-types/PrimitiveField.h	667	fc0186be77c87866c2bbaeeaedde4d21f2189cdbcd2d6ef465ca0e
experimental/cpp-serializer/src/amqp/schema/field-types/RestrictedField.h	693	94d7cd0626ebf912afeda45718a3898aeab62884ea0feadda3dd03
experimental/cpp-serializer/src/amqp/schema/restricted-types/Array.h	1256	7ff4e7e3df52313b823e2089ef40d308a7b722390a509ac9b3bf3d
experimental/cpp-serializer/src/amqp/schema/restricted-types/Enum.h	1224	f44aec4338f2b0c12fc7ca9470380e77c8a501f6c8349b50df5567

experimental/cpp-serializer/src/amqp/schema/restricted-types/List.h	1237	b1282cd43e41c2560fba2e0f33ab988ec787aac59e2cdd68275562
experimental/cpp-serializer/src/amqp/schema/restricted-types/Map.h	1346	443b00d67c69378e95921168ecc80c10050818e2d17823a0e5f919
experimental/cpp-serializer/src/amqp/schema/restricted-types/Restricted.h	3902	12cabe155131406fd298fe864f06a9e33fba6d0c6bca926e651035
experimental/cpp-serializer/src/proton/proton_wrapper.h	2523	c44e829557f6da6dc212ea3ea4fa3b0ac9771d9f14b01f0294e21c
experimental/kryo-hook/src/main/kotlin/net/corda/kryohook/KryoHook.kt	0	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca49599
experimental/netparams/build.gradle	1220	c011fe3d6bf68349eb70cf835e49e6417f234370b87e25a95cc2c4
experimental/netparams/src/main/kotlin/net/corda/netparams/NetParams.kt	10121	b03fdef6e746a6f91252cf3bd8c0cc3f1318120d1de8fa76e3a823
experimental/nodeinfo/build.gradle	1142	e909327e78104c2ca55e0632e4e435cb64f060a2dc68b6c914df75
experimental/nodeinfo/src/main/kotlin/net.corda.nodeinfo/NodeInfo.kt	7528	239d7600c46e2bed6b3894f971fce1decbd9d6ab00fa8864139027
experimental/quasar-hook/build.gradle	950	ca827fe55a63187ebe7fe859979598cf2efe75189edd8f03b5f619
experimental/quasar-hook/src/main/kotlin/net/corda/quasarhook/QuasarInstrumentationHook.kt	11925	f69cee99180eb8eb5fd81c6e738e2130ab474209a7efe040ae07e3
experimental/src/main/kotlin/net/corda/finance/contracts/universal/Arrangement.kt	2244	41c129491c8bbfa5ec73a8fc89ff324163eb898f65f2b54ad2ecb7
experimental/src/main/kotlin/net/corda/finance/contracts/universal/ContractFunctions.kt	1310	d5915e3e0e9a8c01dfdb7c0e7647861e99cf431ce283c957a45301
experimental/src/main/kotlin/net/corda/finance/contracts/universal/Literal.kt	5517	484301804085210e841577579e60aca23fd2652b0616f6a72a8daf
experimental/src/main/kotlin/net/corda/finance/contracts/universal/Perceivable.kt	9279	7c0c17afe54bed7f311fd082aa0485995b90d36e7a00b63cd2fe7a
experimental/src/main/kotlin/net/corda/finance/contracts/universal/PrettyPrint.kt	7164	8789ce79c1b9b84482063c38ac8fae6ceb9dfb9cf9a53163713456
experimental/src/main/kotlin/net/corda/finance/contracts/universal/UniversalContract.kt	16645	1d916f6b3cb5bde0a6a2f8dd76e33c8f389137eaad7f933dd3a9da
experimental/src/main/kotlin/net/corda/finance/contracts/universal/Util.kt	9557	0df852e0d61a9b6c2883536d14d687f05503e550e99c15a660765f
finance/contracts/build.gradle	2636	c6cf093068a8c56a96c7b01e68f95b921993d4f98290f9d6486321
finance/contracts/src/main/java/net/corda/finance/contracts/ICommercialPaperState.java	965	712f13da117f328bdd662aa4b786ecd950ab00deb85d9f35c0854f
finance/contracts/src/main/java/net/corda/finance/contracts/JavaCommercialPaper.java	9863	b07563e6b6817dc4e7d92b622fc6ec2e429a682e74b6a74ef1e0bd
finance/contracts/src/main/kotlin/net/corda/finance/contracts/BusinessCalendar.kt	5719	be6832cee9d508304eb889c6027245f951b8725444955551facecb

finance/contracts/src/main/kotlin/net/corda/finance/contracts/CommercialPaper.kt	8979	c55a9ffa9e5e30ce4b8fef3eae3a405f45b2c52b442c4c3a188bab
finance/contracts/src/main/kotlin/net/corda/finance/contracts/FinanceTypes.kt	11626	0f4b81e8c14d7ed939db093e5d6995849f6ef56b0e5a46fba98931
finance/contracts/src/main/kotlin/net/corda/finance/contracts/asset/Cash.kt	13338	0aa9b04002aa2421956ef2d2bdd13ddac71ca8c3a383a1ddece6ab
finance/contracts/src/main/kotlin/net/corda/finance/contracts/asset/Obligation.kt	28575	376f353e356c40d5d58f7dcf5757fe40a7026e0b7f55124d597f00
finance/contracts/src/main/kotlin/net/corda/finance/contracts/asset/OnLedgerAsset.kt	23116	b0d8804c6d73e0acb3ba11806daecd1afeeddfbfb52ec6e70c439e
finance/contracts/src/main/kotlin/net/corda/finance/contracts/utils/StateSummingUtilities.kt	3366	4ac43625cce48764fd3b9ae7e2463c67726de5540e6da419a7e3e5
finance/contracts/src/main/kotlin/net/corda/finance/schemas/CashSchemaV1.kt	1803	c38acd78966041e4cce2d338df95e3e9c648d12907322ce5622ef1
finance/contracts/src/main/kotlin/net/corda/finance/schemas/CommercialPaperSchemaV1.kt	2467	8b988fc965c4af8b5012d4c5141356c989feb2ade46e4c4fe7530e
finance/workflows/build.gradle	3452	2b405142d26234670b763e6624a51e06ada6395a2e4d15d61cc7e8
finance/workflows/src/main/kotlin/net/corda/finance/Currencies.kt	2418	12dd29b9c545c765248210e63865f185987dbb899c94235be42dc0
finance/workflows/src/main/kotlin/net/corda/finance/flows/AbstractCashFlow.kt	2363	ce3f780b6f52cdc601b77ab6756d9482880a12278e16919d94caa4
finance/workflows/src/main/kotlin/net/corda/finance/flows/CashExitFlow.kt	4450	94d0894f764d88f0962fa0d1b46b46d39f995e6ca2af773baadedd
finance/workflows/src/main/kotlin/net/corda/finance/flows/CashIssueAndPaymentFlow.kt	2791	872f80dc332c45280543e90a4ff94de1d563f60e5e23006816915b
finance/workflows/src/main/kotlin/net/corda/finance/flows/CashIssueFlow.kt	2774	48c7e202af37cb4bb3a58daba62d6444808e25027843555f8642d7
finance/workflows/src/main/kotlin/net/corda/finance/flows/CashPaymentFlow.kt	5239	2567616c0d359e928d5151224a90449156f64874d66fe40692acd0
finance/workflows/src/main/kotlin/net/corda/finance/flows/TwoPartyDealFlow.kt	9451	4599db48d27a6bee1bf135ff9ed84e0ed94b69baaea528a5f0109a
finance/workflows/src/main/kotlin/net/corda/finance/flows/TwoPartyTradeFlow.kt	13142	6e91bed23d49ecda61f18151b9caf0b5f9f4d036205269915c9ffe
finance/workflows/src/main/kotlin/net/corda/finance/internal/CashConfigDataFlow.kt	1859	84591a0dbee7dcfdf7c9bf706cac43b80b2b92fcc52725d9f0f6ed
finance/workflows/src/main/kotlin/net/corda/finance/workflows/CommercialPaperUtils.kt	2785	40184011dc97a6c97826c65a0655268d0e698671bd4655c0479ecc
finance/workflows/src/main/kotlin/net/corda/finance/workflows/GetBalances.kt	3458	66d8c519fb212ea385ed468248b10268f0c4dd95471c0cb3dcefa2
finance/workflows/src/main/kotlin/net/corda/finance/workflows/asset/CashUtils.kt	10520	d78f6a24b2b83da999d4a757c0be326beccf2ada87f5b6ac2537e0

finance/workflows/src/main/kotlin/net/corda/finance/workflows/asset/ObligationUtils.kt	14972	074fde272c233a23ddf1768f39240e3a8df8469d40972e0412d6c4
finance/workflows/src/main/kotlin/net/corda/finance/workflows/asset/selection/AbstractCashSelection.kt	9927	1699b1a0071cbad98b02119e547336dd4c8aaf6f51b5d35b7e4444
finance/workflows/src/main/kotlin/net/corda/finance/workflows/asset/selection/CashSelectionH2Impl.kt	3931	30b43ad56415c3593157af06b54c4f3c26d81343a40d0e4c9b4a49
finance/workflows/src/main/kotlin/net/corda/finance/workflows/asset/selection/CashSelectionPostgreSQLImpl.kt	4330	efa309fa3894dd7b43144bdc0c5e6b434953cce9859b0be7eedc84
finance/workflows/src/main/kotlin/net/corda/finance/workflows/asset/selection/CashSelectionSQLServerImpl.kt	5144	5ab46b9648dbd5fb3110e45e20f1e2c6b3ebf04adf2de7baa304f8
finance/workflows/src/main/kotlin/net/corda/finance/workflows/utils/FinanceWorkflowsUtils.kt	722	4a86a7b8a30e38b0ce4fab1dde56d026eee0794815d83a5b05d6b7
isolated/build.gradle	684	e2b1f537de5bdf2e1b6dd3eccb94be57fc2ab938451cb9300befb8
isolated/src/main/kotlin/net/corda/isolated/contracts/AnotherDummyContract.kt	1254	e2d8169ef5e3af4c164d5b15b35e517a911f987627609665ac0705
isolated/src/main/kotlin/net/corda/isolated/contracts/DummyContractBackdoor.kt	414	f17ff20b7885830f98508d5be673a5a6686468acaeca8be46ad5ab
isolated/src/main/kotlin/net/corda/isolated/workflows/IsolatedIssuanceFlow.kt	889	f2e34c7e9c07b37f2a54288230a62b8fe8d10b5334560de460d76f
legacy411/build.gradle	998	258e14cb2829afa100e7f4b34e2c9d3cd4f94db4b5f28712ebed3d
legacy411/src/main/java/net/corda/legacy/contracts/AnotherDummyContract.java	3831	0d237f8318c68840e587af2a9508b43fe1a1985b1535503a8c03cc
legacy411/src/main/java/net/corda/legacy/contracts/AnotherDummyContractWithSystemEnvVarDefined.java	4189	760e543d100f479dcedefbb9d03f66f053381e1f1d230552735fa6
legacy412/build.gradle	998	ae81f1f05bdfe447bb141b09240d8d75b50dc82d9194f1a5583451
legacy412/src/main/java/net/corda/legacy/contracts/AnotherDummyContract.java	3831	0d237f8318c68840e587af2a9508b43fe1a1985b1535503a8c03cc
legacy412/src/main/java/net/corda/legacy/contracts/AnotherDummyContractWithSystemEnvVarDefined.java	4189	760e543d100f479dcedefbb9d03f66f053381e1f1d230552735fa6
legacy412/src/main/java/net/corda/legacy/workflows/LegacyIssuanceFlow.java	2300	d4faf5ce1615ce946377ac7537de7d511a353b74b6a61a6d900abc
legacy412/src/main/java/net/corda/legacy/workflows/LegacyIssuanceWithSystemEnvVarFlow.java	2401	190ffa0bd4bde2e85ca3465e8ef2540a8c921d512ea9a82cfeb8a8
node-api/build.gradle	4726	231a98b278092da126b8d5fad2881ac52b4a32f588cf1aac5834c3
node-api/src/main/kotlin/net/corda/nodeapi/BrokerRpcSslOptions.kt	138	b50669e4b384ffdcce78f511af20cec15e1151bb9185c8162cea75
node-api/src/main/kotlin/net/corda/nodeapi/RPCApi.kt	19943	2afc4b37af8582f29c5a145e9e31e19c8e7ccb38f3d1b7a6233645
node-api/src/main/kotlin/net/corda/nodeapi/exceptions/RpcExceptions.kt	1886	57cade233dada604d4a1fce0828474aece15f0e0a182dde46bdd65

node-api/src/main/kotlin/net/corda/nodeapi/internal/ArtemisConstants.kt	109	adcfa43cf82fda99bbdfefc6b3b5913d99c8e4f98767abfb196f65
node-api/src/main/kotlin/net/corda/nodeapi/internal/ArtemisMessagingClient.kt	5955	180e1f07094ee9c65516e84767f101d3128c826c552af2ef39db74
node-api/src/main/kotlin/net/corda/nodeapi/internal/ArtemisMessagingComponent.kt	7424	f847f6c95a8ae43d1b5508cff302a4ffd0638d53aaf254ce21c1d6
node-api/src/main/kotlin/net/corda/nodeapi/internal/ArtemisTcpTransport.kt	16430	039574aa6a9051c720d1f0f7e98dedb556e22a5f3c925c4e4c49ec
node-api/src/main/kotlin/net/corda/nodeapi/internal/ArtemisUtils.kt	1828	769d586ce1566f08b09ca84cd5c9d5c8b642f3e59813fc3348c6aa
node-api/src/main/kotlin/net/corda/nodeapi/internal/ClientSessionUtils.kt	297	7d46b89143a51364aab6d893e8d8079dc1ee869ba2765ec8a1603e
node-api/src/main/kotlin/net/corda/nodeapi/internal/ConcurrentBox.kt	732	a75465e3f71f43b02c1cb0317dd06bd0b500fdd92ef03a071321da
node-api/src/main/kotlin/net/corda/nodeapi/internal/ContractsScanning.kt	2549	c34ff443cca3ea0eb366eeb53f52c7615a11d422d34b3fb85ac69c
node-api/src/main/kotlin/net/corda/nodeapi/internal/CordaNettyConnectorFactory.kt	3711	3f9b7658972d2244de21dda7db09664a81c9d36cb7c99eaf248449
node-api/src/main/kotlin/net/corda/nodeapi/internal/DeduplicationChecker.kt	1300	4354cf536ca8400cd859f2c730995ccdfa2f589785b3772a54639d
node-api/src/main/kotlin/net/corda/nodeapi/internal/DevIdentityGenerator.kt	5755	3a383156386e3b40f87884fe0b21c12a6c42f8f4fa9d55a156c1ca
node-api/src/main/kotlin/net/corda/nodeapi/internal/JVMAgentUtilities.kt	963	15b5c1fd82c9b1cc3ce43c70005928a2529009a9e4426b6f02555b
node-api/src/main/kotlin/net/corda/nodeapi/internal/KeyOwningIdentity.kt	1475	74dfa3c37f65d5b14d89c1da670d0437baa809f61a1afd38220c03
node-api/src/main/kotlin/net/corda/nodeapi/internal/KeyStoreConfigHelpers.kt	7869	efc445947784ad7f5b9ea515b1be78e63ba2e9b71be1b2e5629877
node-api/src/main/kotlin/net/corda/nodeapi/internal/MessageSizeChecksInterceptor.kt	2542	5fe27532b472e16276d7bf925bea20ef0d2f9e05eee936ffb0ca1c
node-api/src/main/kotlin/net/corda/nodeapi/internal/MigrationHelpers.kt	1383	6e0f24a0cd83422fe7bf50f7ccb1b8dd78763f7ef3a50b78ec4233
node-api/src/main/kotlin/net/corda/nodeapi/internal/NodeApiUtils.kt	1257	5e7140d9c33c7fe48da6a4ce760befa55c38d2c980a2f96602392f
node-api/src/main/kotlin/net/corda/nodeapi/internal/NodeStatus.kt	124	95888ddef1178ae90990c666d33c5249c9dd6699e12cf1a5fc3719
node-api/src/main/kotlin/net/corda/nodeapi/internal/PublicKeyToOwningIdentityCache.kt	517	60af57949a78d34bc8572d6626489a380b8dba114371a914cb5df2
node-api/src/main/kotlin/net/corda/nodeapi/internal/RoundRobinConnectionPolicy.kt	739	ea2ae667ec40aaedcf5198d3628d19568dbcda57a92e3b927458d6

node-api/src/main/kotlin/net/corda/nodeapi/internal/RpcHelpers.kt	666	ea5719fcde9afa5c0ad1fbcf8d6d19f4ebe4e167cca1a572acfe2e
node-api/src/main/kotlin/net/corda/nodeapi/internal/ShutdownHook.kt	789	14c37064ebcdf88c82153f06ccea1056d3e09ad6c6ba1f66ab0dea
node-api/src/main/kotlin/net/corda/nodeapi/internal/SignedNodeInfo.kt	3202	085027a8cf9fe97133e08b8d90fc1eea5d42072f01491400b0681f
node-api/src/main/kotlin/net/corda/nodeapi/internal/bridging/AMQPBridgeManager.kt	26989	f9a034506f71cab6e2f3f8452c745d542b5ccf947e3f766b77744f
node-api/src/main/kotlin/net/corda/nodeapi/internal/bridging/BridgeControlListener.kt	12806	d5e00ece116b68cf89cf9759be4c57b27d10747e244f3f7a60dc8d
node-api/src/main/kotlin/net/corda/nodeapi/internal/bridging/BridgeControlMessages.kt	3639	199979d525b2bc2f956ffabf020e24699cc535c2934c965fbe9d08
node-api/src/main/kotlin/net/corda/nodeapi/internal/bridging/BridgeManager.kt	758	c637682d0a9625394505d09b5d40b2ca9dee7da372420947c0a56a
node-api/src/main/kotlin/net/corda/nodeapi/internal/bridging/BridgeMetricsService.kt	809	a12b5bf1fa1db48d7f5537fc1639e5291b9438771334eb9a9ec428
node-api/src/main/kotlin/net/corda/nodeapi/internal/bridging/LoopbackBridgeManager.kt	11215	7ef2ccb71fe939fcfb625c309d3b0c887ac7ad497343ea82f1bc3d
node-api/src/main/kotlin/net/corda/nodeapi/internal/config/CertificateStore.kt	3960	fcc650ad6581c222dfa60700b0a3b909a686e1d203c8e138c9d4cb
node-api/src/main/kotlin/net/corda/nodeapi/internal/config/CertificateStoreSupplier.kt	768	3a33374d8fddb8c943b4dd1b2f745bba875f18295174003e9b927f
node-api/src/main/kotlin/net/corda/nodeapi/internal/config/ConfigUtilities.kt	14706	c517c844c8574a2a455edf026287ac4cedac7f319b63eafaa28de6
node-api/src/main/kotlin/net/corda/nodeapi/internal/config/MessagingServerConnectionConfiguration.kt	2726	4068f02e170f51c4c38e8ab39a5ae72c16ba06becf334644eb5637
node-api/src/main/kotlin/net/corda/nodeapi/internal/config/SslConfiguration.kt	1357	01b18b506106f29b66dd1b6bc8a760c76ebb1507cc37f4a564ada1
node-api/src/main/kotlin/net/corda/nodeapi/internal/config/User.kt	290	734dfd492c6bd6c2cf3f5ee8d5568a1b48811a62a2cf2afdd7674d
node-api/src/main/kotlin/net/corda/nodeapi/internal/cordapp/CordappLoader.kt	879	f7ef262510a03bf12c63391f58990f01f5bca453b85a8a64ecdf29
node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/AesEncryption.kt	2901	bc24d67e2bca70a59dedeb2fe45f6460df2d5b6de3c13d3fbe9a4f
node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/ContentSignerBuilder.kt	2777	a6abb731da5babbb19688316a1e57abe0effe049bb8dd6ecc3cae2
node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/KeyStoreUtilities.kt	7813	9daf7d5c3b64eb234fc06f7d4a1532599aed15d54338a87805b2e7
node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/NOT_YET_REGISTERED_MARKER_KEYS_AND_CERTS.kt	678	d0241aa19e69653a96c60cfd85c26c5454ec9df851eeebb2a6943b

node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/X509KeyStore.kt	3935	12b1ad987129a5bfb1b4e045c372304300cc426500f3f8d0d0ddb9
node-api/src/main/kotlin/net/corda/nodeapi/internal/crypto/X509Utilities.kt	26285	48f8f548369b64b5c307d9be1663d16226ce6e26fc98a0e38f4cbd
node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/CryptoService.kt	7711	4ea4fcfa338bf704c3f306cf875c94bb77e452a607e3c4de92e41e
node-api/src/main/kotlin/net/corda/nodeapi/internal/cryptoservice/DefaultCryptoService.kt	11955	85a9aa6e746282b3ea0689a4949de9affe1acbe16a18b8c5efced6
node-api/src/main/kotlin/net/corda/nodeapi/internal/lifecycle/LifecycleStatusHelper.kt	1056	b4fd0984d4e633ea2fdcb4319982c1ebdd7b2b8f44a8ad4124def3
node-api/src/main/kotlin/net/corda/nodeapi/internal/lifecycle/NodeInitialContext.kt	328	a2a3b9d686b444b32fc7916b1022353a937ac73661fbf6bfa9ce85
node-api/src/main/kotlin/net/corda/nodeapi/internal/lifecycle/NodeLifecycleEventsDistributor.kt	5887	0a20c7fa5482a17c5e373bf3b3c8d344e290edc7223890c0468459
node-api/src/main/kotlin/net/corda/nodeapi/internal/lifecycle/NodeLifecycleObserver.kt	2933	5a5d659aa33200483cf12cf8afc5bc7080be1f107cdaf701f32a0f
node-api/src/main/kotlin/net/corda/nodeapi/internal/lifecycle/NodeServicesContext.kt	627	2e9bfd5f179c56952d60ae0876ea72047876463812803e6a366d5e
node-api/src/main/kotlin/net/corda/nodeapi/internal/lifecycle/ServiceLifecycleSupport.kt	1692	2426ba5faca5732d94af048131e57d933c454a7d5798d9fccfa8b6
node-api/src/main/kotlin/net/corda/nodeapi/internal/lifecycle/ServiceStateHelper.kt	1865	243c0cc4e3bf68f755eedbf86a9d4f7887a4f1c5e18b9c11ae0346
node-api/src/main/kotlin/net/corda/nodeapi/internal/network/NetworkBootstrapper.kt	27726	c65d1f1f5f54deed41e6258c718ff198f73f23c1f789031b992cc0
node-api/src/main/kotlin/net/corda/nodeapi/internal/network/NetworkMap.kt	3621	bdc0ddfda18ebf88649bbcfa49b6229e4b218474c1280f81f1e26a
node-api/src/main/kotlin/net/corda/nodeapi/internal/network/NetworkParametersCopier.kt	1610	1301d4567220e127fd79ee566cc72ed71126600752385669a4e476
node-api/src/main/kotlin/net/corda/nodeapi/internal/network/NodeInfoFilesCopier.kt	6502	1967bc83430aa56174c7a1087c717b1e271c70585f6ec715eae8be
node-api/src/main/kotlin/net/corda/nodeapi/internal/network/WhitelistGenerator.kt	2715	2693b1fb70778427485a2264e89449dbf3f1561e2bbfb66359608a
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/AttachmentVersionNumberMigration.kt	5132	910d9b1e51b25b66ac637af98a0886f09e73d0a93b373bb7560f70
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/CordaPersistence.kt	20445	0c66533cf7a2d0d8780199c32d299936f414fbed9c616be291fea4
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/CordaTransactionSupportImpl.kt	986	78257bbcbdedfa097c496d7d6054ec20ccd660abfbb0f8559b49c6
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/DatabaseTransaction.kt	5278	bcd2d21873e8e4ff113f7e75703b562eaeff711a8e6d01e490c56f

node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/HibernateConfiguration.kt	5273	c432c538afb932613189d275726dc2d2a8ed5a75c3f77aa946d1e1
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/HibernateStatistics.kt	6745	15c96e7f093c8f36ead9f9ac2d2d8c7461bc4cdc162ff4a02061d6
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/LiquibaseDatabaseFactory.kt	228	bd03596781b053adaecffd688e44ae1ee12dce858a4cf38389ca81
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/LiquibaseDatabaseFactoryImpl.kt	399	741770e4db270f4959c53571b7fbe2c8a18083fe4b801aaa390e52
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/RestrictedConnection.kt	3216	c0dad0786fbb0517e3accdbd57bdd37bc1c589db9da75422c01851
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/RestrictedDatabaseOperations.kt	1809	fbba7763b21a4cb38c9a87c25880c05300dfe7dbc9b6ef897d92bc
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/RestrictedEntityManager.kt	2565	66f16f78f5ce1749afaba798470f0c965ee7e56cfb95f02be18ae6
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/SchemaMigration.kt	14486	d351a259846f2cb8e5973ee5a315bbc7c245419ad4025c4ae4c07d
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/Slf4jLiquibaseLogService.kt	385	5a104a4d57a5ff440e61d42c9899dfabb1b4638d94b43ec27388e5
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/Slf4jLiquibaseLogger.kt	2277	d9cb6da932e179bfa1d39a17a8b46c4d5b8a53ac079a83f50260e2
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/factory/BaseSessionFactoryFactory.kt	7148	f47892182b00431fc0c5f11e6a9f0d5219ba98d22827f068894aa0
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/factory/CordaSessionFactoryFactory.kt	827	e2439d91cded7575f0687f3e3a0e04bf8ca200cef38ae211f3d768
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/factory/H2SessionFactoryFactory.kt	262	b4af1e24736ed247527c687e95fdf1afaa2c726a82990b04d767c1
node-api/src/main/kotlin/net/corda/nodeapi/internal/persistence/factory/PostgresSessionFactoryFactory.kt	1950	b6bf408bac27f033adae1b387ac2ef0b013726cce62c5e4c0599c8
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/engine/ConnectionStateMachine.kt	25046	0dd3f5c8fd667817ff30e559c2cfc81d4ca0ba5137bd9375bd8ef0
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/engine/EventProcessor.kt	7197	1d0e0654b1b1d1739e57757d8684921121bd708435f5b25800f980
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/engine/NettyWritable.kt	1732	45af4eab3e70f97665a590341f09d6eb33945c2e4cd442ac857529
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/engine/SessionState.kt	694	3bee045591e430838d38965ecbce1f853dea9da17870fe06698366
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/messages/ApplicationMessage.kt	434	2e528b6c6def793d5b072e3b5e1aa95003789a6512ace16184afd3
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/messages/MessageStatus.kt	186	f02ce2653a55fd1f069f3057e94d6bca34a76b59bde886ec3fe199

node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/messages/ReceivedMessage.kt	349	8b5d3fe2eaf8c99d4d1f360ffc395818b0de2803c7248da652d1f1
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/messages/SendableMessage.kt	280	0cbb7a82d0c7fa8a213dd6c593f992fdc287ec026878fbc70c46e1
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/messages/impl/ReceivedMessageImpl.kt	1953	5c24900b565fb81814d9f7056713c5092bc8de35e1224c88458042
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/messages/impl/SendableMessageImpl.kt	1625	34e3ec8220dbb397c78ccfaed3351b66cf695956c9bf876c97c2f9
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/AMQPChannelHandler.kt	14999	7fa97ad10dd6d691d6db4b37fac57022147983f008b2f976bd90b1
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/AMQPClient.kt	17840	b93674b5fffccffd8106c58190b63c6d11192d19b4e7980282df33
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/AMQPConfiguration.kt	2682	1bd9fb7e132d56355badaed7fb2ee5242f885ef8fc29095fa2666e
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/AMQPServer.kt	10362	c6046532473d30a46e0016aac84aae3ee4233195c03f235efc590e
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/AliasProvidingKeyMangerWrapper.kt	2424	0b2ae4e0c7ee4acf53dbbf9dfc401f2e78bce85b484b08faff6acf
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/AllowAllRevocationChecker.kt	1039	d458511d8b183893ba2ec8e70a037b1db4f08733de405d7ec1f4dd
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/CertHoldingKeyManagerFactoryWrapper.kt	3256	2726566c112f1dd9f65414dbd73bd8e237eaa99aea8dc8eca1f2e0
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/ConnectionChange.kt	520	662d3f1bb2c30520f7ebbd0d957077b005d448dcd9f609ae234198
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/ConnectionResult.kt	124	6cbb1a7a6e2d46d17d177b62435fa82d3fcce561714f9f5ea9486d
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/CrlSource.kt	347	1f1de9b1141d4d63bd7677e32c12c8a2ffb62626c01735037275c8
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/ModeSelectingChannel.kt	2974	42ff6637b81f030d55d14ed68a40dde086f7d78cf1f34c70dc8d9a
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/NettyServerEventLogger.kt	2925	68feaf1d7b619d080105c2ddf3c205d8618f2940159d7abd3462d5
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/RevocationConfig.kt	2637	f71528d177878eb080b616e55c92bff06b5e553275f430a3e09b08
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/SNIKeyManager.kt	4293	ffea0c80f2b63981902a6a6f48156545f919d196d139bdc6409522
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/SSLHelper.kt	17035	ff1faebd2ac268593f88d74da43c7114b795acfa02eb2f6a5fa4d9

node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/ServerSNIMatcher.kt	1955	ae4f0a67795701a75f0b0b888e6f0aeb0777942ce2b01341db8e54
node-api/src/main/kotlin/net/corda/nodeapi/internal/protonwrapper/netty/TrustManagerFactoryWrapper.kt	1080	a245a18a725639ca0082826caa4112be3430bf97b84cfbf14298b1
node-api/src/main/kotlin/net/corda/nodeapi/internal/revocation/CertDistPointCrlSource.kt	5271	829dcaa30e5db36629397422af1a08be1ba336980d3c2c596b1a72
node-api/src/main/kotlin/net/corda/nodeapi/internal/revocation/CordaRevocationChecker.kt	5433	781d6b92c287cbf313894d2cf739bcac678098b467e7498fb0d386
node-api/src/main/kotlin/net/corda/nodeapi/internal/rpc/ObservableContextInterface.kt	924	48ac06b2aa7a10dd7b657a1216d26c6efaeb6e99e503323c14fe54
node-api/src/main/kotlin/net/corda/nodeapi/internal/rpc/ObservableSubscription.kt	135	10938d178f67797692bf7584170337d5a93a677c5768c1e92257e5
node-api/src/main/kotlin/net/corda/nodeapi/internal/rpc/client/AMQPClientSerializationScheme.kt	5365	005e2e45bcb200f25e27b7475260e80be48dc44a2df025533fbb25
node-api/src/main/kotlin/net/corda/nodeapi/internal/rpc/client/ObservableContext.kt	1223	c3a4c9151e408116a01366fdceb7f0e7f086bb8c861f99f1af2dba
node-api/src/main/kotlin/net/corda/nodeapi/internal/rpc/client/RpcClientCordaFutureSerializer.kt	1313	560cb8cefeb3c409e38722d94913a462ac8bbf4f72e647d700f017
node-api/src/main/kotlin/net/corda/nodeapi/internal/rpc/client/RpcClientObservableDeSerializer.kt	6244	6469d2a191184cf99c33650b996ff53ff41c82b08e7e0de3867f3e
node-api/src/main/kotlin/net/corda/nodeapi/internal/serialization/amqp/AMQPServerSerializationScheme.kt	2757	7c5a5b4cd9851e66233a4c84422fcbb43d4d75dad240e219b33460
node-api/src/main/kotlin/net/corda/nodeapi/internal/serialization/amqp/RpcServerCordaFutureSerializer.kt	1194	82130bb6c53bc07810432172c977847fa5cc74143f60e7e5634e96
node-api/src/main/kotlin/net/corda/nodeapi/internal/serialization/amqp/RpcServerObservableSerializer.kt	6446	8e5779c553e935f1d0d1fdf5c914d3ebb7ed3fd198d62aa015a812
node-api/src/main/kotlin/net/corda/nodeapi/internal/serialization/kryo/CordaClassResolver.kt	10083	ea49ac4ecd993e26d817eea38ad280de6d34d9d8b004ed2fec6871
node-api/src/main/kotlin/net/corda/nodeapi/internal/serialization/kryo/CordaClosureSerializer.kt	798	5dd307dfe72ee8d90c23b6bf8dbdda7d1d2ed1bfad25b5cf9a55ca
node-api/src/main/kotlin/net/corda/nodeapi/internal/serialization/kryo/CustomIteratorSerializers.kt	6420	706d65a1f49625d377f84590d5a7ba2a1a4049584c497a21774173
node-api/src/main/kotlin/net/corda/nodeapi/internal/serialization/kryo/CustomSerializerCheckpointAdaptor.kt	3772	78f0b957f7c691a771664b1674594e47c35c5038126373ca631a44
node-api/src/main/kotlin/net/corda/nodeapi/internal/serialization/kryo/DefaultKryoCustomizer.kt	14178	355e296beafda8d6011a874cc456876579b41b0d91036bcff994e3
node-api/src/main/kotlin/net/corda/nodeapi/internal/serialization/kryo/IteratorSerializer.kt	1988	bf57a70a3407825217109a72f3a708ec55ad5b42d25ea6e0d091a4
node-api/src/main/kotlin/net/corda/nodeapi/internal/serialization/kryo/Kryo.kt	22017	dfdd2cf5a5bde1cbc1ae5ee9225b37bf10f500f703dbe363f8baa3

node-api/src/main/kotlin/net/corda/nodeapi/internal/serialization/kryo/KryoCheckpointSerializer.kt	9434	e2f2da4a62b477db4f05c7cfb306ab57b2884e33a4a3171534409
node-api/src/main/kotlin/net/corda/nodeapi/internal/serialization/kryo/KryoPool.kt	511	fc07022952c7207bc79124b600b4ffd888eff35ce31ce7e33f0b60
node-api/src/main/kotlin/net/corda/nodeapi/internal/serialization/kryo/KryoStreams.kt	1314	0eae2ae8ce659773a00cebb869297e0dc3dbd6730009996aa638b0
node-api/src/main/kotlin/net/corda/nodeapi/internal/serialization/kryo/SerializeAsTokenSerializer.kt	1526	86bf7baea69914183e3b5f4d8e6660fc33b2db4e97b42b9dcd959d
node-api/src/main/kotlin/net/corda/nodeapi/internal/telemetry/OpenTelemetryComponent.kt	18527	5579e101984ddffce2548bb6100a2559675b0164086e9e03c018eb
node-api/src/main/kotlin/net/corda/nodeapi/notary/NotaryQueries.kt	1130	5d351dc19a35d8ad1191f378de23b0c2474efb39132602caf553fc
node/build.gradle	17758	c54257552dec17c5d7b2f3a11a37ab9673fb347c5a1e2c510e82e2
node/capsule/build.gradle	4489	38a2e9261a5a9abb57a786a11d1104acffb35a6473d2f15c3613fe
node/capsule/src/main/java/CordaCaplet.java	12370	3bee0b8429b0bbf59503df66e9fb362c0dc09757d1c16977b78421
node/src/integration-test-slow/kotlin/net/corda/node/services/rpc/RandomFailingProxy.kt	3371	562e2228bffa121abc91ec7646ec4b56c992ff3b10157de7fe8ec1
node/src/integration-test-slow/kotlin/net/corda/testMessage/MessageState.kt	2611	ba433fd699f2c6f0bca6f25e8f4f336ead084b5db4b695f88ca1df
node/src/integration-test1/kotlin/net/corda/contracts/incompatible/version1/AttachmentContract.kt	1793	1109a8418c775bbc806916f58f1bec75eb863c0c88010bf2f4ff53
node/src/integration-test1/kotlin/net/corda/contracts/incompatible/version2/AttachmentContract.kt	1074	c917c025f7c4243389632c171036a3ce661c6d0727f09244a0a165
node/src/integration-test1/kotlin/net/corda/flows/incompatible/version1/AttachmentFlow.kt	4085	242bc206b440fa22c00e38939862a976050102efd5b43d3aa3e614
node/src/integration-test1/kotlin/net/corda/flows/incompatible/version2/AttachmentFlow.kt	2547	aa21cb6a715bbebf5a3a6ba0dc1955834eda331c67e927d2e83ae4
node/src/integration-test1/kotlin/net/corda/flows/incompatible/version3/AttachmentFlow.kt	2361	634537cff423221b0dab140f8a582a4469aea1d01fbb21a224f2a6
node/src/integration-test2/kotlin/net/corda/contracts/serialization/missing/CustomData.kt	545	296ecee4abdc91e4f4edc52b8f019bbad792b0b9dd7531a360849f
node/src/integration-test2/kotlin/net/corda/contracts/serialization/missing/MissingSerializerContract.kt	1196	5d43556e29b2fb817be4ddcfd1e4098f4c5a50623b06f80b1bb6a6
node/src/integration-test2/kotlin/net/corda/flows/serialization/missing/CustomDataSerializer.kt	476	96f20f6c1250080231677c76e430f7b65140fdf1ede6303ee55bc1
node/src/integration-test2/kotlin/net/corda/flows/serialization/missing/MissingSerializerBuilderFlow.kt	1120	526627293ee317ffcf69e0be8f6932704d328ede126ed784b30267
node/src/integration-test2/kotlin/net/corda/flows/serialization/missing/MissingSerializerFlow.kt	2655	9a7e4ac0217926a72eda070271b91af7c3d6b28de39c253720784c
node/src/integration-test2/kotlin/net/corda/node/flows/QueueWithCountdown.kt	763	640fdec4a87e4efaceec082dac14e8a53ebe715f26d43d366c948f

node/src/integration-test2/kotlin/net/corda/node/services/statemachine/OldFinalityFlow.kt	15942	02304beed282c59c9e5264822e4301ef81b50ef5fffb448b315ad4
node/src/main/kotlin/net/corda/node/Corda.kt	643	758186f320fce0520e4173b13aa069b55c197fcafcbc397a34a15c
node/src/main/kotlin/net/corda/node/CordaClock.kt	2992	2b2b5e4ee91c00abe4225dbee9070c592a2cdb13d061cc100491b4
node/src/main/kotlin/net/corda/node/JolokiaSlf4jAdapter.kt	825	92019764b2d2facccda543c00fb522ef50f04681c936cca325f384
node/src/main/kotlin/net/corda/node/NodeCmdLineOptions.kt	8724	8ac40acf46937ebd290d3982317eed224053a29df64076cffb78d8
node/src/main/kotlin/net/corda/node/SerialFilter.kt	952	d43a338868d8d53054d055cd5ea0d3c606786203d5d9195c090c6f
node/src/main/kotlin/net/corda/node/VersionInfo.kt	825	bfcbbc8c5285dce6a4f10005911fde23cd26c5729c37fabd150a3b
node/src/main/kotlin/net/corda/node/internal/AbstractNode.kt	74185	d9c0f68893f4bd3dbfae83565721d46d95b370ddcc9e08af3ca752
node/src/main/kotlin/net/corda/node/internal/AppServiceHubImpl.kt	6018	c60108442dfce903cfa783c716cbaad3c87af5a5ed512de71baca5
node/src/main/kotlin/net/corda/node/internal/CheckpointVerifier.kt	5811	21382c6fcc2e5dbb5536ee0bb9b1cee00a8b97e3b83c2e737c3485
node/src/main/kotlin/net/corda/node/internal/CordaRPCOpsImpl.kt	21950	d626fabfc1266ce1a91242250a87d0a14fc5139fca2bd778b6b584
node/src/main/kotlin/net/corda/node/internal/DBNetworkParametersStorage.kt	8242	7f5787f1e7630bf89224289849136c7ea20979364d9e9241c6b4c5
node/src/main/kotlin/net/corda/node/internal/DataSourceFactory.kt	1631	b81c8a868b065c54a777dae9eb0e26b8749adbb5fdf653296b9d29
node/src/main/kotlin/net/corda/node/internal/FlowManager.kt	12197	abc79e2ca85704948a09e53659709295ce21f382e171fd64f09aa0
node/src/main/kotlin/net/corda/node/internal/GeneralExceptionHandler.kt	1451	d4ba7cd30b2bbc6564fa4be2a1b415940b73f5651937fe663faf36
node/src/main/kotlin/net/corda/node/internal/InitiatedFlowFactory.kt	681	f385ad272719410a5766006cd0bc28c10f46d119b1f3238a1b3f6c
node/src/main/kotlin/net/corda/node/internal/KeyStoreHandler.kt	11930	9a6a37092d86f587efca601e2a37f2cbb8d499cc470a88a3f64873
node/src/main/kotlin/net/corda/node/internal/LifecycleSupport.kt	336	69e1de27ed0e4397bf8127c6cc129b80fb14da47f8f84d54420e1f
node/src/main/kotlin/net/corda/node/internal/NetworkParametersReader.kt	6220	04adf5cc95bb995595753c81007ee7fdc88bba371073685f7e6c02
node/src/main/kotlin/net/corda/node/internal/Node.kt	32226	ba175b35c7f2465323b78d198f0ec610c1423c1832ddb7141f95de
node/src/main/kotlin/net/corda/node/internal/NodeStartup.kt	32471	502b0194fce023463ab42f382e243b616430d070e3a8259118054e
node/src/main/kotlin/net/corda/node/internal/NodeUniqueIdProvider.kt	558	e32aad2790dbce84bb75b60eab3b2ffad176ad682d2990cf849734

node/src/main/kotlin/net/corda/node/internal/SwapIdentitiesHandler.kt	944	04089bf57039e18527eb9e5da40bcd480c628b73a664b124d995dc
node/src/main/kotlin/net/corda/node/internal/artemis/ArtemisBroker.kt	1084	2b2854d74e3f45cd3c034ef148196dd62022cd1c55a5571fbf1437
node/src/main/kotlin/net/corda/node/internal/artemis/BrokerJaasLoginModule.kt	11052	aad6072f7483ae774359f74e5ed3a5443543221934264c9a9128de
node/src/main/kotlin/net/corda/node/internal/artemis/CertificateChainCheckPolicy.kt	3836	161d03eff145a984b0fcadc0a6d39cfad178672b81b2fc457c7ec2
node/src/main/kotlin/net/corda/node/internal/artemis/ReactiveArtemisConsumer.kt	2947	7bcfb20abae996f112798d7a4ff2f936d21a908485c7373b51df2b
node/src/main/kotlin/net/corda/node/internal/artemis/SecureArtemisConfiguration.kt	790	1b0950ee55de4b7c1abc2a69e88b18b6961f52f6fde26fab44170b
node/src/main/kotlin/net/corda/node/internal/artemis/UserValidationPlugin.kt	2628	39d628598ea1c33ad26b7d22aa609cf97c407c9f544dc6cfe4ef98
node/src/main/kotlin/net/corda/node/internal/attachments/AttachmentTrustInfoRPCOpsImpl.kt	621	298d44e937a447514fcd2c634bc16e4605c940e9870d2a7f7b9e2f
node/src/main/kotlin/net/corda/node/internal/checkpoints/FlowManagerRPCOpsImpl.kt	718	e30e243798ce6189aaa0e8f3c7dc71d227b9e063d548cce17309c7
node/src/main/kotlin/net/corda/node/internal/classloading/Utils.kt	267	7df923204ede285a7bbba3ff14519fa2dfad3c5301c007a2464cb2
node/src/main/kotlin/net/corda/node/internal/cordapp/CordappConfigFileProvider.kt	1294	b4c0742449b3d92ac51dbf48cf14c7e5cab0d675a358ca49f62a74
node/src/main/kotlin/net/corda/node/internal/cordapp/CordappConfigProvider.kt	157	350f2d967f00aa184fb032dfdf3c02e883d7947e94ebfdfa9692e9
node/src/main/kotlin/net/corda/node/internal/cordapp/CordappProviderImpl.kt	6640	10ebf1bfa2f1e3b875f7c83db24883a06480c086ffbb2233ce0786
node/src/main/kotlin/net/corda/node/internal/cordapp/JarScanningCordappLoader.kt	26082	e67e32f3e8479c591f4776f7b6e17343c3ea24194461cbec9c7ed3
node/src/main/kotlin/net/corda/node/internal/cordapp/TypesafeCordappConfig.kt	2681	8460c32868f206dc233bedafea27870808ab0a26051178511bdfe4
node/src/main/kotlin/net/corda/node/internal/cordapp/VirtualCordapps.kt	6238	db08616c4661863ad475eb66ea621caaf477fb50f1b5f9acf47ec2
node/src/main/kotlin/net/corda/node/internal/rpc/proxies/AuthenticatedRpcOpsProxy.kt	3754	848553cfb8c4f97804a31588fbccade4817ec571223d8e75cdcd12
node/src/main/kotlin/net/corda/node/internal/rpc/proxies/ThreadContextAdjustingRpcOpsProxy.kt	1665	53bb15002ceee2f8c35bbeb751d413d01e5509a32cc53260ab9a4e
node/src/main/kotlin/net/corda/node/internal/schemas/NodeInfoSchema.kt	4922	260ee87986c7d8b427f165832926715978e326a205f7760f75cc6a
node/src/main/kotlin/net/corda/node/internal/security/AuthorizingSubject.kt	769	e1d2cce3b6942643d5d4b70316ee40549ef7c58dbb556c529d6153

node/src/main/kotlin/net/corda/node/internal/security/Password.kt	1020	3fb68284765f34767008b2ff8bb3bd23d0c2d8c9522ecb0b6c575f
node/src/main/kotlin/net/corda/node/internal/security/RPCPermissionResolver.kt	8795	5297c8ec682da3bedcef93c307fd5052898c177f6e5d4ee1fb11c1
node/src/main/kotlin/net/corda/node/internal/security/RPCSecurityManager.kt	1454	bc9750ec9f8996d0a9bca36d9fdba696cab856753807aa6041a53c
node/src/main/kotlin/net/corda/node/internal/security/RPCSecurityManagerImpl.kt	10138	f4c4efb0615a623612925b3681b3f2c74e00f3c556bcce24d6561a
node/src/main/kotlin/net/corda/node/internal/security/RPCSecurityManagerWithAdditionalUser.kt	1498	894fd0db7572b50f26a2edf4b705e1d1d6f24ab572c9598c7914bc
node/src/main/kotlin/net/corda/node/internal/shell/InteractiveShell.kt	3486	081c2b26f6fdbab259d9e82d48fcd80f02d25f7d38cb85b00bdc6d
node/src/main/kotlin/net/corda/node/internal/subcommands/ClearNetworkCacheCli.kt	647	001ac08234697524f42699d88290cc46a9782c05a67518c33e1e34
node/src/main/kotlin/net/corda/node/internal/subcommands/GenerateNodeInfoCli.kt	683	99ed11386fbdbe0dfc890340369145c0319a19ef6655d24da8c079
node/src/main/kotlin/net/corda/node/internal/subcommands/GenerateRpcSslCertsCli.kt	4660	8203a3fa565bf52f129091842bf82ecc0fb5f7d74476a5e690fc19
node/src/main/kotlin/net/corda/node/internal/subcommands/InitialRegistrationCli.kt	6479	9eeb42398e38c0ac9d0c67b3b7ee0cb5f1c31b29c3477827c498cb
node/src/main/kotlin/net/corda/node/internal/subcommands/RunMigrationScriptsCli.kt	1383	f58037f92999ae15cdb389be27bf315b6dd8c3168239b42d35b783
node/src/main/kotlin/net/corda/node/internal/subcommands/SynchroniseSchemasCli.kt	636	a0020520e753277edfc5340f45bc0dd6b1eb04da76fe710b48fcb5
node/src/main/kotlin/net/corda/node/internal/subcommands/ValidateConfigurationCli.kt	2153	7cb1fc2cbede30815401c623b4e631cd4f27bb1bee6ea40c222594
node/src/main/kotlin/net/corda/node/migration/CordaMigration.kt	4717	3b1fa2b1cc905865488d7bb887a417e8dd733d9c8d5e3fd727e6d2
node/src/main/kotlin/net/corda/node/migration/MigrationNamedCacheFactory.kt	3544	872014d05c5a56183ed1e809d457e19a7fb57948deb3989fe9f638
node/src/main/kotlin/net/corda/node/migration/PersistentIdentityMigration.kt	3250	40a5112c4c3fe81f227c9159a0f7e3af24f7ff54bbef204840ca28
node/src/main/kotlin/net/corda/node/migration/PersistentIdentityMigrationNewTable.kt	3862	32e46c329ddf6e25ae8730cb97706f4e56c53b1e8b8466db9df189
node/src/main/kotlin/net/corda/node/migration/VaultStateMigration.kt	5388	4a1672502461da70acd5bfc8b60bb3c8df0b1163ab24d47c2f1ec2
node/src/main/kotlin/net/corda/node/services/CoreFlowHandlers.kt	4167	fc05e10f5ebf037473224281ee455f569ad9ae75460a2684d39071
node/src/main/kotlin/net/corda/node/services/DbTransactionsResolver.kt	9024	a8d0b77a29e857598625a415c5f15066210c8936f959cea67d8070

node/src/main/kotlin/net/corda/node/services/EncryptionService.kt	2110	9b51ceb01e3e703ebe041d465318e20ae6fcbec7ab1fe417ad7a82
node/src/main/kotlin/net/corda/node/services/Permissions.kt	1653	a96f91fbd6927c31fc8809f69cb51650a699684ee80efaca544daf
node/src/main/kotlin/net/corda/node/services/api/AuditService.kt	5940	07066b4379667fdc1dd4d4fd4b4ef25b5467103c55a5dd2a7a25c1
node/src/main/kotlin/net/corda/node/services/api/CheckpointStorage.kt	4728	f70f6b59a0211fbeb0293fca3df41abb2c739a41ffb7ab8a0905ba
node/src/main/kotlin/net/corda/node/services/api/IdentityServiceInternal.kt	812	a80b949cfe8d974575cb7428252635344d41c2361c9a94d174e6de
node/src/main/kotlin/net/corda/node/services/api/MonitoringService.kt	426	faa109ec060c12319441f6b7b0cc750cc2abfc1690b3af1a7ab519
node/src/main/kotlin/net/corda/node/services/api/NodePropertiesStore.kt	364	10ab003984bdd5b9c71e0fc5aa0851c21567de7c960063947903a6
node/src/main/kotlin/net/corda/node/services/api/SchedulerService.kt	1077	d349e00e8873755d31a2df3382587f236b3be655d37aabaeb5e262
node/src/main/kotlin/net/corda/node/services/api/SchemaService.kt	932	a401036b84771192b400121ec15b0c1e29ecdab7878595d102811e
node/src/main/kotlin/net/corda/node/services/api/ServiceHubInternal.kt	21729	6a26678278dd92c5247655501adca7f00fbfa81c9351c1533f3931
node/src/main/kotlin/net/corda/node/services/api/VaultServiceInternal.kt	1243	154c25dc9cef883c077dc09c042485c6d4085d777311e02813251a
node/src/main/kotlin/net/corda/node/services/attachments/NodeAttachmentTrustCalculator.kt	7696	9f9052eb8895461a9d722df02b6d65d5964a0a719c99b5bfbb0c5d
node/src/main/kotlin/net/corda/node/services/config/ConfigUtilities.kt	11480	3bc7b613c0e92b98b336d694b7c277d160580e72a964661a8be491
node/src/main/kotlin/net/corda/node/services/config/NodeConfiguration.kt	16440	d6aedc308f107b84bd76f0ae91f776762f85ed52f8c572a4eb2645
node/src/main/kotlin/net/corda/node/services/config/NodeConfigurationImpl.kt	17225	4e05df8474117837e574f07374199299b10f89011141e6a7579984
node/src/main/kotlin/net/corda/node/services/config/ShadowingException.kt	298	29698951edfcba81b99e38bafaa3a2dcef6f774b81b672070603fa
node/src/main/kotlin/net/corda/node/services/config/rpc/NodeRpcOptions.kt	344	a112b615daabb1afc2c65fc521c90b06cbc295ad93eb337362a7de
node/src/main/kotlin/net/corda/node/services/config/schema/parsers/StandardConfigValueParsers.kt	2707	7d8f2328674bad3de48a2342f72d31dda11d8bdc256f60fe476db3
node/src/main/kotlin/net/corda/node/services/config/schema/v1/ConfigSections.kt	22692	4a6c3438289e7e3558f2b64ad676a2d7cafac00bb4f88c0b12c8bb
node/src/main/kotlin/net/corda/node/services/config/shell/SSHDConfiguration.kt	946	621599732602a74c42333d0663d85988d87fc1da9ccc96f24c6328

node/src/main/kotlin/net/corda/node/services/config/shell/ShellConfig.kt	1117	1d340863fb34d4c6389932e97b1d8c979db7d0878b41d4922814a2
node/src/main/kotlin/net/corda/node/services/config/shell/ShellSafetyConfig.kt	1688	63e53084a3697c3ac36346dd94dd35b992b6683562ea878a396793
node/src/main/kotlin/net/corda/node/services/diagnostics/NodeDiagnosticsService.kt	690	58b3f1786f972c2fa9b299565d4fb1d149ac15feaebcf47098ff28
node/src/main/kotlin/net/corda/node/services/events/NodeSchedulerService.kt	20072	cf6e403cd7f6c662fb1a75740b89ecbe79828a51dfdd3867289749
node/src/main/kotlin/net/corda/node/services/events/PersistentScheduledFlowRepository.kt	3036	eaaa3c0273f21c62174201caa46a018575f7b7cb1f4862d64a54c2
node/src/main/kotlin/net/corda/node/services/events/ScheduledActivityObserver.kt	1848	f27fae9ee14cfbdb32e6b6f83f76924b195c58461994d080ecfa7a
node/src/main/kotlin/net/corda/node/services/identity/InMemoryIdentityService.kt	7473	1f64b67baf796928d61d4f31e9cfe660041a5c0f5920777d9fe990
node/src/main/kotlin/net/corda/node/services/identity/PersistentIdentityService.kt	25532	93050a70515cc22eb0569f13da19eec725d61728a4047576ab8e3f
node/src/main/kotlin/net/corda/node/services/keys/BasicHSMKeyManagementService.kt	7963	f480066e249e3e360221908cfae7f673f960919bf8b4cf4a69808f
node/src/main/kotlin/net/corda/node/services/keys/KMSUtils.kt	2915	1aa3e56c2bd30f56bc697d2bd13dee439bd9ffd4349fb59b77527f
node/src/main/kotlin/net/corda/node/services/keys/KeyManagementServiceInternal.kt	1444	e3cc8c18cbb3daab9f0fbe68b1f8d95b5600d361511ddfe709c86e
node/src/main/kotlin/net/corda/node/services/logging/ContextualLoggingUtils.kt	1426	acc9fc5c4994fe33e81b50791c4f90f14a9af180f0b27e84eed64d
node/src/main/kotlin/net/corda/node/services/messaging/ArtemisMessagingServer.kt	17479	556893a36a5ae7c505dc892b195fd96b09a9b6c2c8bd56122886c1
node/src/main/kotlin/net/corda/node/services/messaging/InterceptingActiveMQJAASSecurityManager.kt	8624	124a106c61f7db2477afe59bf5eaf34f46a70f12f7e71d0b635e8f
node/src/main/kotlin/net/corda/node/services/messaging/Messaging.kt	9389	9297a9c53845269eacddddcda211ff71d3097f24fea51a3cdaa2ec
node/src/main/kotlin/net/corda/node/services/messaging/MessagingExecutor.kt	4583	6669f02d044116e86b0191fbbd502a1efdf7cc6351431ead30fc42
node/src/main/kotlin/net/corda/node/services/messaging/NodeNettyAcceptorFactory.kt	9733	99a59ba76120ab19011b0bc8eb206b19d0c0b4d21686af17408545
node/src/main/kotlin/net/corda/node/services/messaging/NodeSSLContextFactory.kt	4406	308c7562f8dcdbe93df6a15c1b0e96bf65b51ce1115279b8e292fa
node/src/main/kotlin/net/corda/node/services/messaging/P2PMessageDeduplicator.kt	5248	1f8cb5b85bb91a7a8be1df09fe577e1ccbe0f83f80c96d1851bbd5
node/src/main/kotlin/net/corda/node/services/messaging/P2PMessagingClient.kt	36779	f32cf2cfa045ef7546a56fec29be71b1ae959ae6cf76fe0b450766

node/src/main/kotlin/net/corda/node/services/network/NetworkMapClient.kt	5591	67b0bb980e0ea560276402fdbfb759763340ca74370b973c2fb964
node/src/main/kotlin/net/corda/node/services/network/NetworkMapUpdater.kt	22802	4a75d3e52b5b96564c76ded010046c4e3754a6d1c3991c26d627c7
node/src/main/kotlin/net/corda/node/services/network/NetworkParameterUpdateListener.kt	401	1ca9641808c568f24605c349f366d23c48e021c588fc0d29f3dca1
node/src/main/kotlin/net/corda/node/services/network/NetworkParametersHotloader.kt	3986	db2315c51d168df359fe86220414e30c25a11d4ea03fe8b75288cb
node/src/main/kotlin/net/corda/node/services/network/NodeInfoWatcher.kt	5883	22b04f6dcee161a043dda210d5e4109e3125399b9e50d944bdb3c1
node/src/main/kotlin/net/corda/node/services/network/NotaryUpdateListener.kt	377	680ca6862aef5a70df1020ad58526cc7edf09adc64c949b8e2f2cc
node/src/main/kotlin/net/corda/node/services/network/PersistentNetworkMapCache.kt	21538	db4393810e16a2b6c62fb52903d34171087c15f614c33eeb5649d4
node/src/main/kotlin/net/corda/node/services/network/PersistentPartyInfoCache.kt	4141	cb9bdfb7533c264ca98b093654202cb0905946bb8ce1ddf4f602d7
node/src/main/kotlin/net/corda/node/services/persistence/AbstractPartyDescriptor.kt	2676	a7dbe1c30c1cf15e85c6cc0cdefee0fae1b27094cf9b468d298714
node/src/main/kotlin/net/corda/node/services/persistence/AbstractPartyToX500NameAsStringConverter.kt	1644	64bb9ae0acc1b77acfd9bd5dca071931cfcf2f4eea2ecce273b93e
node/src/main/kotlin/net/corda/node/services/persistence/AesDbEncryptionService.kt	6684	90aed52472192bddf0664081d593bf3c4fd6f6ff31da29ca2e613b
node/src/main/kotlin/net/corda/node/services/persistence/AttachmentStorageInternal.kt	2038	0a726048790ac9197ab1b41762c48300ac143c632dec50a8ae6046
node/src/main/kotlin/net/corda/node/services/persistence/DBCheckpointPerformanceRecorder.kt	2960	b814af024bf41d2ff78ddfcec128209c6205ca12d8a0016e29cfbb
node/src/main/kotlin/net/corda/node/services/persistence/DBCheckpointStorage.kt	33554	776d72234969bad1b61fa9b97b54c1bac88193fa78be67b90219af
node/src/main/kotlin/net/corda/node/services/persistence/DBTransactionMappingStorage.kt	2363	c83c7a76ea15f382e331d04dc4784bbe433a770d180cec4cf157d4
node/src/main/kotlin/net/corda/node/services/persistence/DBTransactionStorage.kt	19887	4a2af71876421e9334c4950abfbc8f41a1ede4f49d93b2222346e7
node/src/main/kotlin/net/corda/node/services/persistence/DBTransactionStorageLedgerRecovery.kt	10482	d358e9e4b678617a3287bc2a6c2d951dc7fe3d1dd3bf5f11d1e6f0
node/src/main/kotlin/net/corda/node/services/persistence/HashedDistributionList.kt	5546	0bc868cc72d972c82fac23e5c182c1ccddde4b2fc5740dc35adc6d
node/src/main/kotlin/net/corda/node/services/persistence/NodeAttachmentService.kt	28418	3e3f34212183496e615fa8870208f4d2c99d7095fc93c4cd08d913
node/src/main/kotlin/net/corda/node/services/persistence/NodePropertiesPersistentStore.kt	2779	42261881e25224b4ecb4a5bd94182aaa946206856feb40573ae1a7

node/src/main/kotlin/net/corda/node/services/persistence/PublicKeyHashToExternalId.kt	703	937667e8aa2e8ee441593ad91368d62e697740a3270be65900e0e8
node/src/main/kotlin/net/corda/node/services/persistence/PublicKeyToOwningIdentityCacheImpl.kt	3036	20cc2401d140c6294315d78f1990640c9954d91d3b7d8268c93050
node/src/main/kotlin/net/corda/node/services/persistence/PublicKeyToTextConverter.kt	766	183686e68852febda6bb7080b1df049fcaaab923815c2ce672f5ce
node/src/main/kotlin/net/corda/node/services/persistence/WritablePublicKeyToOwningIdentityCache.kt	543	224f405fe2ffb4e5380c05a0107f964efc0113a5d64c21b8e5fe36
node/src/main/kotlin/net/corda/node/services/rpc/ArtemisRpcBroker.kt	6509	1633268f64e2498521c2d7ddd66f854230e278044c8b3fd320d380
node/src/main/kotlin/net/corda/node/services/rpc/CheckpointDumperImpl.kt	28035	b942a91430a39b1007db1121cf8ef456952086848ec075c05c39e9
node/src/main/kotlin/net/corda/node/services/rpc/InternalRPCMessagingClient.kt	2391	cc85a7b412626acf2f6e73caec0f89c71f4d8e9897bc865eecc45c
node/src/main/kotlin/net/corda/node/services/rpc/RPCServer.kt	30494	4055f84c3fc4663690592ba1f6b9c00968065db5051ab5cf944f58
node/src/main/kotlin/net/corda/node/services/rpc/RolesAdderOnLogin.kt	1527	7473eaa699f00d0be9c558002bc06ca781d0294c116eddc29d055e
node/src/main/kotlin/net/corda/node/services/rpc/RpcAuthContext.kt	316	e4e6311ee687fb4294b71b35fe54c81cd99a9f050d21e6586c9e00
node/src/main/kotlin/net/corda/node/services/rpc/RpcBrokerConfiguration.kt	7804	42649b5d01d81c295bd41c70365bf98d3515199dc428411bb5b9c9
node/src/main/kotlin/net/corda/node/services/schema/NodeSchemaService.kt	6118	9796b14235d5d53d56591ec7f5d9bfe8f9b5c607ef0c0f889065e9
node/src/main/kotlin/net/corda/node/services/schema/PersistentStateService.kt	2177	2944c51cfd71808b25ea072cc9c6608e2d0f21b7fef68cc705add6
node/src/main/kotlin/net/corda/node/services/statemachine/Action.kt	7328	4f1a6ca31531ff838448de192a5bd0a2414998d3a3fcfe0f0eebee
node/src/main/kotlin/net/corda/node/services/statemachine/ActionExecutor.kt	354	cd37bea725ad285d6ad54e0d9c3df05fa1e03763c0ca7b16463675
node/src/main/kotlin/net/corda/node/services/statemachine/ActionExecutorImpl.kt	11467	5b380a07bc2ae446eaebdd142d16844b10fed42668fe929e20494a
node/src/main/kotlin/net/corda/node/services/statemachine/ActionFutureExecutor.kt	4239	438fdb80423d7e306812605f72f6a87b5f161115d9a9b4e6398ca6
node/src/main/kotlin/net/corda/node/services/statemachine/CountUpDownLatch.kt	1681	598ff0422d6fcd04a82fce16e579d631f0184463a02c3bebae4970
node/src/main/kotlin/net/corda/node/services/statemachine/DeduplicationId.kt	2803	8a95b9ac4ce01a8848fc880f62cba8651235ddaee12cdea6d21fa3
node/src/main/kotlin/net/corda/node/services/statemachine/Event.kt	7634	eea7ad53b5b641da1e699a3c98c6dffeb20dd98f299629f79c06bc

node/src/main/kotlin/net/corda/node/services/statemachine/FiberUtils.kt	1125	0bc9140b4b4d82bfa351f6c4c8c08bc998162e815b3ebbd9eb5299
node/src/main/kotlin/net/corda/node/services/statemachine/FlowCreator.kt	15241	a3b83623f187f550e72acd14623e17dccbc90835689b0b6b980941
node/src/main/kotlin/net/corda/node/services/statemachine/FlowDefaultUncaughtExceptionHandler.kt	3938	515c931a1e4f3c5bff032e072989a077fd20bd14395a2a823e253c
node/src/main/kotlin/net/corda/node/services/statemachine/FlowFiber.kt	490	57a1e433d4d39b006495a80a80e5e9f050d83c93208548d8030701
node/src/main/kotlin/net/corda/node/services/statemachine/FlowLogicRefFactoryImpl.kt	12040	fc5206cdad20863b617451140edcb3fd0ea7a952be3ec04118f356
node/src/main/kotlin/net/corda/node/services/statemachine/FlowMessaging.kt	5886	0fbed827ad29d0d87c3e3e034fea8ad91c0eb6b27b4ed9b978bc24
node/src/main/kotlin/net/corda/node/services/statemachine/FlowMonitor.kt	4294	8c7b5796e2c39edacdd9ce33c939ddead023c31a74d14f30a0766a
node/src/main/kotlin/net/corda/node/services/statemachine/FlowOperator.kt	8020	c186f8a80495f75fff9aadb30c0feede4718986b456e746fc26486
node/src/main/kotlin/net/corda/node/services/statemachine/FlowSessionImpl.kt	4602	68ee9124e4e0f50ddc2479d483cae154f94300ab07c195578c1378
node/src/main/kotlin/net/corda/node/services/statemachine/FlowStackSnapshotFactory.kt	1483	46376afafc7de910b3d9d595a5b9f1877b127d470f93d1aa4488f9
node/src/main/kotlin/net/corda/node/services/statemachine/FlowStateMachineImpl.kt	30776	2c82d52b07330db5f03624396330c5572f4233769ed6c81c5de101
node/src/main/kotlin/net/corda/node/services/statemachine/FlowStateMachineUtils.kt	626	9c00ec83b04413630f71e5dcd658452715d698fcb8bff558cb6e9b
node/src/main/kotlin/net/corda/node/services/statemachine/FlowTimeoutException.kt	316	6f8619621b98e6e9fe8ad3fda5d4ec091df1312f43deeac7c7cea4
node/src/main/kotlin/net/corda/node/services/statemachine/FlowTimeoutScheduler.kt	4366	e4af2c7b7cb3ae2b3cb0cf52cb974b5125a60dfaa68c1ca8acc60c
node/src/main/kotlin/net/corda/node/services/statemachine/SessionMessage.kt	5199	be050cc9a63ffb557f38df5d77dfcdeb5b1425d480b0420c5dbce7
node/src/main/kotlin/net/corda/node/services/statemachine/SessionRejectException.kt	669	09c49bfbcf2c386dca69549bf14700561114e329d6f11759ffda33
node/src/main/kotlin/net/corda/node/services/statemachine/SingleThreadedStateMachineManager.kt	59755	6602c5b3bc6f45b7ac8085c0bbc7ca3801138f3188c2ecf9fe5bd8
node/src/main/kotlin/net/corda/node/services/statemachine/StaffedFlowHospital.kt	35665	761be75b44403c6b7e0e6161c50970240ae9612c74aae22bc8cb93
node/src/main/kotlin/net/corda/node/services/statemachine/StateMachineInnerState.kt	2185	f991a877a7c12fc6c18c2247576d55c6b394d9c5ec3db049552a9a
node/src/main/kotlin/net/corda/node/services/statemachine/StateMachineInstanceId.kt	175	cfcd98a40a8733edb00c3c676f49f6eac56b91da0c6d8e5b4f9850

node/src/main/kotlin/net/corda/node/services/statemachine/StateMachineManager.kt	7408	59e233a9209ac10d9b0c3fb23192ad30ae112ad68230777e0668e2
node/src/main/kotlin/net/corda/node/services/statemachine/StateMachineShutdownLogger.kt	2508	4d3f4d18b358eff68f22612bddcce1b2c50580f5e4138d73e42e0f
node/src/main/kotlin/net/corda/node/services/statemachine/StateMachineState.kt	20257	be2eaa0d379f8fe824bc0d867235ff6c1ef564128a118fbf346bfe
node/src/main/kotlin/net/corda/node/services/statemachine/StateTransitionExceptions.kt	1063	6e83fe7b1531671bb973d917f71c3082575609753ef800249f8955
node/src/main/kotlin/net/corda/node/services/statemachine/SubFlow.kt	3456	fbcde6f9e9feeb81abbf64447e51629dbfee064d5958514d2e6525
node/src/main/kotlin/net/corda/node/services/statemachine/TransitionExecutor.kt	928	5cad8b39ca6077ce136ccd757a924750eae5a7540867099bf39c4f
node/src/main/kotlin/net/corda/node/services/statemachine/TransitionExecutorImpl.kt	6271	1f35d37671bcc6f91e2985a17aaef9aee7b1c05f67f57007fdf77a
node/src/main/kotlin/net/corda/node/services/statemachine/interceptors/DumpHistoryOnErrorInterceptor.kt	2870	783fb5c4efc5ff2f2942dfd19573351ac5cf43ed9e53f11b5031cf
node/src/main/kotlin/net/corda/node/services/statemachine/interceptors/HospitalisingInterceptor.kt	2563	dbae7e5cccb9ae9e009b2f931a19fc480f8c6d14c708df6ccca8bd
node/src/main/kotlin/net/corda/node/services/statemachine/interceptors/MetricInterceptor.kt	1579	ed4834fa35267fbee688ce8106a534d5a84bbe2334a773077b4733
node/src/main/kotlin/net/corda/node/services/statemachine/interceptors/PrintingInterceptor.kt	1534	6d42da89046e40fdf85aedf0dda8294fb8c7b021abf4b348765069
node/src/main/kotlin/net/corda/node/services/statemachine/interceptors/TransitionDiagnosticRecord.kt	2113	bcb221e5ffcde4eb5dd3a73f448c293cc5e137e2ce1bc16f9fd5e7
node/src/main/kotlin/net/corda/node/services/statemachine/transitions/DeliverSessionMessageTransition.kt	10404	d5b2ddb56fdaedf694b847a5aea37fc882a5c2d6692fea25da68d5
node/src/main/kotlin/net/corda/node/services/statemachine/transitions/DoRemainingWorkTransition.kt	1907	b54f52ddaeaccc84b0e90838e68135575260b19fe1a47932576f20
node/src/main/kotlin/net/corda/node/services/statemachine/transitions/ErrorFlowTransition.kt	8266	76d62cea7e7ed91143714b8f5eacb557bf2833945335dd9e1a6e7e
node/src/main/kotlin/net/corda/node/services/statemachine/transitions/KilledFlowTransition.kt	6448	f2a396c71e65ae834a0da4d02ac45a837f5a3e6b158bdcb3a682ed
node/src/main/kotlin/net/corda/node/services/statemachine/transitions/StartedFlowTransition.kt	29602	de19453e70a8fc579257216be7f6c06f11f795e6f5c8db6301a270
node/src/main/kotlin/net/corda/node/services/statemachine/transitions/StateMachine.kt	534	d3b5afc79c12426a8dfd31eed572d793ef2f80ddff8b9904ddfe2d
node/src/main/kotlin/net/corda/node/services/statemachine/transitions/TopLevelTransition.kt	19305	b7e34c31abea581ed7ca4ab3f530bf7335e844ace247c4bb98be46

node/src/main/kotlin/net/corda/node/services/statemachine/transitions/Transition.kt	1942	2fedb72c10c71b45a291eaa9a1d0ef22e1da4960ce96a89e529c00
node/src/main/kotlin/net/corda/node/services/statemachine/transitions/TransitionBuilder.kt	3784	6303fcb54e779f6a91034e22ed391b5720b1a8aebaed6358bd543b
node/src/main/kotlin/net/corda/node/services/statemachine/transitions/TransitionResult.kt	1693	d462ba91dab4d0a643965d4453f4b6e141f9a338afd2e8a962f855
node/src/main/kotlin/net/corda/node/services/statemachine/transitions/UnstartedFlowTransition.kt	4340	159bf1d8fa59df2def1bd86b04f32e6aa479bcb3cbc88c0fb25630
node/src/main/kotlin/net/corda/node/services/transactions/NonValidatingNotaryFlow.kt	6022	d87dd38fdcb1f39f54739be3e1f9ac9e2f9a1bfead11167683678c
node/src/main/kotlin/net/corda/node/services/transactions/PathManager.kt	1520	b3038272bf8b2886d4c902ef5e48dffaeda22e2a11402acf550423
node/src/main/kotlin/net/corda/node/services/transactions/PersistentUniquenessProvider.kt	14475	5fdb98f1972300205e22484cadf233e3189c1f586a7c7875c2bc9d
node/src/main/kotlin/net/corda/node/services/transactions/ValidatingNotaryFlow.kt	3051	6561feb132dab94c54257c69f82c25ef71382199027316ea2770ec
node/src/main/kotlin/net/corda/node/services/transactions/ValidatingNotaryService.kt	0	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca49599
node/src/main/kotlin/net/corda/node/services/upgrade/ContractUpgradeServiceImpl.kt	2531	1e9d17163728c685c79c0f2852975d2f2b0b5420f1ba21ecbbfffb
node/src/main/kotlin/net/corda/node/services/vault/HibernateQueryCriteriaParser.kt	48506	7e7d44f4e81053d8fe32b8de9dddc61d31daecc2ba2404b880edc1
node/src/main/kotlin/net/corda/node/services/vault/NodeVaultService.kt	52990	3752203b93a1ced2358562b1f1702cfc64676765bc22181814983f
node/src/main/kotlin/net/corda/node/services/vault/VaultSchema.kt	8664	6a8b23860e6e0f7dd5bae1cc8c4ccf0735bdfcf764476ada00f867
node/src/main/kotlin/net/corda/node/utilities/AddressUtils.kt	1069	7c0425977b1d38a62c64b5c78aacc4b5d18c875b3545629ff49f28
node/src/main/kotlin/net/corda/node/utilities/AffinityExecutor.kt	3081	5a6c111c3239cfe37485ffb546c58995124938bddf101cdd497ae3
node/src/main/kotlin/net/corda/node/utilities/AppendOnlyPersistentMap.kt	24433	5bb1ac9c8afa615a5b9a5d663917a02d560ea3118b2190f0b8330a
node/src/main/kotlin/net/corda/node/utilities/CertificatesUtils.kt	1801	586ec0ce49d4cdb05db45d0fc5b708e717280ce7532a5292dd4b18
node/src/main/kotlin/net/corda/node/utilities/DemoClock.kt	752	e6b60b1aaf931b6a881b1f4b374e9bf5761b19a9015ca4d6f5fc63
node/src/main/kotlin/net/corda/node/utilities/InfrequentlyMutatedCache.kt	4682	5d8bccba30a65832700feb1c42c5565e03d23cce01f07b1254bde4
node/src/main/kotlin/net/corda/node/utilities/NamedThreadFactory.kt	852	a5f6d94194a4816fabef4e64af8fa9d8ca58b4668b17a012009f11

node/src/main/kotlin/net/corda/node/utilities/NodeBuildProperties.kt	630	3d46f2bb3419bb9ab91a01862f29d79ac90d9cf1d327e321ae6884
node/src/main/kotlin/net/corda/node/utilities/NodeNamedCache.kt	6277	859d2acbd3b898f04aaaf94aba810ece1e582e488a1e44e0ac258e
node/src/main/kotlin/net/corda/node/utilities/NonInvalidatingCache.kt	2092	cb0b30ff573b48b8beeb2b34a896ab0300c155fcb764b1006f9c48
node/src/main/kotlin/net/corda/node/utilities/NonInvalidatingUnboundCache.kt	1997	ceb16f720c846140d419079af17e97e10745763690b2bf806e4d31
node/src/main/kotlin/net/corda/node/utilities/NotaryLoader.kt	5830	86d394f5e18ca4d1d42b8d236cd5d42ed96876c19065d437c14bfe
node/src/main/kotlin/net/corda/node/utilities/ObjectDiffer.kt	5303	d0f82472ba59ef453fe00af2a15772d182efe530a4d6b5ddce2567
node/src/main/kotlin/net/corda/node/utilities/PersistentMap.kt	8961	3dd5f9770b25a5d8157a01ad158bd92864d9babd3d0817b0e64343
node/src/main/kotlin/net/corda/node/utilities/TimedFlowUtils.kt	405	76324cdc8376732dcae3b49f7f41a5db1d8f3f0b1e012155088a91
node/src/main/kotlin/net/corda/node/utilities/artemis/ArtemisStartupUtil.kt	648	53b4bc1df2f5081fd45645780c6fdc1def90ca785d2ea677d659a6
node/src/main/kotlin/net/corda/node/utilities/errorAndTerminate.kt	680	8096067be733bdbc8b7af75c71f0b2af9decc8a29c8a153c16fd9c
node/src/main/kotlin/net/corda/node/utilities/logging/AsyncLoggerContextSelectorNoThreadLocal.kt	1801	d5d4f0f657868e3423bc111b228774db9104d14840382dcecd3f70
node/src/main/kotlin/net/corda/node/utilities/registration/HTTPNetworkRegistrationService.kt	3568	c3cab1b88698dc3ff7aa894ff2195b28e50204f5bbf29aabc842f6
node/src/main/kotlin/net/corda/node/utilities/registration/NetworkRegistrationHelper.kt	27738	79c6c65d63f659002a0013ea64193585a6249c9e38df25b7b8cfa3
node/src/main/kotlin/net/corda/node/utilities/registration/NetworkRegistrationService.kt	838	63d464ea912e67ee367e0ab0ba8260bfdf18a4c4a43e058f784626
node/src/main/kotlin/net/corda/node/verification/ExternalVerifierHandleImpl.kt	13306	974705796c8b38860fe2c51b42c6a8d710b40742f01dcfac87a108
node/src/main/kotlin/net/corda/node/verification/NoDbAccessVerifier.kt	337	6d7d7b482ffe2d1e7cceab76f9954b08f2aa38935c4daf1193e59d
node/src/main/kotlin/net/corda/notary/common/BatchSigning.kt	2518	57febf76b1582c3bd605bbbdc0774b48d739f686f2a073b141e50a
node/src/main/kotlin/net/corda/notary/common/ValidationModeNotaryService.kt	1252	ba38d0e63bbbe1ffa40dc86ffe29a594b6107245c349f0c76ed707
node/src/main/kotlin/net/corda/notary/experimental/bftsmart/BFTSmart.kt	18631	330801014065201e806df08577657d8f8ef52f6612390621a124ee
node/src/main/kotlin/net/corda/notary/experimental/bftsmart/BFTSmartConfigInternal.kt	4792	4781c7c8e7e33b3ea58841b855920316270ff9e5f1b9293bccfc51

node/src/main/kotlin/net/corda/notary/experimental/bftsmart/BFTSmartNotaryService.kt	10146	377d8e801151497b1831d1654c4d9ea7693916dc036bd6f13cd93e
node/src/main/kotlin/net/corda/notary/experimental/bftsmart/Schema.kt	736	08f80a1dd9e9e23aeb980cfe8bd2fbeba80a07ecceaca2b3c82b3f
node/src/main/kotlin/net/corda/notary/experimental/raft/RaftConfig.kt	819	5ca02deccbe338147ec382cff12d9265f38734642160078a7bbfdb
node/src/main/kotlin/net/corda/notary/experimental/raft/RaftNotaryService.kt	1972	1455b3b6bd11aecb6815fdfa6f9b8e2fadf81b81920fc5da9168f7
node/src/main/kotlin/net/corda/notary/experimental/raft/RaftTransactionCommitLog.kt	12482	790e4e3fb6d68804700efd303c6ae8268cda41535aafd7897d423b
node/src/main/kotlin/net/corda/notary/experimental/raft/RaftUniquenessProvider.kt	10753	b2577f2a943457ddaa5abb021cb1498a9037d682414de959311fbe
node/src/main/kotlin/net/corda/notary/experimental/raft/Schema.kt	717	b067756e780d7f4d0385208af66dc01a6895cc30514b338e1a569c
node/src/main/kotlin/net/corda/notary/jpa/JPANotaryConfiguration.kt	249	14e2775da73b460de7a7ba118edabbe23501e54ad2da43f95f6387
node/src/main/kotlin/net/corda/notary/jpa/JPANotaryService.kt	2132	14d1d47d5b238d36a30ee2d9449a105e84ccf89d52a3785f4f6dd4
node/src/main/kotlin/net/corda/notary/jpa/JPAUniquenessProvider.kt	17042	c147e77fc6ac94c841d7a8e5c86847964cce56c96e96ab24d6206e
node/src/main/kotlin/net/corda/notary/jpa/Schema.kt	543	cf95295e1d128c7d3fc764fa4db708f73ffeef6bb2e53c9b163ed6
opentelemetry/build.gradle	732	68e91fb2e30d63378904538981a569e337841d476702a0b9d1e6a0
opentelemetry/opentelemetry-driver/build.gradle	611	c467527d979ebf32512c84a5a14f9ae6dd215b03a92c4e29113730
opentelemetry/src/main/kotlin/net/corda/opentelemetrydriver/OpenTelemetryDriver.kt	2241	8610e0a2026d049b4ec7d6525170a297a8c629ce3850f7cd86433b
release-tools/jiraReleaseChecker.py	8918	35459025b5b507de898a921fe8621f46415259356808e4c202ec98
serialization-1.2/build.gradle	1065	58b2c477db3d515197c3f634447e41fd9ca1de8433abc6976b85c3
serialization/build.gradle	2455	df547345417df26d4f379ab01b2112253f021093edacb5ad53dacd
serialization/src/main/java/net/corda/serialization/internal/amqp/custom/CacheKey.java	757	e0b800a8dfa718f3a3ad224e8378b2b082ee38cbb43fbc4c4d54f0
serialization/src/main/kotlin/net/corda/serialization/internal/AllButBlacklisted.kt	5808	3cd3f237dd5b11e9e10498fdba14b510383764016e32a10c3356b1
serialization/src/main/kotlin/net/corda/serialization/internal/ByteBufferStreams.kt	2084	60fbef07f726b5410674e781d3157b8e26c5cc689889bcd944d806
serialization/src/main/kotlin/net/corda/serialization/internal/CheckpointSerializationScheme.kt	2438	c4d738a3e9bbbfeddcc3b2dab5baca350f68a282c5346cc62d5ec5
serialization/src/main/kotlin/net/corda/serialization/internal/ClassWhitelists.kt	1897	8967b1df63aab3fad7c21ff063c36bfbafedc7c8e8b44e8490363e
serialization/src/main/kotlin/net/corda/serialization/internal/ClientContexts.kt	743	5fe7255401dd6c65dceed0c33d2f55f12b579b0165ffc1e1f987de

serialization/src/main/kotlin/net/corda/serialization/internal/DefaultWhitelist.kt	3049	8116ee750842f20cc13b56fade3891f12d3a2100730f1201c80c60
serialization/src/main/kotlin/net/corda/serialization/internal/GeneratedAttachment.kt	279	d9978ac18a602c707d827e59df1ea9028dca996455ff32c18fbef0
serialization/src/main/kotlin/net/corda/serialization/internal/NotSerializableExceptions.kt	621	d864c869c62045130978b14698716adb1422efc30c9f544fe69a20
serialization/src/main/kotlin/net/corda/serialization/internal/OrdinalIO.kt	1083	d7552984b0fab8c38743d742d72a10c9284cc8a842240d40c3e11f
serialization/src/main/kotlin/net/corda/serialization/internal/SerializationFormat.kt	2812	b7be47ee525f87ab13a03666fc822cea9318d57087379ec84482b3
serialization/src/main/kotlin/net/corda/serialization/internal/SerializationScheme.kt	8418	a82e218de280c408f38468a3bed13a1c308213ec432161065af9f3
serialization/src/main/kotlin/net/corda/serialization/internal/SerializationUtils.kt	279	a1411a20e4a9f44813407dc106449f2f892cbc9adac49204cc1241
serialization/src/main/kotlin/net/corda/serialization/internal/SerializeAsTokenContextImpl.kt	5574	7880be21f9288731cf551e2f9a5163aa39c82c6ae57e155abca014
serialization/src/main/kotlin/net/corda/serialization/internal/ServerContexts.kt	1184	7e116bfdbbdaf18cd56f5c52218e39b107f07598bc379fabd7dc5a
serialization/src/main/kotlin/net/corda/serialization/internal/SharedContexts.kt	824	0f78fdd6cd0a31773a99095d2756b3a5e651e05f066e3adc12af6c
serialization/src/main/kotlin/net/corda/serialization/internal/UseCaseAwareness.kt	998	dfa3b8e94fdaafddc7b40dd8dd3a691d5dc1cfae11601272259f13
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/AMQPDescriptorRegistry.kt	972	e81a199ca9715e614dbedd9a1e801c2e9e7cbe51a0727a58991cbf
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/AMQPExceptions.kt	3020	2e9e94fd82606694f636a291cdf404ce769d8f2103689436db306f
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/AMQPPrimitiveSerializer.kt	1326	c6fb49d509877848f29097d0b396bc5f427068eb9840e7cbc9e2d3
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/AMQPRemoteTypeModel.kt	10702	879ff97e3db31511149539edcff09267f9ec2ae3a10a11df4e72bf
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/AMQPSerializationScheme.kt	11033	5ba2cf6c111a055d6e4cbfd61eec5230acafe0b6f64d1ba989449a
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/AMQPSerializationThreadContext.kt	193	599371ee597bc91bfa32175190200314ac9aaa28ca965cd83df757
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/AMQPSerializer.kt	1522	a2b3f6caff2dbbb28258cbe8953a40ed4e522c993e21f648620758
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/AMQPSerializerFactories.kt	945	afb95cdf47eb3a03312f355da176c0b2362abf7b369d9d4a891156
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/AMQPStreams.kt	1013	d5a37a6f160e696901dade205556a49f25592387ffeb47dc0c2530

serialization/src/main/kotlin/net/corda/serialization/internal/amqp/AMQPTypeIdentifierParser.kt	8321	1f129e37d57f21f368baffd7ab4926e3bf222085a012a7f8e8a9b6
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/AMQPTypeIdentifiers.kt	3119	d122bfd9c09a26e875cea6a4d29ec1d5415504c7726a9e2c11ae76
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/AccessOrderLinkedHashMap.kt	340	e338687c88143fb82ed46c6d16c5803b1f22af35286375a9cbbebf
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/ArraySerializer.kt	10064	4789ff4950d919e29993a810eba1381196837b17ffcc866d2b69b6
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/CollectionSerializer.kt	6322	fa479a7b47c905537516b3473eb7524a54762f44732595328a9891
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/ComposableTypePropertySerializer.kt	11147	56262e3bb5211f7e50c7308e8a763d7be2ddc26a3e15be00f898a8
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/CorDappCustomSerializer.kt	4296	2e396005835d915b30337a6451abad636e0b13cfcabf27f2644068
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/CustomSerializer.kt	10863	5ae8d64e0ad2b04ec420bb413e6b4d4e40a109c1169c3087eddabe
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/CustomSerializerRegistry.kt	8875	f69082124c86461568370bf4a530f5574c1518d638aa9d7fc3a6e3
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/DescriptorBasedSerializerRegistry.kt	1205	377648177745cd5272a83632a0fe7d697609fcd4e60806e2e15414
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/DeserializationInput.kt	13164	7adbdfe2c74c8e251ff5e13a6d9371b7368d0895ac00b6f9296cb7
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/EnumEvolutionSerializer.kt	3382	8267875b2f60606a5f93dee7ff3d57817467b81e8f60236d786ae8
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/EnumSerializer.kt	2257	dea19d12986422e34a429ace934b718dea57dd48f24faa54efd390
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/Envelope.kt	3619	1fc24b41ab546a97452b018e85f97368bcdbe3ca248d68fe013b4e
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/EvolutionSerializerFactory.kt	10498	0719ddd7e21ace6f158b0b411074b6ce9339976b5ceafb738fc921
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/InterningSerializer.kt	1180	fa08039f510ce5e40d1e4b92df2d3506950c3e02a86d3523b01f8b
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/LocalSerializerFactory.kt	14571	8ec4bde4f8826c37e39d336d4822a782c6902ee04738958bbed762
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/MapSerializer.kt	8412	202a58e6bf746f3c3e18a72f0b17ed3e1e9b2df3fcea57ebb94e8d
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/ObjectBuilder.kt	13003	a23b4fb9fc5fd0073e2f2c638cc5ebab84ca14fa037f3f29ea606a
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/ObjectSerializer.kt	12368	e5aedc57dccde5aa73248a9c4e63ab36dec0c1cce851df3d5a3a67

serialization/src/main/kotlin/net/corda/serialization/internal/amqp/OutputStreamWritableBuffer.kt	2300	0d63a6aec0b68c1dc4af8ce5ac1a8973d8ce0299c11c9034702090
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/PropertyDescriptor.kt	10941	21d4a279ec246caba38dafee9ea8a20069e98af250c676e5dc07de
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/RemoteSerializerFactory.kt	7887	a1412ab3d09813165dc177777c4126602580361fb31f32b01ab1b8
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/Schema.kt	16267	f55b2c55fa0f96a446e4709595ba8b5b497d7ea57915ae0fdd4846
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/SerializationHelper.kt	5073	64e0a2d7e7587d0de1a727565f41168c995bb17841c6c84ba1a30b
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/SerializationOutput.kt	7096	7ac8c080cd30e4df97ef8ecdc82e26cd14aa96640b80b6b36e9284
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/SerializerFactory.kt	1919	7cf8dff341f378618f1162c760841fd3bab25684f7f39283ea92cd
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/SerializerFactoryBuilder.kt	6012	d48fbdab9308ca2a821e2df4bd30e31153fac881adc2f87600f9b9
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/SingletonSerializer.kt	1682	88b39d50055b403d117d2f8ef8f72c98ef4387401760f21fa61b9d
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/SupportedTransforms.kt	3148	e855b09bf4591cb96f6719745e8e1af4d23d994c430a04da0c8338
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/TransformTypes.kt	3292	acaf16567c22195367a515e9c503a969eb7058e203c2aa5d4c10f4
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/TransformsSchema.kt	14464	761fb42fe5ce5513cca246e028056f29d072e3255c71ae81aee77c
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/TypeNotationGenerator.kt	3112	a0a5921afd5d6d00689d4f642113c99bc4987430cc9830104f2be8
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/TypeParameterUtils.kt	3856	2adf2b2bd35520f7bdfd90da611e52e37451370171621ae3dc6408
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/WhitelistBasedTypeModelConfiguration.kt	2504	1572a2ec7277a6844e66b290ff2afddc94e2fd7ca919d6c07b30a6
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/BigDecimalSerializer.kt	516	899211eece8fc790d9862227ed3f8a5608487cc3763714c42142ee
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/BigIntegerSerializer.kt	516	3044d121a1a5915e0e71f7b8b7c4a3b4236b53dd78a551f7f6488b
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/BitSetSerializer.kt	795	d5d4cbbbb4dadd5f0dc0bd16c3e1452ba4005d2608b7d67cd7e81b
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/CertPathSerializer.kt	1973	86024ca9097a8a0a74e40db0a1218357f8f1874852b0fb70e96540
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/ClassSerializer.kt	1789	407592d2413d4d50d6561cf491cab6a5a9b58ed9d1afa5166dd5e6

serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/ContractAttachmentSerializer.kt	1999	577e2bfd556f97e6fe95cdae55fbc86aa77fe209662f06d1156303
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/CurrencySerializer.kt	462	3f1c26ccbe64abe1d7f463ddce76be090acbd4b53fbd41511aa965
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/DurationSerializer.kt	806	89d29f0143378d12582b06f4634ed223ee4a50f9a11a8b46415ebb
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/EnumSetSerializer.kt	1565	aca295e5c665161fc1c4a8d305d3d62ef133bc1b60a8cdde27730f
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/InputStreamSerializer.kt	2117	7d250ad2bf81abbb083fca9d4f087900915f13ed88303cceb70547
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/InstantSerializer.kt	825	ba479a45ff3656ff218d5d210fd1f00dd09e06007ab4d461ec81aa
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/LocalDateSerializer.kt	876	a4cb0e2342943434bbb42ee9c7d09b96886823040af54ebc535f81
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/LocalDateTimeSerializer.kt	1105	450a73e328861b8a15891128de5243a692508f325464c253cbddeb
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/LocalTimeSerializer.kt	1058	3623299391d8938bf7ebcdb50b319005c33d9555465189f81c1721
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/MonthDaySerializer.kt	802	e3e044ffbc2b39f0cad70742d8c6776a9c1ef296df80ef9dc25919
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/OffsetDateTimeSerializer.kt	1148	9f28bfe246bd17b6b67c91b3e1cb5462e6bbc58d3cea26cf0bdcdf
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/OffsetTimeSerializer.kt	1064	d982780d9ee0a49f35d3d0e7bee88175ab220512bbbbcda8a1b4f3
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/OpaqueBytesSubSequenceSerializer.kt	1056	4880b149626e1bae007c9d136bdaa8d6afd8b37287e2bd655ffa1d
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/OptionalSerializer.kt	855	9690141756238b52f6300351217f332ba46caa782807dde0464602
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/PeriodSerializer.kt	786	25d1c8919e92b44987329e771925db7309d2fb135d1744018070bc
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/PrivateKeySerializer.kt	1426	9164cd51a5c6d625a22cf7149087482d66d5722c885185a9001a8d
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/PublicKeySerializer.kt	2183	f3dfed15a5cd994dbf4034d8e884715c4cef33f7563d8999e31744
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/RxNotificationSerializer.kt	1003	e4eac76e99ef7b1de6f8bac86f2507da69a58af2457c09adc89d12
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/SignableDataSerializer.kt	4353	92d4e133f381c0b475660cfcd9017df77886caa3c96d2be008baf5
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/SignatureMetadataSerializer.kt	4882	763e1ae6c1f90ff86ac7170a0bfcb124ca770ef529e6ae4c12a211

serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/SimpleStringSerializer.kt	315	cec066f4c59699e6ab2341b8efb83137f7fae30f998052c01d5450
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/StringBufferSerializer.kt	258	d929a93d022923c8b9a9438cbdc7d98007ab6c12f8159420b3fee1
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/ThrowableSerializer.kt	6057	4030e1910e26d8b3306c661ca81f1f8a7f331fa54ebbd81d4db621
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/X509CRLSerializer.kt	1711	58a421ee34bb5ed21616fefbe876be0e7fd3a91803a2cc267aaf37
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/X509CertificateSerializer.kt	1810	5fa9b9437a733df737d33264a2c56036eecec4feeeea4a6ede23b4
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/YearMonthSerializer.kt	810	0e4c2efb85d7ccf0a2475ab0e7e5a21ef3f2ce5f3c807a1de01c4a
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/YearSerializer.kt	673	af157ee7213600d687d167125f171cc65058afd11dd97cf2500c95
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/ZoneIdSerializer.kt	781	2f7902d0db6bad3b464c47f7325dc6d403ae8ea92707ec2631c269
serialization/src/main/kotlin/net/corda/serialization/internal/amqp/custom/ZonedDateTimeSerializer.kt	1233	099ee2d4276d8b13256b27c33182f26f574173b7c14e7130073a71
serialization/src/main/kotlin/net/corda/serialization/internal/carpenter/ClassCarpenter.kt	21712	7cc06b7e6d666ccf5f3683d6fcf1a61fb43b7cc584616c47e98a26
serialization/src/main/kotlin/net/corda/serialization/internal/carpenter/Exceptions.kt	1007	912b4427f59a62312bf6f643aad17a965ce904b535f0a436eda3d3
serialization/src/main/kotlin/net/corda/serialization/internal/carpenter/Schema.kt	4006	e00755dacd6c213ca04a42cc061a40a4987a531fd3c47c610bfe0f
serialization/src/main/kotlin/net/corda/serialization/internal/carpenter/SchemaFields.kt	4178	28a3154005cb6339eb49ad7077950595d3b9f4b5161d01cbc7e6d1
serialization/src/main/kotlin/net/corda/serialization/internal/model/CarpentryDependencyGraph.kt	6014	a5af57a0cb7a82399eff8b18a014691ae786ff4a6ed5cf5ff0d073
serialization/src/main/kotlin/net/corda/serialization/internal/model/EnumTransforms.kt	7257	98629737f9d30b2a6f2426c8d1c6ea363701791910d44a42c7d969
serialization/src/main/kotlin/net/corda/serialization/internal/model/LocalPropertyInformation.kt	3824	2242ee5eed897fdcdad317d8f3b47e1c64fb661da0a0479712ce6a
serialization/src/main/kotlin/net/corda/serialization/internal/model/LocalTypeInformation.kt	20331	df7352c572795cb4e044864f6ce4705fa490f072d148476c208b44
serialization/src/main/kotlin/net/corda/serialization/internal/model/LocalTypeInformationBuilder.kt	29647	60c66121361c346909e6da5c82daf30159cb862f1a3f662ee44c84
serialization/src/main/kotlin/net/corda/serialization/internal/model/LocalTypeModel.kt	6087	a73f195397987978c0a222f46ed130c23fb8fe44bd35fc04421310
serialization/src/main/kotlin/net/corda/serialization/internal/model/RemoteTypeCarpenter.kt	4140	f4a56c7670d3f7e11106bc364655b2b890ed25bab9d41a71e67b59

serialization/src/main/kotlin/net/corda/serialization/internal/model/RemoteTypeInformation.kt	8846	c09ff537d68a30f581f0ecf71991198daaa1af182ab42911461ef8
serialization/src/main/kotlin/net/corda/serialization/internal/model/TypeIdentifier.kt	12194	28ba81953dd1bc0f0b120cdf88fb0bfd227a5305ee1e328fc80cb3
serialization/src/main/kotlin/net/corda/serialization/internal/model/TypeLoader.kt	2830	479f5a4faceed3a2b7211f47db97cd43b50e1943b62d3ac95e4fab
serialization/src/main/kotlin/net/corda/serialization/internal/model/TypeModellingFingerPrinter.kt	12096	249da54057150da1671d33b5a21a456d1fc961a5a1a1895cb27854
serialization/src/main/kotlin/net/corda/serialization/internal/verifier/CustomSerializationSchemeAdapter.kt	3423	f3dc58114b470ef2de25ac51e63cdb761f2e579792aa86aa2b93be
serialization/src/main/kotlin/net/corda/serialization/internal/verifier/ExternalVerifierTypes.kt	5184	01d99e6498f2c5a49e06141444a4ba60c02ce4793079286619fbcb
settings.gradle	4598	9047eca9db9dc20decf28314375215181926ccf75b0d34865cbca8
tools/blobinspector/build.gradle	1915	31f57918506eaf8d24314e8c8e0c85e1c6de97a73caf8c7cd5afd8
tools/blobinspector/src/main/kotlin/net/corda/blobinspector/BlobInspector.kt	6931	d952853a04e5544278d1d84b501999fa2ef9b74a26399802137d69
tools/bootstrapper/build.gradle	1948	b03165050c3aa541b463b0860d4d1aa60057ce34ab16176065efe5
tools/bootstrapper/src/main/kotlin/net/corda/bootstrapper/Main.kt	6186	8597dea0e49485d6d642a924905651e476e93523cfc5c217246c44
tools/checkpoint-agent/build.gradle	1255	c48a25011dcaff2d043d2cb0b93d21e2bcb5d71a8452da13a1ac58
tools/checkpoint-agent/src/main/kotlin/net/corda/tools/CheckpointAgent.kt	24488	be60233d22e768d6d883a277362f9b2278262e49781f19ba6d15e6
tools/cliutils/build.gradle	885	fb60ce9364ee6b8ad64e011ae2cb16939ccf50d8303002ca8a44c1
tools/cliutils/src/main/kotlin/net/corda/cliutils/CordaCliWrapper.kt	8909	8000c357a95bbed83e351d02f3dbca5149fc33264fa5473782bb96
tools/cliutils/src/main/kotlin/net/corda/cliutils/CordaVersionProvider.kt	459	fd321136e6c756dacd8d7a1f4d988c144bf8007f373d62a0311da6
tools/cliutils/src/main/kotlin/net/corda/cliutils/ExitCodes.kt	151	a37afac14231eb65542ac8ea2fb2b66c5a48cc043de2c361a38d64
tools/cliutils/src/main/kotlin/net/corda/cliutils/InstallShellExtensionsParser.kt	8235	1dd963163b7b0a0882c9c84e53edb03aac88dd40c72999c050046a
tools/demobench/build.gradle	8189	835e997749f81f05220384460657557b7006ce957c62cd251c8c5a
tools/demobench/package/bugfixes/java/sun/swing/JLightweightFrame.java	17866	dd0579ef47e2f9124ed46bae473f637a22bad3f62c2a674965fedd
tools/demobench/src/main/java/net/corda/demobench/config/LoggingConfig.java	1174	c8f717536a14b68c3c6c4d66e690700a880106bc2e4b7bba0f0df8
tools/demobench/src/main/kotlin/net/corda/demobench/DemoBench.kt	2999	e6dfef4e4a8328b03207671832b0f3f5daf3faa0d2d78f4ecb9420
tools/demobench/src/main/kotlin/net/corda/demobench/explorer/Explorer.kt	5130	8a2f0a098585f941860753c134f1665e124b0decefbe5d7a9f44ca

tools/demobench/src/main/kotlin/net/corda/demobench/model/DemoBenchNodeInfoFilesCopier.kt	1453	ffb1de39afb6af03e7142cf7ee3dd511b84e48ae2d614c79af411a
tools/demobench/src/main/kotlin/net/corda/demobench/model/HasCordapps.kt	114	f81d0ab42e0fa4309fabb507fc8100b961ffd79e0ebd2862e84672
tools/demobench/src/main/kotlin/net/corda/demobench/model/InstallFactory.kt	1538	65ae86e0669d2ec412a26c58311bc0ab8c2b6a9c75c7f4f0a21039
tools/demobench/src/main/kotlin/net/corda/demobench/model/JVMConfig.kt	1752	d531d1fe4a0369d7e5868db30a33ebb7de0cc63940ec8951d8245b
tools/demobench/src/main/kotlin/net/corda/demobench/model/NodeConfig.kt	5518	ea085b186d3266139bef28a03cded44068972e3b3e62dc26858975
tools/demobench/src/main/kotlin/net/corda/demobench/model/NodeController.kt	9859	443236bb62a1957fc023bd070c0819ca7a4fcbf2c618ef6d6e21a8
tools/demobench/src/main/kotlin/net/corda/demobench/model/NodeData.kt	2523	878897108fa7e07bb240f67764108cc93ac94a623dec0975b42366
tools/demobench/src/main/kotlin/net/corda/demobench/model/NodeRpcSettings.kt	202	6ba905e85a5a6d29a69b9d6c92e80bde8642bbbe578b9192abecd3
tools/demobench/src/main/kotlin/net/corda/demobench/model/NodeState.kt	96	16339648061b1660ccce7719e150009c11775951d97b107afd9868
tools/demobench/src/main/kotlin/net/corda/demobench/plugin/CordappController.kt	2133	f96015bb45e7d79045c3aee37de70b3c46c6ef0e42c1b1dc0532cc
tools/demobench/src/main/kotlin/net/corda/demobench/profile/ProfileController.kt	5673	b04192ec781cf0ce966ee439c54e3a2919646d23fef334fab7733d
tools/demobench/src/main/kotlin/net/corda/demobench/pty/PtyProcessTtyConnector.kt	905	60896e00a9018584b70a0b4183d2b65b4baf781b2986df1d4d430c
tools/demobench/src/main/kotlin/net/corda/demobench/pty/R3Pty.kt	2643	445cfc3f987c8db5d6e9f623e00b9ce1f2b7c638c8c499dfe2f1b7
tools/demobench/src/main/kotlin/net/corda/demobench/pty/ZeroFilter.kt	1808	e57973372b0858d9e0dfaa97c6c48d829dcc222ec0ac02109dbfbe
tools/demobench/src/main/kotlin/net/corda/demobench/rpc/NodeRPC.kt	2787	55334fca47cf92428ba1d9bf1b40dda1707a1030370a35ba37e66b
tools/demobench/src/main/kotlin/net/corda/demobench/ui/CloseableTab.kt	734	92fe838e691e2794e4ef1b4f7ee5301ee20c88e74d3e9a55ca5fb0
tools/demobench/src/main/kotlin/net/corda/demobench/ui/PropertyLabel.kt	995	4cdd8a587e42e516feef1df79a4207413a0af662649007035b9e9c
tools/demobench/src/main/kotlin/net/corda/demobench/views/DemoBenchView.kt	3676	24da8cf9a34ae6885b1822c19d8a0a9322e8e90f660bb0b2676f01
tools/demobench/src/main/kotlin/net/corda/demobench/views/NodeTabView.kt	12738	80acd200e6dcab171e9963578282a5a39addd08f2dc69e9b04700d
tools/demobench/src/main/kotlin/net/corda/demobench/views/NodeTerminalView.kt	10721	15b1e14f8487bb095abcf4a0545e13b0a6be2db678526f0a0a5db5

tools/demobench/src/main/kotlin/net/corda/demobench/web/DBViewer.kt	1352	fd941284341b4b55417ef046f17985b288dd043b8c47c3f4a675a3
tools/demobench/src/main/kotlin/net/corda/demobench/web/WebServer.kt	3647	6bf04f5716184f72b2344a7f6bccd4ace48d8a65fdb0de349a22d3
tools/demobench/src/main/kotlin/net/corda/demobench/web/WebServerController.kt	488	5d55315222ad1819bbaac87b36e4109c84fc2c5037d14128b66350
tools/demobench/src/main/kotlin/org/h2/server/web/LocalWebServer.kt	562	8ece0c03be42abda58321dd8b0c24166fcf5ac110161e423d8e4af
tools/error-tool/build.gradle	740	04e8d96a9cca1f069e25aa0151e2b2c7ed5bf52b4217612d7aea89
tools/error-tool/src/main/kotlin/net/corda/errorUtilities/ErrorResourceUtilities.kt	1538	9b00877ef7006401ed65b177cd787ffdd89dc6158bac6a022d0f71
tools/error-tool/src/main/kotlin/net/corda/errorUtilities/ErrorTool.kt	972	d716fac08403155d27ccc3a4a420965db09538547366a252613dd5
tools/error-tool/src/main/kotlin/net/corda/errorUtilities/ErrorToolCLIUtilities.kt	792	5cfbfcc5fe12594e3698209de32b927f00875f6601bef8066cc633
tools/error-tool/src/main/kotlin/net/corda/errorUtilities/ErrorToolExceptions.kt	399	8b13d0fdfb8d19bfaead6d5836eaf42b45f5245b936885070a93e1
tools/error-tool/src/main/kotlin/net/corda/errorUtilities/docsTable/DocsTableCLI.kt	2846	84395eb53b2e17610f7f77c38e8c501928707aed71c62c90c1d6f2
tools/error-tool/src/main/kotlin/net/corda/errorUtilities/docsTable/DocsTableGenerator.kt	2310	3be415f1db0637c206253178d11e6024210bbd27c1011f43b70f32
tools/error-tool/src/main/kotlin/net/corda/errorUtilities/resourceGenerator/ResourceGenerator.kt	3257	132eb54b573154921f1659e12c89aabfacf3553a175f20c72d0fda
tools/error-tool/src/main/kotlin/net/corda/errorUtilities/resourceGenerator/ResourceGeneratorCLI.kt	2997	3facfcb7b5bbd03d6afe2bd10f6ebd1c6dbab175ed129f4362b1da
tools/explorer/build.gradle	3123	6d72f6c1008a4c7aee771200f38ff24ea1d09fc041fad36272cb6d
tools/explorer/capsule/build.gradle	1408	0d4e44f75654d2c2f8613c052ac67c5d45c09b242000b6e35f05e1
tools/explorer/src/main/java/ExplorerCaplet.java	1630	a6a48306c6a99fe8aa128f6b051d9bb51769618096e34038703ab3
tools/explorer/src/main/kotlin/net/corda/explorer/AmountDiff.kt	658	69788d2cc583e6093747acb60c271392c362da22cefa21310111e8
tools/explorer/src/main/kotlin/net/corda/explorer/Main.kt	4308	9eb2f4c6f40e3f32595e24a83b6b9dfe5d22f9a4485ff9cb3e291e
tools/explorer/src/main/kotlin/net/corda/explorer/formatters/AmountFormatter.kt	614	31bf4388ea0ccdc82a10126b660aa8a89c7c859664520ed85b0c24
tools/explorer/src/main/kotlin/net/corda/explorer/formatters/Formatter.kt	102	e7d45f361a4bf7b85ffa795050949aa97e7ef6899950b9c1b144f9
tools/explorer/src/main/kotlin/net/corda/explorer/formatters/NumberFormatter.kt	285	4c9df6efe132aa3adbd36ca23f79131b3f446d612301705b6b2490
tools/explorer/src/main/kotlin/net/corda/explorer/formatters/PartyNameFormatter.kt	379	1579102a15721a1abcd2ebaf3a31c9e43e954ac8a5cc1fbca48d13

tools/explorer/src/main/kotlin/net/corda/explorer/identicon/IdenticonRenderer.kt	10051	d29d274e601fb7bbeddf8294adf7b3ac67039f43daf8726a3c9bbd
tools/explorer/src/main/kotlin/net/corda/explorer/model/CordaViewModel.kt	1415	cec7f6a7ea45a2a67db3f584871493271623ee06983cf7797781b3
tools/explorer/src/main/kotlin/net/corda/explorer/model/IssuerModel.kt	1337	718200d0b59292e4e35da0957a97d7899fa159a718c0cbfa7b2312
tools/explorer/src/main/kotlin/net/corda/explorer/model/ReportingCurrencyModel.kt	1106	d27cba362cd898a3adc3871ade77a79eefbe96d41f9833cfa1ac52
tools/explorer/src/main/kotlin/net/corda/explorer/model/SettingsModel.kt	3838	e846fb35ec053d80a1111359944393fbf18dd7d10e4f39c2549c43
tools/explorer/src/main/kotlin/net/corda/explorer/model/TransactionTypes.kt	215	1852a840e7febe65a130d0c6b029044128fcf14a8fe5f5c7a762bc
tools/explorer/src/main/kotlin/net/corda/explorer/ui/AdvancedExceptionDialog.kt	2654	2bdf98b5999cf90e18033275944b3ea9d06ca2470a9047c52da966
tools/explorer/src/main/kotlin/net/corda/explorer/ui/ListViewUtilities.kt	1057	246dac96423ed8d1556d6b6ec7ea602a994e0288c376d86b0cd744
tools/explorer/src/main/kotlin/net/corda/explorer/ui/SingleRowSelection.kt	187	98c2295488d24499503b03a90f2e3ac986aa4eb8fe82e69ef24db6
tools/explorer/src/main/kotlin/net/corda/explorer/ui/TableViewUtilities.kt	2705	913a6e61935fd6f2d058c34d4e48cde7775255b1b169ef8a371097
tools/explorer/src/main/kotlin/net/corda/explorer/ui/TreeTableViewUtilities.kt	2257	08119b93e064c68313f7b1e7f5647107b1431ca31bc1068fd799c7
tools/explorer/src/main/kotlin/net/corda/explorer/views/Dashboard.kt	2225	5bf2f5e1f83a277a46067d3059b04c168aedbeaf54bf7133ff2834
tools/explorer/src/main/kotlin/net/corda/explorer/views/Formatter.kt	1297	c8598750fa17609bd6a2a1b15a2104896cb7bf7f20c85cdc3da94d
tools/explorer/src/main/kotlin/net/corda/explorer/views/GuiUtilities.kt	3415	037c68f73df17536c35335723488f60b0fb587d2c4fea504f3a2b2
tools/explorer/src/main/kotlin/net/corda/explorer/views/LoginView.kt	4393	1163d0cc9cacb80f19e8bc3a1f527f1f86cbe6eae4cddc62d3fe20
tools/explorer/src/main/kotlin/net/corda/explorer/views/MainView.kt	4739	0585d9970fb96f8ecd69ecb1e86a8db77ec6f6735d3dcdace4cabf
tools/explorer/src/main/kotlin/net/corda/explorer/views/Network.kt	13505	e9ba468c5a019bbbc0e4ce51c78562339dc95f925f65c57c4fca53
tools/explorer/src/main/kotlin/net/corda/explorer/views/SearchField.kt	2993	3fd62d6430becd89ef24b35a3524b72a2c6015b5a7a978d2e28ab1
tools/explorer/src/main/kotlin/net/corda/explorer/views/Settings.kt	2646	2cef758baec2974d5e7794ac7ae48b7cebc55ba7b4f5df2bb3eb0f
tools/explorer/src/main/kotlin/net/corda/explorer/views/TransactionViewer.kt	16916	c02c64f6a6448283574daf0e9082d810d216ede9a61e9c944a41bf

tools/explorer/src/main/kotlin/net/corda/explorer/views/cordapps/cash/CashViewer.kt	16959	9e913b67f3c1a2f3733ae189d3d6e3632371328743d41aa543031b
tools/explorer/src/main/kotlin/net/corda/explorer/views/cordapps/cash/NewTransaction.kt	13014	14c41f989d7c94cc337fc0935806a2d7ff9fe3c4099eb67df9c501
tools/graphs/build.gradle	2827	5f9d08c75e488975e5578d34fe8bc5a829d7464085e6b5e5e5b355
tools/loadtest/build.gradle	1926	8b63dcce116c01ed0b44db5666c827981a46c69b0bfe4f688ad465
tools/loadtest/src/main/kotlin/net/corda/loadtest/ConnectionManager.kt	4577	8b9fa42086b269e6801b94981f18fb0d51809bbd608025fa427e89
tools/loadtest/src/main/kotlin/net/corda/loadtest/Disruption.kt	4026	1147650af34cbdb88ecc525d98db96af84abada94069297e82c4cd
tools/loadtest/src/main/kotlin/net/corda/loadtest/LoadTestConfiguration.kt	2242	9ff6500515c3ebfd0c84830e5229edf66df30e0813e0bd149909d5
tools/loadtest/src/main/kotlin/net/corda/loadtest/Main.kt	8484	04e731567c954c6b9d40aad806996daccae95554fbec2843a94b1c
tools/loadtest/src/main/kotlin/net/corda/loadtest/NodeConnection.kt	6801	77ccdaf1e82d5f353c808a579eb84fd47fcc27f4d65542da2348a4
tools/network-builder/build.gradle	2927	f2de20da5009d06e2862fd60ca5f4fa3324fd1adc0d801064eb36d
tools/network-builder/src/main/java/net/corda/networkbuilder/GuiUtils.java	2146	53982b64ba329e99b953c8cb18d0c07c250fee2d1c2c486d56ff86
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/Constants.kt	2100	bc176d0dbf78c9e69898c1aa47266b673b6932cd2a296e428c1cf7
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/Main.kt	1763	044328a051f296366b8f6480b6a6fa5e2e01853079eb0ca41404b1
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/NetworkBuilder.kt	9759	e98220ec9e5d190853797be5bf2958fbaa2c42d3b583f28d861c45
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/backends/AzureBackend.kt	3624	a5fd37d14b6ef2f887897e6d41f2014be4c75638a5c2b4e187ed03
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/backends/Backend.kt	1307	727f3faa18dd99a09363271396388202c80b74767ed32ca55f6b4d
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/backends/DockerBackend.kt	938	51d2d9e9551259ab64fee307b83af01e9498f6ac685a50f17bc55b
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/cli/CommandLineInterface.kt	3216	896937a94c6987a3db4a52817d287d69fbc3a2f767e721a7d2704c
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/cli/CommandParsers.kt	2084	1f964c4e244d0f3fe3823644cc86c23b28cc84851113b56362827b
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/containers/instance/InstanceInfo.kt	324	a5e15e832c528003158e545b298193ae3a1fd6d316dde30f99948f
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/containers/instance/Instantiator.kt	555	0871641d42016d2f6da4ead99a8a20b2ba2688bff3f68c74a6d2bf

tools/network-builder/src/main/kotlin/net/corda/networkbuilder/containers/instance/azure/AzureInstantiator.kt	4566	e29f0e95f19646641e1486b0c9fd9ce8679204e4abd4789847dd76
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/containers/instance/docker/DockerInstantiator.kt	4513	ac0e6efe54515ab1dc83082386210203d44334c2a3563d44e8ff17
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/containers/push/ContainerPusher.kt	338	7ff17e7f09fbb0c6c5eb4e9b27e1d3fbbbbca0d50ceb14eb03e3c9
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/containers/push/azure/AzureContainerPusher.kt	2517	281f1ac5a6294a0f1806dc3bdbb9a8a44f9262c0e154baf32a8712
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/containers/push/azure/AzureRegistryLocator.kt	2304	abdce410bead1d9987dc9f5229f9d3e1dc39b92bbc722e50be3d57
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/containers/push/docker/DockerContainerPusher.kt	664	6dde69fbd90b0fb26f217b43c88a7449a95c69fcb11bb31f20e672
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/context/Context.kt	2448	999f26dff4c07eb1e497ee822f57c6ee524f72fcf9c15f1ca5ebc0
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/docker/DockerUtils.kt	1422	3eb413227344150e86ca6f74f1c40e718f5d404faf53c2e69cd4e5
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/gui/BootstrapperView.kt	15943	fabe4ba38a37e1eec28ffe90d20adf3d16e97ab25243f024ef1dc2
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/gui/Gui.kt	264	a98835612efacbb9c54840626048c1e49b08ee4bc2676a91c37bad
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/nodes/BuiltNode.kt	826	f97ffe56bb6915ff32862584529a637e35509bc0c955fff3ed2eab
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/nodes/CopiedNode.kt	1311	dee108dac6b6f620e39689ff0e12bb250fecd943efc55b7be3cb06
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/nodes/FoundNode.kt	1511	9f913f6ad58e834621c06a7a0962abaf77d484835776a9e3b73a03
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/nodes/NodeAdder.kt	1273	951aee72786cfa6d8fff249952595d7ef5723b2545ec7cfddd8c77
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/nodes/NodeBuilder.kt	3277	fb35c9c14d5849114a840429ea79cedf6e9b61cdab6ffa5c320c73
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/nodes/NodeCopier.kt	4434	91562f893a6b36bd249fd1272c50c061adec7c47278e3b416e2513
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/nodes/NodeFinder.kt	1274	a3019ca1ee492e0a9692f669b035a79ebcb1e56553dfa575bde28b
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/nodes/NodeInstance.kt	987	868cdc6e280d53c12a8564d803bb166ea4b9ac1df1fe000d156604
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/nodes/NodeInstanceRequest.kt	1227	1bab4debe6ba75398abd44e29623d08cfbc9b544aa205b85ac3d53

tools/network-builder/src/main/kotlin/net/corda/networkbuilder/nodes/NodeInstantiator.kt	4113	33d15f786a00d8524772fca939f1285c233cff1e2a57d0581014f9
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/nodes/NodePusher.kt	755	6963eee6750f9c2d59ae17e85e7bd4f881ac1602d56ea062034d8c
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/nodes/PushedNode.kt	942	96d204ab824e3562a1d2642cca5179a483b35e723fbce6ce391139
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/notaries/CopiedNotary.kt	520	7c93ecb24c61c06351a8e85d5fd4a9a7da0de4b622a678283f392f
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/notaries/NotaryCopier.kt	3319	61febca96ceb4663e254d5fce33519cb528e61e08468adfb825a07
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/serialization/SerializationHelper.kt	1617	748699fd26cc39bf61d3edfa1eac65d884cf490da1fb89c3e086c6
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/volumes/Volume.kt	2340	66867565adba3eb3f55c6185048e795005515e001f2af54436c9cd
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/volumes/azure/AzureSmbVolume.kt	4184	574e82031326451b177c84a2f069fba9328817b3e0d7a91202c1a8
tools/network-builder/src/main/kotlin/net/corda/networkbuilder/volumes/docker/LocalVolume.kt	1448	c680dbef7c2424b69178b209119078ba8b91bf5dcd3952efb36e6a
tools/worldmap/build.gradle	389	f587624413033345276d1aa4111de9d0770c7e95be2c642c67224f
tools/worldmap/src/main/kotlin/net/corda/worldmap/PhysicalLocationStructures.kt	3746	6315204e802cbbb2cb022f61d53c067d4d0983e2645b8fd18999bc
verifier/build.gradle	483	cf9162bd9e61a4376b38f83bea33463eebf9367a4972a698bc406d
verifier/src/main/kotlin/net/corda/verifier/ExternalVerificationContext.kt	2122	9b6253a538f48007db7af9b6dda1caca5ff5a120f4fe4c0b784965
verifier/src/main/kotlin/net/corda/verifier/ExternalVerifier.kt	10654	ef02fd188478ab90545dde8876ad803b24385aa913a3f242f7ac3b
verifier/src/main/kotlin/net/corda/verifier/ExternalVerifierNamedCacheFactory.kt	1572	7328ddf6d8888e336ed6e161ae5d198f754ff1c9b8a4babdb9e236
verifier/src/main/kotlin/net/corda/verifier/Main.kt	1527	8e16e9e607ce45cbfc75a60931006f8b0f58ddc35cab314056fba1